Confidence Transport for Relinearized Curvature in Contextual Bandits

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Abstract

Relinearized generalized Gauss–Newton and Fisher metrics use current model Jacobians, while bandit confidence sequences are naturally built from predictable collection-time features. We study when confidence can be transported between these geometries. A logarithmic path length for positive-definite metrics gives a two-sided exponential comparison without a small-drift assumption or fixed tangent subspace. Combining this comparison with a two-sided approximate-operator certificate yields an estimator-agnostic, solver-agnostic UCB theorem for measurable action spaces and approximate score maximization. Its regret closes through the sequential frozen information gain rather than an uncontrolled dynamic-width term. For differentiable squared-loss models, a corrected center makes the representation path arbitrary and predictable: no nonlinear optimizer certificate is needed. The construction also permits explicit historical and current misspecification envelopes and recovers the standard finite-dimensional LinUCB rate in the frozen linear exact-solve limit. The result is conditional on computable path, operator, confidence, and solver certificates; it is not a guarantee for unrestricted neural training, a complete exponential-family theorem, or verified floating-point certification.

1 Introduction

A differentiable contextual-bandit model supplies a current prediction query, typically a Jacobian $\mathbf{q}_t(a) = \nabla_\theta \mu_{\boldsymbol{\theta}_t}(x_t, a)$. The corresponding current GGN or Fisher width is geometrically natural for action selection. Statistical confidence, however, is built from the queries that were predictable when past rewards were collected. Relinearizing every replay item at $\boldsymbol{\theta}_t$ changes all summands at once and destroys the sequential rank-one update behind the usual self-normalized argument. This mismatch appears in neural UCB, neural-linear, and Laplace-style exploration (Zhou et al., 2020; Riquelme et al., 2018; Ritter et al., 2018; Daxberger et al., 2021).

The central question is not whether one curvature approximation is uniformly better. It is when a confidence statement in the predictable frozen metric can be transported to a current relinearized metric, and then to the operator and linear solver used by the policy. The required chain is

$$\begin{aligned} \text{reference confidence} &\longrightarrow \text{current metric}, \\ \text{current metric} &\longrightarrow \text{algorithmic operator}, \\ \text{algorithmic operator} &\longrightarrow \text{certified score}. \end{aligned}$$

Each arrow needs its own operational certificate.

Our primary transport quantity is the integrated normalized speed of an SPD path. For a path $V(\tau)$ it is $\int_0^1 \|V^{-1/2}\dot{V}V^{-1/2}\|_{\mathrm{op}} d\tau$. Differentiating $\log(v^\top V(\tau)v)$ for a fixed vector gives an exponential Loewner sandwich for every finite path length. Unlike the previous additive endpoint argument, the result does not require a smallness condition. A rectangular factor-path bound covers relinearized scalar GGN factors and, conditionally on an absolutely continuous output-weight factor, vector-output Fisher geometry.

The transported-UCB theorem is abstract at the confidence and solver interfaces. It permits a measurable action domain, a predictable approximate score oracle, action-dependent model bias, a certified upper width over the score domain, and solver sharpness only at the played action. A two-sided algorithmic-operator comparison then transports the selected width back to the sequential frozen potential. This lower comparison is essential. When only a one-sided operator certificate is

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available, the previous dynamic-width theorem remains valid but does not give the new closed guarantee.

For differentiable squared-loss models, we make the corrected center primary. It combines the current nonlinear prediction with a frozen-feature ridge correction. The resulting identity holds for any predictable representation sequence in the certified region, not only an ERM or approximate stationary point. Historical linearization and misspecification enter the confidence radius through an explicit energy; current errors enter additively. Constant nonvanishing misspecification can still make regret linear.

Contributions. **(1)** A logarithmic metric-transport lemma and factor-path certificates for moving, relinearized curvature, including a Hessian certificate that allows arbitrary gradient rotation and requires no small-drift premise (Lemmas 3 and 4 and Corollary 5). **(2)** A transported-UCB regret theorem with arbitrary predictable centers, measurable approximate score maximization, generic certified solver widths, two-sided approximate operators, and a closed frozen-potential bound (Theorem 1). **(3)** A corrected-center squared-loss specialization with approximate realizability and a primitive finite-dimensional rate that reduces to LinUCB in the frozen linear exact-solve case (Corollaries 7 and 8).

The contribution is the confidence transport and its contextual-bandit instantiation, not generic conjugate gradients, a generic UCB-to-regret reduction, or uniform superiority of full curvature. Earlier numerical work is scoped separately in Appendix M; it does not instantiate the new path and two-sided-operator premises.

2 Abstract confidence transport

The analysis separates the statistical confidence metric from the current relinearized metric and from the operator used by the algorithm. This separation is the main object of the paper; the squared-loss construction in Section 4 is one way to instantiate it.

Protocol and timing. Let $(\mathsf{A}, \mathcal{B})$ be a measurable action space. At round t, the learner observes a context and a nonempty measurable action domain $\mathcal{A}_t \subseteq \mathsf{A}$. Write $\mathcal{H}_t^-$ for the sigma-algebra before fresh round-t algorithmic randomization, and $\mathcal{H}_t = \mathcal{H}_t^- \vee \sigma(\Omega_t)$ after that randomization but before the reward. All quantities used to select the action are $\mathcal{H}_t$-measurable. We assume the action-indexed scores have measurable, finite suprema and that the score oracle below returns a measurable action. This is automatic for finite actions with a deterministic tie rule; a bare measurable action space alone does not guarantee a measurable selector.

Let $\mathbf{q}_t(a) \in \mathbb{R}^d$ be the prediction query. The three metrics are

$$\begin{aligned} \bar{\mathbf{V}}_1 &= \lambda \mathbf{I}, \\ \bar{\mathbf{V}}_{t+1} &= \bar{\mathbf{V}}_t + \sigma^{-2} \mathbf{q}_t(a_t) \mathbf{q}_t(a_t)^\top, \qquad (1) \\ \mathbf{V}_t &= \text{the exact current relinearized metric}, \\ \mathcal{C}_t &= \text{the algorithmic SPD metric}. \qquad (2) \end{aligned}$$

Here $\lambda > 0$. Define the corresponding widths

$$\begin{aligned} \bar{s}_t^2(a) &= \mathbf{q}_t(a)^\top \bar{\mathbf{V}}_t^{-1} \mathbf{q}_t(a), \\ s_t^2(a) &= \mathbf{q}_t(a)^\top \mathbf{V}_t^{-1} \mathbf{q}_t(a), \\ s_{\mathcal{C},t}^2(a) &= \mathbf{q}_t(a)^\top \mathcal{C}_t^{-1} \mathbf{q}_t(a). \end{aligned} \qquad (3)$$

The frozen information gain is

$$\gamma_T := \log \frac{\det \bar{\mathbf{V}}_{T+1}}{\det \bar{\mathbf{V}}_1} = \sum_{t=1}^{T} \log\bigl(1 + \sigma^{-2} \bar{s}_t^2(a_t)\bigr). \qquad (4)$$

Set $\gamma_0 = 0$, and define

$$\mathrm{reg}_t := \sup_{a \in \mathcal{A}_t} \mu_t^*(a) - \mu_t^*(a_t), \qquad R_T = \sum_{t=1}^{T} \mathrm{reg}_t. \qquad (5)$$

Assumption 1 (Reference confidence and operational certificates)**.** *For predictable $\beta_t \geq 0$, center $m_t(a)$, and action-dependent envelope $b_t(a) \geq 0$, a simultaneous confidence event satisfies*

$$|\mu_t^*(a) - m_t(a)| \leq \beta_t \bar{s}_t(a) + b_t(a) \qquad \text{for every } a \in \mathcal{A}_t. \qquad (6)$$

There is an absolutely continuous SPD path $\mathbf{V}_t(\tau)$ from $\bar{\mathbf{V}}_t$ to $\mathbf{V}_t$ with a predictable certificate

$$\begin{aligned} \mathcal{D}_t &:= \int_0^1 \left\| \mathbf{V}_t(\tau)^{-1/2} \dot{\mathbf{V}}_t(\tau) \mathbf{V}_t(\tau)^{-1/2} \right\|_{\mathrm{op}} d\tau \\ &\leq \bar{D}_t < \infty. \end{aligned} \qquad (7)$$

The algorithmic operator has a predictable two-sided certificate

$$0 < \kappa_{-,t} \leq \kappa_{+,t} < \infty, \qquad \kappa_{-,t} \mathbf{V}_t \preceq \mathcal{C}_t \preceq \kappa_{+,t} \mathbf{V}_t. \qquad (8)$$

Finally, the solver returns $\hat{s}_t(a) \geq 0$ such that

$$\begin{aligned} s_{\mathcal{C},t}(a) &\leq \hat{s}_t(a) && \text{for every } a \in \mathcal{A}_t, \\ \hat{s}_t(a_t) &\leq \alpha_t s_{\mathcal{C},t}(a_t), && \alpha_t \geq 1. \end{aligned} \qquad (9)$$

The first condition is an all-domain optimism certificate. The second is required only at the played action.

Use the transported score

$$U_t(a) = m_t(a) + \beta_t e^{\bar{D}_t/2} \sqrt{\kappa_{+,t}}\, \hat{s}_t(a) + b_t(a), \qquad (10)$$

and let a predictable score oracle return a_t satisfying

$$U_t(a_t) \geq \sup_{a \in \mathcal{A}_t} U_t(a) - \xi_t, \qquad \xi_t \geq 0. \tag{11}$$

Regret is defined against the mean-reward supremum, so an optimal action need not exist.

Theorem 1 (Corrected-center, estimator- and solver-agnostic transport)**.** *Suppose Assumption 1 holds, the score obeys* (10)–(11)*, and* $\|\mathbf{q}_t(a)\|_2 \leq G$*. On the joint confidence and certificate event, the instantaneous regret satisfies*

$$\mathrm{reg}_t \leq \beta_t \left[1 + \alpha_t e^{\bar{D}_t} \sqrt{\frac{\kappa_{+,t}}{\kappa_{-,t}}}\right] \bar{s}_t(a_t) + 2b_t(a_t) + \xi_t. \tag{12}$$

Consequently,

$$\begin{aligned} R_T \leq &\sqrt{\left(\sigma^2 + \frac{G^2}{\lambda}\right) \gamma_T \sum_{t=1}^{T} \beta_t^2 \left[1 + \alpha_t e^{\bar{D}_t} \sqrt{\frac{\kappa_{+,t}}{\kappa_{-,t}}}\right]^2} \\ &+ 2 \sum_{t=1}^{T} b_t(a_t) + \sum_{t=1}^{T} \xi_t. \end{aligned} \tag{13}$$

In particular, the simpler bound

$$\begin{aligned} R_T \leq 2 &\sqrt{\left(\sigma^2 + \frac{G^2}{\lambda}\right) \gamma_T \sum_{t=1}^{T} \alpha_t^2 \beta_t^2 e^{2\bar{D}_t} \frac{\kappa_{+,t}}{\kappa_{-,t}}} \\ &+ 2 \sum_{t=1}^{T} b_t(a_t) + \sum_{t=1}^{T} \xi_t \end{aligned} \tag{14}$$

holds. The bias envelope enters twice and the approximate-oracle error enters once. If the confidence event has probability at least $1 - \delta$ *and the joint certificate event has probability at least* $1 - \delta_{\mathrm{cert}}$*, these bounds hold with probability at least* $1 - \delta - \delta_{\mathrm{cert}}$*.*

Corollary 2 (Finite full enumeration)**.** *If* $\mathcal{A}_t$ *is finite and the learner maximizes* (10) *with a deterministic tie rule, then* (11) *holds with* $\xi_t = 0$*, and all score-supremum and selector measurability conditions in Theorem 1 are automatic.*

The headline theorem contains no optimizer residual, CG-specific quantity, or dynamic-width complexity. Its price is explicit: closing through the frozen potential requires the lower operator factor $\kappa_{-,t}$. The previous one-sided theorem remains useful when that factor is unavailable and is retained in Appendix B.

3 Logarithmic transport for relinearized curvature

Lemma 3 (Logarithmic comparison along an SPD path)**.** *Let* $\mathbf{V} : [0,1] \to \mathbb{S}^d_{++}$ *be entrywise absolutely continuous and let* $\dot{\mathbf{V}}(\tau)$ *denote its almost-everywhere derivative. If*

$$\mathcal{D}(\mathbf{V}) = \int_0^1 \|\mathbf{V}(\tau)^{-1/2} \dot{\mathbf{V}}(\tau) \mathbf{V}(\tau)^{-1/2}\|_{\mathrm{op}}\, d\tau, \tag{15}$$

then

$$e^{-\mathcal{D}(\mathbf{V})} \mathbf{V}(0) \preceq \mathbf{V}(1) \preceq e^{\mathcal{D}(\mathbf{V})} \mathbf{V}(0). \tag{16}$$

No commutativity or small-path assumption is required.

Algorithm 1 Corrected-center transported UCB

Observe the context and measurable action domain $\mathcal{A}_t$.
Choose any predictable representation parameter $\boldsymbol{\theta}_t$ in the certified region.
Form the query map $\mathbf{q}_t(a)$, corrected center $m_t^{\mathrm{corr}}(a)$, and bias envelope $b_t(a)$.
Construct $\mathcal{C}_t$ and obtain certificates $\bar{D}_t$, $\kappa_{-,t}$, and $\kappa_{+,t}$.
Obtain solver upper widths $\hat{s}_t(a)$ over the score domain.
Return an ξ_t-approximate maximizer of the score in (10).
Observe the reward; update the frozen ridge estimator and (1).

Lemma 4 (Rectangular factor-path certificate)**.** *Let* $\mathbf{B} : [0,1] \to \mathbb{R}^{m \times d}$ *be absolutely continuous and set* $\mathbf{V}(\tau) = \lambda \mathbf{I} + \mathbf{B}(\tau)^\top \mathbf{B}(\tau)$*, with* $\lambda > 0$*. Then*

$$\mathcal{D}(\mathbf{V}) \leq 2 \int_0^1 \|\dot{\mathbf{B}}(\tau) \mathbf{V}(\tau)^{-1/2}\|_{\mathrm{op}}\, d\tau. \tag{17}$$

Equivalently, if $\dot{\mathbf{B}}(\tau)^\top \dot{\mathbf{B}}(\tau) \preceq \nu_t(\tau)^2 \mathbf{V}(\tau)$ *almost everywhere, then* $\mathcal{D}(\mathbf{V}) \leq 2 \int_0^1 \nu_t(\tau) d\tau$*.*

For scalar differentiable means, take the sth row of $\mathbf{B}_t(\tau)$ to be

$$\sigma^{-1} \nabla_\theta \mu_{\boldsymbol{\theta}_s + \tau(\boldsymbol{\theta}_t - \boldsymbol{\theta}_s)}(x_s, a_s)^\top. \tag{18}$$

This interpolates from collection-time frozen curvature to the exact current relinearized curvature. Predictably weighted or heteroscedastic squared loss uses row $\sqrt{w_s}\, \nabla \mu^\top$ for a nonnegative collection-time weight w_s. If the weight is fixed along the interpolation, the same proof only rescales the corresponding row. If it changes with the path, the derivative of $\sqrt{w_s(\tau)}$ is an additional obligation.

Corollary 5 (Hessian path certificate)**.** *Suppose the parameter region is convex and* $\|\nabla^2_\theta \mu_\theta(x_s, a_s)\|_{\mathrm{op}} \leq L_g$ *along every segment in* (18)*. With*

$$Q_t = \sum_{s<t} \|\boldsymbol{\theta}_t - \boldsymbol{\theta}_s\|_2^2, \tag{19}$$

the exact current and frozen metrics satisfy

$$e^{-\bar{D}_t}\bar{\mathbf{V}}_t \preceq \mathbf{V}_t \preceq e^{\bar{D}_t}\bar{\mathbf{V}}_t, \qquad \bar{D}_t = \frac{2L_g\sqrt{Q_t}}{\sigma\sqrt{\lambda}}. \quad (20)$$

This remains valid when $\bar{D}_t \geq 1$. At $t = 1$, the factor has zero rows, $Q_1 = 0$, and both metrics equal $\lambda\mathbf{I}$.

The relative condition in Lemma 4 can be much sharper than the ridge-only bound in Corollary 5: excitation in the current metric attenuates motion in already-observed directions. It permits arbitrary gradient rotation and requires neither a fixed subspace nor known rank.

Proposition 6 (Width transport and approximate operators)**.** *Under* (7) *and* (8)*,*

$$\bar{s}_t^2(a) \leq e^{\bar{D}_t}\kappa_{+,t}s_{\mathcal{C},t}^2(a), \quad s_{\mathcal{C},t}^2(a) \leq e^{\bar{D}_t}\kappa_{-,t}^{-1}\bar{s}_t^2(a). \quad (21)$$

Exact current curvature has $\kappa_{-,t} = \kappa_{+,t} = 1$. More generally, the operational relative-error certificate

$$\|\mathbf{V}_t^{-1/2}(\mathcal{C}_t - \mathbf{V}_t)\mathbf{V}_t^{-1/2}\|_{\mathrm{op}} \leq \varepsilon_t < 1 \quad (22)$$

gives $\kappa_{-,t} = 1 - \varepsilon_t$ and $\kappa_{+,t} = 1 + \varepsilon_t$. An upper comparison alone does not imply a lower factor.

For vector outputs, let $J_s(\theta) \in \mathbb{R}^{p_s \times d}$ and supply an absolutely continuous factor $R_s(\theta) \in \mathbb{R}^{r_s \times p_s}$ satisfying $R_s^\top R_s = W_s \succeq 0$. Stacking $B_s = R_sJ_s \in \mathbb{R}^{r_s \times d}$ gives $\lambda I + \sum_s J_s^\top W_s J_s$. Along a path, $\dot{B}_s = \dot{R}_sJ_s + R_s\dot{J}_s$; both terms must be controlled. This covers the geometry of weighted Fisher/GGN factors but does not by itself prove a likelihood confidence sequence or an exponential-family regret theorem.

4 Corrected-center squared-loss specialization

We now construct (6) without assuming that the representation parameter minimizes a nonlinear loss. Let $\sigma > 0$ and let the reward obey $r_t = \mu_t^*(a_t) + \eta_t$, where, for every $u \in \mathbb{R}$,

$$\mathbb{E}[\exp(u\eta_t) \mid \mathcal{H}_t] \leq \exp(u^2\sigma^2/2). \quad (23)$$

Thus the noise condition is imposed after the action is selected and before the reward is revealed. Fix before data collection a reference parameter $\boldsymbol{\theta}^\circ \in \Theta$ with $\|\boldsymbol{\theta}^\circ\|_2 \leq S$, and allow approximate realizability

$$\mu^*(z) = \mu_{\boldsymbol{\theta}^\circ}(z) + m^\circ(z), qquad|m^\circ(z)| \leq \epsilon_{\mathrm{mis}}(z). \quad (24)$$

The predictable sequence $\boldsymbol{\theta}_t \in \Theta$ is otherwise arbitrary. Set $\mathbf{q}_t(a) = \nabla_\theta\mu_{\boldsymbol{\theta}_t}(x_t, a)$ and define collection-time pseudo-responses and the frozen ridge estimator by

$$y_s = r_s - \mu_{\boldsymbol{\theta}_s}(x_s, a_s) + \mathbf{q}_s(a_s)^\top\boldsymbol{\theta}_s, \quad (25)$$

$$\hat{\boldsymbol{\theta}}_t^{\mathrm{lin}} = \bar{\mathbf{V}}_t^{-1}\sigma^{-2}\sum_{s<t}\mathbf{q}_s(a_s)y_s. \quad (26)$$

The corrected center is

$$m_t^{\mathrm{corr}}(a) = \mu_{\boldsymbol{\theta}_t}(x_t, a) + \mathbf{q}_t(a)^\top(\hat{\boldsymbol{\theta}}_t^{\mathrm{lin}} - \boldsymbol{\theta}_t). \quad (27)$$

Fix $\delta \in (0, 1)$. Let ρ_s° be the Taylor remainder of $\mu_{\boldsymbol{\theta}^\circ}$ around $\boldsymbol{\theta}_s$, and supply envelopes

$$|\rho_s^\circ| \leq \epsilon_{\mathrm{lin}}(s), \qquad |m^\circ(x_s, a_s)| \leq \epsilon_{\mathrm{mis}}(s), \quad (28)$$

with the corresponding actionwise current envelopes $\epsilon_{\mathrm{lin,t}}(a)$ and $\epsilon_{\mathrm{mis,t}}(a)$. Each envelope is deterministic or measurable before the reward at the round where it is used; the complete historical sum below is known before selecting a_t. Define

$$\beta_t^{\mathrm{corr}} = \sqrt{\gamma_{t-1} + 2\log(1/\delta)} + \sqrt{\lambda}S + \frac{1}{\sigma}\sqrt{\sum_{s<t}[\epsilon_{\mathrm{lin}}(s) + \epsilon_{\mathrm{mis}}(s)]^2}. \quad (29)$$

Corollary 7 (Corrected-center confidence without optimizer assumptions)**.** *With probability at least $1 - \delta$, simultaneously over rounds and actions,*

$$|\mu_t^*(a) - m_t^{\mathrm{corr}}(a)| \leq \beta_t^{\mathrm{corr}}\bar{s}_t(a) + \epsilon_{\mathrm{lin,t}}(a) + \epsilon_{\mathrm{mis,t}}(a). \quad (30)$$

The identity and bound hold for any predictable $\boldsymbol{\theta}_t$ in the certified region. They require no exact ERM, approximate stationarity, optimizer residual, or optimizer convergence rate.

The historical deterministic term in (29) follows from $\Phi_t^\top\bar{\mathbf{V}}_t^{-1}\Phi_t \preceq I$ for $\Phi_t = [\mathbf{q}_1/\sigma, \ldots, \mathbf{q}_{t-1}/\sigma]$. Its scaling is $1/\sigma$, not $1/\sigma^2$. Fixed nonvanishing misspecification generally creates a linear current-bias term and can also make the historical radius vacuous. This is a graceful-degradation statement, not a robust misspecified-bandit guarantee.

5 Closed finite-dimensional rates and local alternatives

Let $r_T = \min\{d, T\}$. Rank–trace concavity gives the primitive closure

$$\gamma_T \leq \Gamma_{T,d} := r_T\log\left(1 + \frac{TG^2}{r_T\lambda\sigma^2}\right). \quad (31)$$

No fixed tangent subspace is required.

Corollary 8 (Finite-dimensional corrected-center rate)**.** *Suppose $\bar{D}_t \le D$, $\alpha_t \le \alpha$, and $\kappa_{+,t}/\kappa_{-,t} \le K$ for all t. Let*

$$B_T = \sum_{s<T} [\epsilon_{\mathrm{lin}}(s) + \epsilon_{\mathrm{mis}}(s)]^2,$$

$$E_T^{\mathrm{cur}} = \sum_{t=1}^{T} [\epsilon_{\mathrm{lin,t}}(a_t) + \epsilon_{\mathrm{mis,t}}(a_t)].$$

Then Theorem 1 and Corollary 7 imply

$$\begin{aligned} R_T \le\ & 2\alpha e^{D}\sqrt{K} \\ & \times \left[\sqrt{\Gamma_{T,d} + 2\log(1/\delta)} + \sqrt{\lambda}S + \frac{\sqrt{B_T}}{\sigma}\right] \\ & \times \sqrt{\left(\sigma^2 + \frac{G^2}{\lambda}\right) T\Gamma_{T,d}} \\ & + 2E_T^{\mathrm{cur}} + \sum_{t=1}^{T} \xi_t. \end{aligned} \tag{32}$$

With exact realizability and a frozen linear model, $B_T = E_T^{\mathrm{cur}} = 0$. For fixed d and problem constants, the leading term is then $O(e^D\sqrt{K}\,d\sqrt{T}\log T)$. Exact solves set $\alpha = 1$, exact current curvature sets $K = 1$, and exact score maximization sets $\xi_t = 0$, recovering the standard LinUCB determinant-potential rate up to displayed constants.

The previous additive stacked-design result remains a useful local corollary. If $\chi_t < 1$ and

$$(1-\chi_t)^2\bar{\mathbf{V}}_t \preceq \mathbf{V}_t \preceq (1+\chi_t)^2\bar{\mathbf{V}}_t, \tag{33}$$

use the local score obtained from (10) by replacing $e^{\bar{D}_t/2}$ with $1+\chi_t$.

Corollary 9 (Local small-drift transport)**.** *Under the remaining hypotheses of Theorem 1, the local score satisfies*

$$\begin{aligned} R_T \le\ & \sqrt{\left(\sigma^2 + \frac{G^2}{\lambda}\right)\gamma_T \sum_{t=1}^{T}\beta_t^2\left[1 + \alpha_t\frac{1+\chi_t}{1-\chi_t}\sqrt{\frac{\kappa_{+,t}}{\kappa_{-,t}}}\right]^2} \\ & + 2\sum_{t=1}^{T} b_t(a_t) + \sum_{t=1}^{T}\xi_t. \end{aligned} \tag{34}$$

Neither the local factor nor the exponential factor uniformly dominates the other because they summarize different paths.

Scope. The abstract theorem applies to any estimator that supplies (6). A complete exponential-family specialization would additionally require a likelihood-score confidence event, prediction conversion, predictable collection-time Fisher weights, control of weight and Jacobian transport, and an information-potential bound. The factor geometry above does not establish those statistical steps, so no such theorem is claimed here.

Route	Assumption	two-way factor	smallness	rotation
Logarithmic path	normalized SPD path length $\bar{D}_t$	$e^{\bar{D}_t}$	no	yes
Additive endpoint	whitened design error χ_t	$(1+\chi_t)/(1-\chi_t)$	yes	yes
Scalar link	fixed features, derivative ratios	ratio bound	lower ratio	scalar only
Exact current	$\mathcal{C}_t = \mathbf{V}_t$	1	no	yes
Approximate current	two-sided $\kappa_\pm$ certificate	$\sqrt{\kappa_+/\kappa_-}$	$\kappa_- > 0$	yes

Table 1: Transport certificates. "Rotation" records whether replay gradients may change direction.

6 Conclusion

Relinearized exploration can be analyzed without identifying the current metric with the predictable confidence metric. The logarithmic path comparison gives a two-sided transport certificate for every finite path length, and the transported-UCB theorem composes it with approximate operators, generic solver upper widths, and approximate score maximization. A lower operator certificate is what closes the selected width through the sequential frozen potential; the one-sided dynamic theorem remains the appropriate fallback when that premise is unavailable.

The corrected center removes nonlinear optimizer convergence from the concrete squared-loss confidence proof. Representation learning may follow any predictable path inside the certified region. The remaining obligations are visible rather than implicit: historical and current approximation error, metric path length, two-sided operator distortion, solver validity, and score oracle accuracy. In the frozen linear exact-solve limit these terms reduce to the usual LinUCB determinant-potential guarantee.

Limitations. The abstract theorem assumes a valid reference confidence event, measurable score supremum and approximate selector, a finite predictable path certificate, and a two-sided operator comparison. The squared-loss specialization requires a fixed ex-ante reference model and predictable linearization and misspecification envelopes. Constant misspecification can yield linear regret. The corrected center also needs access to the frozen-feature ridge estimator. The factor-path geometry does not by itself prove an exponential-family confidence theorem, and no unrestricted neural-training guarantee is claimed. Ordinary floating-point residual checks are diagnostics rather than verified numerical certificates. Legacy audits are summarized in Appendix M; none instantiates the revised theorem event.

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A Proofs for confidence transport

Proof of Theorem 1. Lemma 3 and (7) imply

$$\bar{s}_t(a) \le e^{\bar{D}_t/2} s_t(a), \qquad s_t(a) \le e^{\bar{D}_t/2} \bar{s}_t(a). \tag{35}$$

Inverting (8) gives

$$s_t(a) \le \sqrt{\kappa_{+,t}} s_{\mathcal{C},t}(a), \qquad s_{\mathcal{C},t}(a) \le \kappa_{-,t}^{-1/2} s_t(a). \tag{36}$$

Thus, for every scored action,

$$\mu_t^*(a) \le m_t(a) + \beta_t \bar{s}_t(a) + b_t(a) \le U_t(a),$$

where the final step uses the first inequalities in (35)–(36) and the all-domain solver upper certificate. Taking suprema does not require an optimizer to exist, so

$$\sup_{a \in \mathcal{A}_t} \mu_t^*(a) \le \sup_{a \in \mathcal{A}_t} U_t(a) \le U_t(a_t) + \xi_t.$$

The lower side of confidence at the played action gives

$$m_t(a_t) - \mu_t^*(a_t) \le \beta_t \bar{s}_t(a_t) + b_t(a_t).$$

Substituting the score and combining the last two displays yields

$$\begin{aligned} \mathrm{reg}_t &\le \beta_t \bar{s}_t(a_t) + \beta_t e^{\bar{D}_t/2} \sqrt{\kappa_{+,t}} \hat{s}_t(a_t) + 2b_t(a_t) + \xi_t \\ &\le \beta_t \left[1 + \alpha_t e^{\bar{D}_t} \sqrt{\frac{\kappa_{+,t}}{\kappa_{-,t}}} \right] \bar{s}_t(a_t) + 2b_t(a_t) + \xi_t, \end{aligned}$$

where the second line uses played-action sharpness and then the second inequalities in (36)–(35). This proves (12).

Let $c_t = \beta_t[1 + \alpha_t e^{\bar{D}_t} \sqrt{\kappa_{+,t}/\kappa_{-,t}}]$. Summation and one Cauchy–Schwarz application give

$$R_T - 2\sum_t b_t(a_t) - \sum_t \xi_t \le \sqrt{\sum_t c_t^2} \sqrt{\sum_t \bar{s}_t^2(a_t)}.$$

Since $\bar{s}_t^2(a_t) \le G^2/\lambda$, the scalar inequality $x \le (1+M)\log(1+x)$ for $0 \le x \le M$, with $x = \sigma^{-2}\bar{s}_t^2(a_t)$ and $M = G^2/(\lambda\sigma^2)$, yields

$$\sum_{t=1}^{T} \bar{s}_t^2(a_t) \le \left(\sigma^2 + \frac{G^2}{\lambda} \right) \gamma_T. \tag{37}$$

This proves (13). Finally, $\alpha_t e^{\bar{D}_t} \sqrt{\kappa_{+,t}/\kappa_{-,t}} \ge 1$, hence $1 + x \le 2x$ gives (14). The probability statement follows by a union bound; independence is not required. □

Proof of Corollary 2. A finite maximum is attained. Applying the fixed deterministic tie rule to the measurable score vector gives a measurable selected action and zero oracle error. □

Proof of Lemma 3. Fix $v \neq 0$ and set $f_v(\tau) = v^\top \mathbf{V}(\tau) v$. The path is continuous and positive definite on a compact interval, so f_v is absolutely continuous and bounded away from zero. Thus $\log f_v$ is absolutely continuous and, almost everywhere,

$$\frac{d}{d\tau} \log f_v(\tau) = \frac{v^\top \dot{\mathbf{V}}(\tau) v}{v^\top \mathbf{V}(\tau) v}.$$

With $w = \mathbf{V}(\tau)^{1/2} v$, the absolute value of this derivative is at most $\|\mathbf{V}(\tau)^{-1/2} \dot{\mathbf{V}}(\tau) \mathbf{V}(\tau)^{-1/2}\|_{\mathrm{op}}$. Integrating gives

$$e^{-\mathcal{D}(\mathbf{V})} v^\top \mathbf{V}(0) v \le v^\top \mathbf{V}(1) v \le e^{\mathcal{D}(\mathbf{V})} v^\top \mathbf{V}(0) v.$$

The inequality holds for every v, which is exactly (16). Inverting the sandwich gives the corresponding inverse-quadratic-form comparison. The proof never differentiates a matrix square root. □

Proof of Lemma 4. Almost everywhere, $\dot{\mathbf{V}} = \dot{\mathbf{B}}^\top \mathbf{B} + \mathbf{B}^\top \dot{\mathbf{B}}$. Put $E = \dot{\mathbf{B}}\mathbf{V}^{-1/2}$ and $H = \mathbf{B}\mathbf{V}^{-1/2}$. Then

$$\mathbf{V}^{-1/2}\dot{\mathbf{V}}\mathbf{V}^{-1/2} = E^\top H + H^\top E.$$

Moreover,

$$H^\top H = \mathbf{V}^{-1/2}\mathbf{B}^\top \mathbf{B}\mathbf{V}^{-1/2} = \mathbf{I} - \lambda \mathbf{V}^{-1} \preceq \mathbf{I},$$

so $\|H\|_{\mathrm{op}} \le 1$. Therefore the normalized SPD speed is at most $2\|E\|_{\mathrm{op}}$, which proves (17) after integration. For the PSD form, congruence by $\mathbf{V}^{-1/2}$ gives

$$\|\dot{\mathbf{B}}\mathbf{V}^{-1/2}\|_{\mathrm{op}}^2 = \|\mathbf{V}^{-1/2}\dot{\mathbf{B}}^\top \dot{\mathbf{B}}\mathbf{V}^{-1/2}\|_{\mathrm{op}} \le \nu_t(\tau)^2.$$

□

Proof of Corollary 5. The sth factor-row derivative is

$$\sigma^{-1}\left[\nabla_\theta^2 \mu_{\boldsymbol{\theta}_s + \tau(\boldsymbol{\theta}_t - \boldsymbol{\theta}_s)}(x_s, a_s)(\boldsymbol{\theta}_t - \boldsymbol{\theta}_s)\right]^\top.$$

Hence $\|\dot{\mathbf{B}}_t(\tau)\|_F^2 \le L_g^2 Q_t/\sigma^2$. Since $\mathbf{V}_t(\tau) \succeq \lambda \mathbf{I}$,

$$\dot{\mathbf{B}}_t(\tau)^\top \dot{\mathbf{B}}_t(\tau) \preceq \frac{L_g^2 Q_t}{\sigma^2 \lambda}\mathbf{V}_t(\tau).$$

Apply Lemma 4 with the constant $\nu_t = L_g\sqrt{Q_t}/(\sigma\sqrt{\lambda})$. □

Proof of Proposition 6. The logarithmic sandwich and inversion give

$$\bar{s}_t^2(a) \le e^{\bar{D}_t} s_t^2(a), \qquad s_t^2(a) \le e^{\bar{D}_t}\bar{s}_t^2(a).$$

Inverting the operator sandwich gives

$$s_t^2(a) \le \kappa_{+,t} s_{\mathcal{C},t}^2(a), \qquad s_{\mathcal{C},t}^2(a) \le \kappa_{-,t}^{-1} s_t^2(a).$$

Composition proves (21). For (22), congruence gives $-\varepsilon_t \mathbf{I} \preceq \mathbf{V}_t^{-1/2}(\mathcal{C}_t - \mathbf{V}_t)\mathbf{V}_t^{-1/2} \preceq \varepsilon_t \mathbf{I}$; undoing the congruence gives the claimed factors. □

Proof of Corollary 7. Taylor expansion around $\boldsymbol{\theta}_s$ gives

$$\mu_{\boldsymbol{\theta}^\circ}(x_s, a_s) = \mu_{\boldsymbol{\theta}_s}(x_s, a_s) + \mathbf{q}_s(a_s)^\top(\boldsymbol{\theta}^\circ - \boldsymbol{\theta}_s) + \rho_s^\circ.$$

Combining this with (24) and the reward model turns (25) into

$$y_s = \mathbf{q}_s(a_s)^\top \boldsymbol{\theta}^\circ + \eta_s + e_s, \qquad e_s = \rho_s^\circ + m^\circ(x_s, a_s).$$

The ridge normal equations therefore give

$$\bar{\mathbf{V}}_t(\hat{\boldsymbol{\theta}}_t^{\mathrm{lin}} - \boldsymbol{\theta}^\circ) = -\lambda\boldsymbol{\theta}^\circ + \sum_{s<t}\frac{\mathbf{q}_s(a_s)}{\sigma}\frac{\eta_s}{\sigma} + \frac{1}{\sigma}\Phi_t e, \tag{38}$$

where $\Phi_t = [\mathbf{q}_1(a_1)/\sigma, \ldots, \mathbf{q}_{t-1}(a_{t-1})/\sigma]$. The self-normalized martingale inequality bounds the middle term in $\bar{\mathbf{V}}_t^{-1}$ norm by $\sqrt{\gamma_{t-1} + 2\log(1/\delta)}$ simultaneously over t. The ridge term is at most $\sqrt{\lambda}S$. Finally, $\bar{\mathbf{V}}_t = \lambda\mathbf{I} + \Phi_t\Phi_t^\top$ implies $\Phi_t^\top \bar{\mathbf{V}}_t^{-1}\Phi_t \preceq \mathbf{I}$, so

$$\left\|\frac{1}{\sigma}\Phi_t e\right\|_{\bar{\mathbf{V}}_t^{-1}} \le \frac{\|e\|_2}{\sigma} \le \frac{1}{\sigma}\sqrt{\sum_{s<t}[\epsilon_{\mathrm{lin}}(s) + \epsilon_{\mathrm{mis}}(s)]^2}.$$

Cauchy–Schwarz in the $\bar{\mathbf{V}}_t$ metric bounds $|\mathbf{q}_t(a)^\top(\boldsymbol{\theta}^\circ - \hat{\boldsymbol{\theta}}_t^{\mathrm{lin}})|$ by $\beta_t^{\mathrm{corr}}\bar{s}_t(a)$.

For the prediction itself, direct algebra gives the exact identity

$$\mu_t^*(a) - m_t^{\mathrm{corr}}(a) = \mathbf{q}_t(a)^\top(\boldsymbol{\theta}^\circ - \hat{\boldsymbol{\theta}}_t^{\mathrm{lin}}) + \rho_t^\circ(a) + m^\circ(x_t, a).$$

Applying the current actionwise envelopes proves (30). No optimality property of $\boldsymbol{\theta}_t$ appears in the argument. □

Proof of Corollary 8. Let $A_T = \sigma^{-2}\sum_{t=1}^{T}\mathbf{q}_t(a_t)\mathbf{q}_t(a_t)^\top$. Its rank is at most $r_T = \min\{d, T\}$ and its trace is at most TG^2/σ^2. Concavity of $x \mapsto \log(1+x)$ over the nonzero eigenvalues gives

$$\gamma_T = \log\det(\mathbf{I} + A_T/\lambda) \le r_T \log\left(1 + \frac{TG^2}{r_T\lambda\sigma^2}\right).$$

The radius in (29) is bounded by the bracketed term in (32). Substitute this bound, the uniform transport factors, and (31) into (14). The current bias envelope is exactly $b_t(a_t) = \epsilon_{\mathrm{lin,t}}(a_t) + \epsilon_{\mathrm{mis,t}}(a_t)$. □

Proof of Corollary 9. Inverting (33) gives

$$\bar{s}_t(a) \le (1+\chi_t)s_t(a), \qquad s_t(a) \le (1-\chi_t)^{-1}\bar{s}_t(a).$$

The first inequality, the upper operator comparison, and solver upper validity make the local score optimistic. At the played action, solver sharpness, the lower operator comparison, and the second inequality bound its width term by

$$\alpha_t \frac{1+\chi_t}{1-\chi_t}\sqrt{\frac{\kappa_{+,t}}{\kappa_{-,t}}}\,\bar{s}_t(a_t).$$

Repeat the proof of Theorem 1 and apply (37). □

A.1 A concrete solver certificate

Lemma 10 (Inverse-quadratic interval from the original residual)**.** *Let $\mathcal{C} = \mathcal{C}^\top \succeq \lambda_{\mathcal{C}}\mathbf{I}$ with $\lambda_{\mathcal{C}} > 0$. For an approximate solution $\tilde{\mathbf{u}}$ to $\mathcal{C}\mathbf{u} = \mathbf{q}$, recompute the original residual $\mathbf{r} = \mathbf{q} - \mathcal{C}\tilde{\mathbf{u}}$. In exact arithmetic,*

$$\mathbf{q}^\top\mathcal{C}^{-1}\mathbf{q} = \mathbf{q}^\top\tilde{\mathbf{u}} + \tilde{\mathbf{u}}^\top\mathbf{r} + \mathbf{r}^\top\mathcal{C}^{-1}\mathbf{r}. \tag{39}$$

Consequently,

$$L = \mathbf{q}^\top\tilde{\mathbf{u}} + \tilde{\mathbf{u}}^\top\mathbf{r} \le \mathbf{q}^\top\mathcal{C}^{-1}\mathbf{q} \le L + \frac{\|\mathbf{r}\|_2^2}{\lambda_{\mathcal{C}}} = U. \tag{40}$$

Proof. The exact solution is $\mathcal{C}^{-1}\mathbf{q} = \tilde{\mathbf{u}} + \mathcal{C}^{-1}\mathbf{r}$. Substitute this expression and $\mathbf{q} = \mathcal{C}\tilde{\mathbf{u}} + \mathbf{r}$ into the quadratic form. The remainder is $\mathbf{r}^\top\mathcal{C}^{-1}\mathbf{r}$, which lies between zero and $\|\mathbf{r}\|_2^2/\lambda_{\mathcal{C}}$. □

Set $\hat{s} = \sqrt{U}$. If the played lower endpoint is positive, the observable factor $\alpha = \sqrt{U/L}$ verifies played-action sharpness. If $L = 0 < U$, no finite multiplicative factor follows; the solve must continue or an additive solver theorem must be used. For $\mathbf{q} = 0$, skip the solve and return $\hat{s} = 0$.

Corollary 11 (CG and PCG as solver-interface instances)**.** *Suppose CG or PCG supplies, in exact arithmetic, an original-system energy certificate $\|\mathbf{u} - \tilde{\mathbf{u}}\|_{\mathcal{C}} \le \varepsilon\|\mathbf{u}\|_{\mathcal{C}}$ with $\varepsilon < 1$. Then*

$$(1-\varepsilon)s_{\mathcal{C}}^2 \le \mathbf{q}^\top\tilde{\mathbf{u}} \le (1+\varepsilon)s_{\mathcal{C}}^2.$$

Thus

$$\hat{s} = \sqrt{\frac{\mathbf{q}^\top\tilde{\mathbf{u}}}{1-\varepsilon}}, \qquad \alpha = \sqrt{\frac{1+\varepsilon}{1-\varepsilon}}$$

instantiate (9)*. A preconditioner may change the recurrence, but certificate checks must use the original operator.*

Proof. Let $\mathbf{u} = \mathcal{C}^{-1}\mathbf{q}$. Since $s_{\mathcal{C}}^2 = \mathbf{q}^\top\mathbf{u} = \|\mathbf{u}\|_{\mathcal{C}}^2$,

$$\left|\mathbf{q}^\top\tilde{\mathbf{u}} - s_{\mathcal{C}}^2\right| = |\langle\mathbf{u}, \tilde{\mathbf{u}} - \mathbf{u}\rangle_{\mathcal{C}}| \le \|\mathbf{u}\|_{\mathcal{C}}\|\tilde{\mathbf{u}} - \mathbf{u}\|_{\mathcal{C}} \le \varepsilon s_{\mathcal{C}}^2.$$

This proves the two-sided quadratic-form comparison. Dividing its lower side by $1-\varepsilon$ gives upper validity, while combining both sides gives the displayed sharpness factor. For PCG the argument is unchanged once the certificate is expressed in the original $\mathcal{C}$-energy norm. □

Neither Lemma 10 nor Corollary 11 turns an ordinary floating-point residual into a verified enclosure. Literal numerical certification requires interval arithmetic, directed rounding with derived norm bounds, or another verified residual enclosure. Float64 point checks remain engineering diagnostics.

A.2 Measurability, randomized certificates, and edge cases

For certificate source j, it is sufficient to allocate a predictable failure budget before drawing its random object:

$$\Pr(\mathcal{E}_{j,t}^c \mid \mathcal{H}_t^-) \le \delta_{j,t}, \qquad \sum_{j,t} \delta_{j,t} \le \delta_{\text{cert}} \quad \text{almost surely.}$$

The tower property and a union bound give simultaneous validity at least $1 - \delta_{\text{cert}}$ without independence. On an uncountable score domain, the solver upper event must already be simultaneous. A pointwise certificate cannot be union-bounded after adaptive selection unless it is independently verified at the selected action with the correct conditioning. Realized terminal spectral quantities may be used for post-hoc analysis, but they cannot be inserted into an earlier action-time radius unless a predictable prefix envelope was supplied at that time.

The logarithmic theorem remains correct for finite $\bar{D}_t \ge 1$, but an exponentially large factor can be vacuous. Rotating ill-conditioned metrics can have large path length even with fixed eigenvalues; conversely, a closed path can have positive length and equal endpoints. The direct SPD length is canonical, whereas a factor-path bound depends on the chosen factorization. For data-dependent Fisher weights, an explicit absolutely continuous factor is required: a singular principal square root need not be differentiable. A uniform positive spectral floor is essential; positive damping supplies it automatically.

B One-sided dynamic fallback: original setup

This appendix preserves the earlier Gaussian/squared-loss, finite-action route. It remains applicable when only one-sided operator transfer is available, but its dynamic realized-width term is not the paper's headline closed guarantee. The optimizer-centered policy is retained as a secondary specialization.

We present the problem setup for the *Gaussian/squared-loss* instantiation, which is the setting analyzed in Theorem 12. An extension to exponential-family likelihoods is discussed in Appendix L.

Protocol. At each round $t = 1, \dots, T$ the learner observes a context $\mathbf{x}_t \in \mathcal{X}$ and a finite action set $\mathcal{A}_t$. It selects $a_t \in \mathcal{A}_t$ and observes a stochastic reward

$$r_t \;=\; \mu^*(\mathbf{x}_t, a_t) + \eta_t, \qquad \eta_t \text{ is } \sigma\text{-sub-Gaussian,} \tag{41}$$

where $\mu^* : \mathcal{X} \times \mathcal{A} \to \mathbb{R}$ is the unknown mean reward. The goal is to minimize cumulative regret $R_T = \sum_{t=1}^{T} \big[\mu^*(\mathbf{x}_t, a_t^*) - \mu^*(\mathbf{x}_t, a_t)\big]$, where $a_t^* = \arg\max_{a \in \mathcal{A}_t} \mu^*(\mathbf{x}_t, a)$. We assume $|\mathcal{A}_t| \le A_{\max}$ deterministically for all t.

Reward model and parameterization convention. We fix a pretrained initialization $\boldsymbol{\theta}_0 \in \mathbb{R}^d$ and parameterize the model by the *displacement* $\boldsymbol{\theta} \in \mathbb{R}^d$ from it: the deployed predictor is $\mu_{\boldsymbol{\theta}_0 + \boldsymbol{\theta}}(\mathbf{x}, a)$, and we abbreviate $\mu_{\boldsymbol{\theta}}(\mathbf{x}, a) := \mu_{\boldsymbol{\theta}_0 + \boldsymbol{\theta}}(\mathbf{x}, a)$ throughout. The initial displacement is $\boldsymbol{\theta}_1 = \mathbf{0}$ (the deployed model starts at the pretrained weights $\boldsymbol{\theta}_0$). With this convention the ridge penalty below is a penalty on the *displacement from the pretrained weights*, not a zero-centered prior on the raw weights, and the realizability radius S bounds $\|\boldsymbol{\theta}^\star\| = \|\boldsymbol{\theta}^\star_{\text{orig}} - \boldsymbol{\theta}_0\|$, the distance of the true parameter from the pretrained initialization. Define the regularized loss at round t:

$$\mathcal{L}_t(\boldsymbol{\theta}) \;:=\; \frac{1}{2\sigma^2} \sum_{s=1}^{t-1} \big(r_s - \mu_{\boldsymbol{\theta}}(\mathbf{x}_s, a_s)\big)^2 \;+\; \frac{\lambda}{2} \|\boldsymbol{\theta}\|^2. \tag{42}$$

We write $\boldsymbol{\theta}_t$ for the parameter returned by the warm-started optimizer on $\mathcal{L}_t$. Let $\mathcal{D}_{t-1} := \{(\mathbf{x}_s, a_s, r_s) : s < t\}$ and $N_t := |\mathcal{D}_{t-1}| = t - 1$. Equation (42) is a *full-history* objective. The curvature replay set $\mathcal{S}_t \subseteq \{1, \dots, t-1\}$ and its size $m_t := |\mathcal{S}_t|$ affect $\mathcal{C}_t$, but do not replace $\mathcal{D}_{t-1}$ in (42). Thus a generic full-batch evaluation of $\nabla \mathcal{L}_t$ replays N_t observations even when $m_t \ll N_t$. In convex settings (e.g. a frozen backbone with a linear head), under exact optimization this coincides with the global minimizer; in nonconvex settings $\boldsymbol{\theta}_t$ may be a local minimizer or approximate stationary point. The gap between $\boldsymbol{\theta}_t$ and an idealized global minimizer is absorbed by the action-scaled centering certificate (Assumption 5).

Gradient features and curvature matrices. Define the *gradient feature* (Jacobian of the mean reward) at the current parameter estimate:

$$\mathbf{g}_t(\mathbf{x}, a) \;:=\; \nabla_{\boldsymbol{\theta}}\, \mu_{\boldsymbol{\theta}_t}(\mathbf{x}, a) \;\in\; \mathbb{R}^d. \tag{43}$$

The algorithm uses the *relinearized curvature matrix*

$$\mathbf{C}_t \;=\; \lambda\,\mathbf{I} \;+\; \frac{1}{\sigma^2}\sum_{s=1}^{t-1} \mathbf{g}_t(\mathbf{x}_s, a_s)\, \mathbf{g}_t(\mathbf{x}_s, a_s)^\top, \tag{44}$$

the *damped GGN curvature* for the Gaussian/squared-loss model at the current parameter $\boldsymbol{\theta}_t$. (For the Gaussian working model underlying this squared-loss quasi-likelihood, the GGN coincides with the model Fisher information; see the remark below.) While $\mathbf{C}_t$ is the natural object for computing predictive variances at $\boldsymbol{\theta}_t$, it re-evaluates all past Jacobians at $\boldsymbol{\theta}_t$ and therefore does *not* admit a sequential rank-one update—all summands change when $\boldsymbol{\theta}_t$ changes.

For the regret analysis we introduce the *frozen-feature Gram matrix*:

$$\bar{\mathbf{C}}_t \;:=\; \lambda\,\mathbf{I} \;+\; \frac{1}{\sigma^2}\sum_{s=1}^{t-1} \mathbf{g}_s(\mathbf{x}_s, a_s)\, \mathbf{g}_s(\mathbf{x}_s, a_s)^\top, \tag{45}$$

where each $\mathbf{g}_s(\mathbf{x}_s, a_s) = \nabla_{\boldsymbol{\theta}}\mu_{\boldsymbol{\theta}_s}(\mathbf{x}_s, a_s)$ is evaluated at the parameter $\boldsymbol{\theta}_s$ that was current when the sample was collected. This matrix satisfies the recursive update

$$\bar{\mathbf{C}}_{t+1} = \bar{\mathbf{C}}_t + \sigma^{-2}\mathbf{g}_t(\mathbf{x}_t, a_t)\mathbf{g}_t(\mathbf{x}_t, a_t)^\top, \tag{46}$$

and, critically, each realized design vector $\mathbf{g}_s(\mathbf{x}_s, a_s)$ is measurable with respect to the pre-reward sigma-algebra $\mathcal{G}_s = \mathcal{H}_s \vee \sigma(a_s)$ of (144), which is the filtration used in the self-normalized bound. Both the additive update (46) and this measurability are structural requirements of the self-normalized martingale bound and the elliptic potential lemma invoked in Appendix D. The algorithm computes UCB widths using the algorithmic curvature $\mathcal{C}_t$ (via CG; the exact case is $\mathcal{C}_t = \mathbf{C}_t$), but the confidence analysis is conducted in terms of $\bar{\mathbf{C}}_t$; the one-sided optimism-transfer factor of Assumption 6, certified by Lemma 30, provides the bridge.

C Original-center CC-UCB specialization

C.1 CC-UCB Algorithm

For a predictable SPD oracle $\mathcal{C}_t \succeq \lambda\mathbf{I}$, set $\omega_t = \bar{\beta}_t + \bar{\psi}_t$ for the original nonlinear center. Algorithm 2 is the analyzed full-enumeration policy. The same fixed curvature operator is reused across separate per-action CG solves; a solution for one action is not reused as the solution for another action.

Algorithm 2 Curvature-Calibrated UCB (CC-UCB)

Observe context $\mathbf{x}_t$ and finite action set $\mathcal{A}_t$.
Construct a fixed predictable SPD oracle $\mathbf{v} \mapsto \mathcal{C}_t\mathbf{v}$ with $\mathcal{C}_t \succeq \lambda\mathbf{I}$.
for each $a \in \mathcal{A}_t$ **do**
  Compute $\widehat{\mu}_t(a) = \mu_{\boldsymbol{\theta}_t}(\mathbf{x}_t, a)$ and $\mathbf{g}_t(a) = \nabla_{\boldsymbol{\theta}}\mu_{\boldsymbol{\theta}_t}(\mathbf{x}_t, a)$.
  If $\mathbf{g}_t(a) \neq \mathbf{0}$, solve $\mathcal{C}_t\mathbf{u} = \mathbf{g}_t(a)$ separately by CG until the certified tolerance holds; otherwise set $\tilde{\mathbf{u}} = \mathbf{0}$.
  Set $\tilde{s}_t^2(a) = \mathbf{g}_t(a)^\top\tilde{\mathbf{u}}$.
end for
Select and play $a_t \in \arg\max_{a\in\mathcal{A}_t}\{\widehat{\mu}_t(a) + \omega_t\sqrt{u_t(a)/(1-\bar{\varepsilon}_t)}\sqrt{\tilde{s}_t^2(a)}\}$.
Observe r_t, append $(\mathbf{x}_t, a_t, r_t)$, and update $\boldsymbol{\theta}_{t+1}$.

The oracle is accessed only through matrix–vector products. At each round the agent computes, for every candidate action a, the gradient feature $\mathbf{g}_t(a) = \nabla_{\boldsymbol{\theta}}\mu_{\boldsymbol{\theta}_t}(\mathbf{x}_t, a)$ and the *curvature-calibrated variance proxy*, the generic algorithmic width,

$$s^2_{\mathcal{C},t}(a) \;=\; \mathbf{g}_t(a)^\top\mathcal{C}_t^{-1}\,\mathbf{g}_t(a). \tag{47}$$

Under the GGN/Fisher-linearized Gaussian approximation with covariance $\mathcal{C}_t^{-1}$, the quantity $s^2_{\mathcal{C},t}(a)$ is the metric-dependent predictive variance of the linearized reward at $(\mathbf{x}_t, a)$. The *exact* full-history instantiation of CC-UCB takes $\mathcal{C}_t = \mathbf{C}_t$, the relinearized curvature (44); weighted replay, subsampling, sliding windows, periodic refreshes, and randomized sketches are special implementations of $\mathcal{C}_t$ (§F.3), covered by the theory only when the corresponding one-sided transfer factor is certified.

Because $\mathcal{C}_t \succ 0$ by construction, conjugate gradients (CG) is guaranteed to converge on the system $\mathcal{C}_t\mathbf{u} = \mathbf{g}_t(a)$, provided the operator is held *fixed* across all iterations of the solve (the randomized sketch/subsample is drawn into Ω_t before action selection; §B). For each action with $\mathbf{g}_t(a) \neq \mathbf{0}$ we run CG to the per-action relative energy-norm error

$$\varepsilon_{t,a} \;:=\; \frac{\|\mathbf{u}_{t,a} - \tilde{\mathbf{u}}_{t,a}\|_{\mathcal{C}_t}}{\|\mathbf{u}_{t,a}\|_{\mathcal{C}_t}}, \qquad \mathbf{u}_{t,a} := \mathcal{C}_t^{-1}\mathbf{g}_t(a), \tag{48}$$

yielding the computable proxy $\tilde{s}^2_t(a) = \mathbf{g}_t(a)^\top \tilde{\mathbf{u}}_{t,a}$; if $\mathbf{g}_t(a) = \mathbf{0}$ we set $\tilde{s}^2_t(a) = 0$ and skip the solve (leaving the residual ratio $\|r_I\|/\|\mathbf{g}\|$ undefined), and by convention put $\varepsilon_{t,a} := 0$ in this case, so that $\varepsilon_{t,a}$ is defined for every $a \in \mathcal{A}_t$ and (129) holds trivially there. Lemma 37 shows $\tilde{s}^2_t(a)$ approximates $s^2_{\mathcal{C},t}(a)$ within a multiplicative $(1 \pm \varepsilon_{t,a})$ factor. To ensure the UCB bonus *upper-bounds* the confidence width $\omega_t\, \bar{s}_t(a)$ despite the approximation, the algorithm inflates the width by $\sqrt{u_t(a)/(1-\bar{\varepsilon}_t)}$ (line 8), where $\bar{\varepsilon}_t \in (0, 1)$ is a *per-round* known tolerance enforced uniformly over all actions,

$$\max_{a\in\mathcal{A}_t} \varepsilon_{t,a} \;\le\; \bar{\varepsilon}_t \;<\; 1, \tag{49}$$

and $u_t(a) \ge 1$ is the one-sided transfer factor (Assumption 6).

D One-sided dynamic analysis

The motivating scalar-link structure is formalized in Lemma 31, but the proof dependencies run in the opposite direction: we first state a master transfer theorem, then its exact-current, spectral-tail, and gap-dependent closures, and finally the operational and scalar-link specializations. The one-sided fallback theorem analyzes CC-UCB (Algorithm 2, with full action enumeration $K{=}|\mathcal{A}_t|$) under the Gaussian/squared-loss setup and Assumptions 2 to 6. It is not a full exponential-family or generalized-linear-bandit proof; the extension in Appendix L defines curvature only.

For a general predictable SPD sequence $\mathcal{C}_t$, confidence is proved in the frozen Gram $\bar{\mathbf{C}}_t$, whose update and measurability support self-normalized concentration. A one-sided factor $u_t(a)$ transfers optimism to $\mathcal{C}_t$, CG contributes α_t, and nonmonotonicity remains in the realized width complexity. Appendix F.5 gives exact determinant bookkeeping, not a primitive sublinear bound. Later specializations make the nonlinear inputs observable, but their smallness still requires a restricted parameter path.

Core conditions and links. Assume conditionally σ-sub-Gaussian noise, $\|\mathbf{g}_t(\mathbf{x}_s, a)\| \le G$, realizability by $\|\boldsymbol{\theta}^*\| \le S$, and a local expansion with round-t remainder at most $\varepsilon_{\mathrm{lin}}(t)$. For the original center assume the action-scaled discrepancy is at most $\bar{\psi}_t\bar{s}_t(a)$. The operator $\mathcal{C}_t$ is predictable, SPD, satisfies $\mathcal{C}_t \succeq \lambda\mathbf{I}$ and $\mathcal{C}_1 = \lambda\mathbf{I}$, and remains fixed throughout every CG solve. Define

$$\begin{aligned} &\bar{s}^2_t(a) = \mathbf{g}_t(a)^\top \bar{\mathbf{C}}_t^{-1}\mathbf{g}_t(a), \quad s^2_{\mathcal{C},t}(a) = \mathbf{g}_t(a)^\top \mathcal{C}_t^{-1}\mathbf{g}_t(a), \\ &\tilde{s}^2_t(a) = \mathbf{g}_t(a)^\top \tilde{\mathbf{u}}_t(a). \end{aligned} \tag{50}$$

The analysis requires only the one-sided transfer and uniform CG relations

$$\begin{aligned} \bar{s}^2_t(a) &\le u_t(a) s^2_{\mathcal{C},t}(a), \\ (1-\bar{\varepsilon}_t)s^2_{\mathcal{C},t}(a) &\le \tilde{s}^2_t(a) \le (1+\bar{\varepsilon}_t)s^2_{\mathcal{C},t}(a), \end{aligned} \tag{51}$$

for every candidate action, where $u_t(a) \ge 1$, $\bar{\varepsilon}_t < 1$, and $\alpha_t = \sqrt{(1+\bar{\varepsilon}_t)/(1-\bar{\varepsilon}_t)}$. Sufficient feature-drift and approximate-operator constructions are given in Lemma 30 and Theorem 45; no lower spectral transfer is needed.

Dynamic width complexity. The realized quantity used below is

$$\Lambda^{\mathcal{C}}_T := \sum_{t=1}^{T} \log\bigl(1 + \sigma^{-2} s^2_{\mathcal{C},t}(a_t)\bigr)\,. \tag{52}$$

The bounded-width cap gives

$$\sum_{t=1}^{T} s^2_{\mathcal{C},t}(a_t) \le (\sigma^2 + G^2/\lambda)\Lambda^{\mathcal{C}}_T. \tag{53}$$

For a general predictable nonmonotone SPD sequence, we do not derive a primitive sublinear bound on $\Lambda^{\mathcal{C}}_T$. Appendix Appendix F.5 gives an exact determinant bookkeeping identity, but its upper bound is informative only when both the terminal log determinant and determinant-decreasing refresh charge are independently controlled, for example under the rank and spectral-floor hypotheses of Lemma 35.

D.1 Regret Bound

Let $\mathcal{E}_{\mathrm{conf}}$ denote the simultaneous self-normalized confidence event and let $\mathcal{E}_{\mathrm{cert}}$ collect the predictable linearization, centering, transfer, condition-number, CG, and randomized-operator certificates. The appendix defines each component and its failure allocation; write their total failure probability as δ_{cert}.

Theorem 12 (Dynamic transfer regret bound for CC-UCB (Gaussian/squared-loss))**.** *Under the conditions above and Assumptions 2 to 6, suppose $|\mathcal{A}_t| \le A_{\max}$, the optimal action is present, and Algorithm 2 enumerates all actions using a predictable schedule $\bar\beta_t \ge \beta_t$ and uniform per-round CG tolerance. Let $E_T = \sum_{t=1}^{T} \varepsilon_{\mathrm{lin}}(t)$, $\omega_t = \bar\beta_t + \bar\psi_t$, $\alpha_t = \sqrt{(1+\bar\varepsilon_t)/(1-\bar\varepsilon_t)}$, and $S_T = \sum_{t=1}^{T} \alpha_t^2 u_t(a_t)\omega_t^2$. On $\mathcal{E}_{\mathrm{conf}} \cap \mathcal{E}_{\mathrm{cert}}$,*

$$R_T \;\le\; 2\sqrt{(\sigma^2 + G^2/\lambda)\,\Lambda^{\mathcal{C}}_T\,S_T} \;+\; 2\,E_T, \tag{54}$$

with probability at least $1-\delta-\delta_{\mathrm{cert}}$. Deterministic or almost-sure certificates contribute zero to δ_{cert}. For the corrected center, the same statement holds without Assumption 5 after replacing ω_t by $\bar\beta_t$.

In particular, the bound is sublinear whenever $\Lambda^{\mathcal{C}}_T S_T = o(T^2)$ and $E_T = o(T)$. The exact-curvature instantiation is $\mathcal{C}_t = \mathbf{C}_t$ with $u_t = (1+\bar\chi_t)^2$ from Lemma 30; approximate operators are covered through Theorem 45. The proof is in Appendix H.

Proof sketch. Self-normalized confidence plus the centering and linearization terms gives $|\mu^*(a) - \mu_{\boldsymbol{\theta}_t}(a)| \le \omega_t \bar s_t(a) + \varepsilon_{\mathrm{lin}}(t)$. The one-sided relation and CG lower bound make the computed bonus optimistic, so instantaneous regret is at most $2w_t(a_t) + 2\varepsilon_{\mathrm{lin}}(t)$. The CG upper bound gives $w_t(a_t) \le \alpha_t\omega_t\sqrt{u_t(a_t)}s_{\mathcal{C},t}(a_t)$. Summing, applying Cauchy–Schwarz only once, and using (53) yields (54); hence time-varying factors remain inside S_T rather than being replaced by horizon maxima. The corrected center replaces ω_t by $\bar\beta_t$ and removes the centering certificate, but requires exact or certified frozen-feature access (Corollary 21).

Corollary 13 (Exact-current regret through the frozen potential)**.** *Adopt Theorem 12 with exact current curvature $\mathcal{C}_t = \mathbf{C}_t$. Suppose a predictable certificate satisfies $\chi_t \le \bar\chi_t < 1$ for every t, and use the valid transfer factor $u_t = (1+\bar\chi_t)^2$. Then, on the event of Theorem 12,*

$$R_T \le 2\left\{(\sigma^2 + G^2/\lambda)\,\gamma_T \times \sum_{t=1}^{T} \alpha_t^2\omega_t^2 \left(\frac{1+\bar\chi_t}{1-\bar\chi_t}\right)^2\right\}^{1/2} + 2E_T. \tag{55}$$

For the corrected center of Corollary 21, the same statement holds without Assumption 5 after replacing ω_t by $\bar\beta_t$. This removes the nonmonotone dynamic-width term only in the small-drift, exact-current regime.

Proof. The sharpened sandwich in Corollary 32 and Loewner inversion give

$$s_t(a) \le \frac{\bar s_t(a)}{1-\bar\chi_t}.$$

The per-round proof of Theorem 12, together with $\sqrt{u_t} = 1+\bar\chi_t$, therefore gives

$$\mathrm{regret}_t \le 2\alpha_t\omega_t \frac{1+\bar\chi_t}{1-\bar\chi_t}\bar s_t(a_t) + 2\varepsilon_{\mathrm{lin}}(t).$$

Sum and apply Cauchy–Schwarz once. The frozen rank-one update and the usual elliptic-potential inequality give $\sum_t \bar{s}_t^2(a_t) \le (\sigma^2 + G^2/\lambda)\gamma_T$, proving the display. The corrected-center claim follows from the prediction identity in Corollary 21. $\square$

For an integer $0 \le r \le d$, define

$$\begin{aligned} \Delta_{T,r} &:= \sum_{i>r} \nu_i, \qquad r_T := \min\{r, T\}, \\ \Gamma_{T,r}^{\mathrm{tail}} &:= r_T \log\left(1 + \frac{TG^2}{r_T \lambda \sigma^2}\right) + \frac{\Delta_{T,r}}{\lambda}, \end{aligned} \tag{56}$$

where $\nu_1 \ge \cdots \ge \nu_d \ge 0$ are the eigenvalues of $\bar{\mathbf{C}}_{T+1} - \lambda \mathbf{I}$ and the first term is defined to be zero when $r_T = 0$. Lemma 19 proves $\gamma_T \le \Gamma_{T,r}^{\mathrm{tail}}$. Writing $\tau_T = \sum_i \nu_i$ and $q_{T,r} = \min\{d - r, \max\{T - r, 0\}\}$, the same lemma proves the sharper realized split bound

$$\begin{aligned} \Gamma_{T,r}^{\mathrm{split}} := r_T \log&\left(1 + \frac{\tau_T - \Delta_{T,r}}{r_T \lambda}\right) \\ &+ q_{T,r} \log\left(1 + \frac{\Delta_{T,r}}{q_{T,r}\lambda}\right), \end{aligned} \tag{57}$$

where a term with zero multiplier is defined as zero.

Corollary 14 (Approximate-rank exact-current GGN)**.** *Under Corollary 13, on the same confidence event the following holds simultaneously for every $r \in \{0, \ldots, d\}$:*

$$\begin{aligned} R_T \le 2\Bigg\{ &(\sigma^2 + G^2/\lambda)\, \Gamma_{T,r}^{\mathrm{tail}} \\ &\times \sum_{t=1}^{T} \alpha_t^2 \omega_t^2 \left(\frac{1 + \bar{\chi}_t}{1 - \bar{\chi}_t}\right)^2 \Bigg\}^{1/2} + 2E_T. \end{aligned} \tag{58}$$

The same display holds with $\Gamma_{T,r}^{\mathrm{tail}}$ replaced by the smaller $\Gamma_{T,r}^{\mathrm{split}}$ from (57)*. This is a realized-complexity statement: the policy may retain the existing observable $\hat{\gamma}_{t-1}$ schedule, but a terminal realized $\Delta_{T,r}$ is not used in any action-time radius. Because the deterministic spectral inequality holds for every r, the terminal report may minimize $\Gamma_{T,r}^{\mathrm{tail}}$ over r a posteriori, provided that choice does not alter any past action, radius, or other action-time quantity.*

Operationally, if before round t a supplied deterministic or $\mathcal{H}_t$-measurable envelope $\bar{\Delta}_{t-1,r}$ upper-bounds the spectral tail of $\bar{\mathbf{C}}_t - \lambda\mathbf{I}$, the old predictable envelope below remains valid:

$$\begin{aligned} r_{t-1} &:= \min\{r, t-1\}, \\ \bar{\Gamma}_{t-1,r}^{\mathrm{tail}} &:= r_{t-1} \log\left(1 + \frac{(t-1)G^2}{r_{t-1}\lambda\sigma^2}\right) + \frac{\bar{\Delta}_{t-1,r}}{\lambda}. \end{aligned}$$

again taking the first term as zero when $r_{t-1} = 0$. Replacing γ_{t-1} in the constructive confidence radius by this predictable upper bound is valid. In that version, (58) *holds with $\Gamma_{T,r}^{\mathrm{tail}}$ replaced by any supplied terminal envelope $\bar{\Gamma}_{T,r}^{\mathrm{tail}}$ and with ω_t formed from the resulting radius.*

A refined predictable alternative uses the observable frozen trace $\tau_{t-1} = \mathrm{tr}(\bar{\mathbf{C}}_t - \lambda\mathbf{I}) = \sigma^{-2}\sum_{s<t} \|\mathbf{g}_s\|^2$. With $q_{t-1,r} = \min\{d - r, \max\{t - 1 - r, 0\}\}$, set

$$\begin{aligned} \bar{\Gamma}_{t-1,r}^{\mathrm{split}} := r_{t-1} \log&\left(1 + \frac{\tau_{t-1}}{r_{t-1}\lambda}\right) \\ &+ q_{t-1,r} \log\left(1 + \frac{\bar{\Delta}_{t-1,r}}{q_{t-1,r}\lambda}\right), \end{aligned} \tag{59}$$

again omitting zero-multiplier terms. The head term deliberately uses the full trace: subtracting an upper *tail envelope could reduce the head term and invalidate an upper bound. Thus $\bar{\Gamma}_{t-1,r}^{\mathrm{split}} \ge \gamma_{t-1}$ is a valid action-time schedule, although it can be looser than the realized split expression.*

No exact tangent subspace is assumed. A small spectral tail can contain decision-relevant directions and does not supply their basis, so full-dimensional CG may still be necessary. Exact rank is recovered when $\Delta_{T,r} = 0$; the supplied-subspace result Corollary 27 is then an additional computational special case. The corrected-center version again replaces ω_t by $\bar{\beta}_t$.

Proof. Apply Lemma 19 to substitute either $\gamma_T \le \Gamma^{\mathrm{split}}_{T,r} \le \Gamma^{\mathrm{tail}}_{T,r}$ in (55). The prefix version of the same lemma makes either $\bar{\Gamma}^{\mathrm{tail}}_{t-1,r}$ or $\bar{\Gamma}^{\mathrm{split}}_{t-1,r}$ a predictable upper bound on γ_{t-1}, so Lemma 40 and Corollary 41 justify the operational radius. No eigenspace is used by either argument. □

Corollary 15 (Gap-dependent exact-current regret)**.** *Adopt Corollary 13. Define the realized gap $\Delta_t := \mu^*(\mathbf{x}_t, a_t^*) - \mu^*(\mathbf{x}_t, a_t)$ and suppose every suboptimal candidate action at every round has gap at least a deterministic (or almost-sure) $\Delta_{\min} > 0$. This gap is an analysis condition and is not used to tune the policy; if it holds only on a high-probability event, that event must be added to the certificate failure allocation. Suppose also $\varepsilon_{\mathrm{lin}}(t) \le \Delta_{\min}/4$ for all t, and set*

$$H_t^{\mathrm{gap}} := \alpha_t \omega_t \frac{1+\bar{\chi}_t}{1-\bar{\chi}_t}, \qquad H_{\max}^{\mathrm{gap}} := \max_{t \le T} H_t^{\mathrm{gap}}.$$

Then, on the event of Theorem 12,

$$R_T \le \frac{16 (H_{\max}^{\mathrm{gap}})^2}{\Delta_{\min}} \left(\sigma^2 + \frac{G^2}{\lambda} \right) \gamma_T. \tag{60}$$

Consequently γ_T in this display may be replaced by the exact-rank bound $\Gamma_{T,r}$ of (66), the simple tail bound $\Gamma^{\mathrm{tail}}_{T,r}$ of (56), or the refined $\Gamma^{\mathrm{split}}_{T,r}$ of (57), under the corresponding premises. The corrected-center version replaces ω_t by $\bar{\beta}_t$ as usual. This result is not a minimax-optimality claim: unlike the gap-free result, it requires a uniform positive gap and a per-round linearization error smaller than one quarter of that gap.

Proof. The per-round argument in Corollary 13 gives

$$\Delta_t \le 2 H_t^{\mathrm{gap}} \bar{s}_t(a_t) + 2\varepsilon_{\mathrm{lin}}(t).$$

On a suboptimal round, $\Delta_t \ge \Delta_{\min}$ and hence $2\varepsilon_{\mathrm{lin}}(t) \le \Delta_t/2$. Rearranging and using $H_t^{\mathrm{gap}} \le H_{\max}^{\mathrm{gap}}$ gives $\Delta_t \le 4 H_{\max}^{\mathrm{gap}} \bar{s}_t(a_t)$. Squaring this inequality and using $\Delta_t \ge \Delta_{\min}$ yields

$$\Delta_t \le \frac{\Delta_t^2}{\Delta_{\min}} \le \frac{16 (H_{\max}^{\mathrm{gap}})^2}{\Delta_{\min}} \bar{s}_t^2(a_t).$$

Optimal rounds contribute zero. Sum over t and apply $\sum_t \bar{s}_t^2(a_t) \le (\sigma^2 + G^2/\lambda)\gamma_T$ to prove the display; the three substitutions follow from Lemmas 18 and 19. □

E Additional conditional rates

E.1 Online nonlinear certificates

At the start of round t, let $n_t = t-1$, and let $\bar{\boldsymbol{\theta}}_{<t}$ and $J_{<t} = \sum_{s<t} \|\boldsymbol{\theta}_s - \bar{\boldsymbol{\theta}}_{<t}\|^2$ be maintained by a vector Welford update. With collection residuals $c_s = \mu_{\boldsymbol{\theta}_s}(x_s, a_s) - r_s$, define

$$Q_t = J_{<t} + n_t \|\boldsymbol{\theta}_t - \bar{\boldsymbol{\theta}}_{<t}\|^2, \qquad R_t^{\mathrm{col}} = \sum_{s<t} c_s^2. \tag{61}$$

Both are pre-action quantities and $Q_t = \sum_{s<t} \|\boldsymbol{\theta}_t - \boldsymbol{\theta}_s\|^2$ exactly. If prediction gradients are L_g-Lipschitz on the certified region, set

$$\bar{\chi}_t = \frac{L_g \sqrt{Q_t}}{\sigma \sqrt{\lambda}}, \qquad u_t = \kappa_{+,t}(1+\bar{\chi}_t)^2, \tag{62}$$

where $\kappa_{+,t} = 1$ for exact full curvature and for an unrescaled current-parameter subset, and otherwise is a supplied one-sided operator certificate. If $\|\nabla\mathcal{L}_t(\boldsymbol{\theta}_t)\| \le \zeta_t$, an observable centering schedule is

$$\begin{aligned}\bar{\psi}_t &= \frac{\zeta_t + \bar{M}_t}{\sqrt{\lambda}},\\ \bar{M}_t &= \sigma^{-2}\Big[L_g\sqrt{R_t^{\mathrm{col}}Q_t} + \frac{3}{2}GL_gQ_t + \frac{1}{2}L_g^2Q_t^{3/2}\Big].\end{aligned} \tag{63}$$

For a known radius-R trust region, the final numerator term improves to $L_g^2RQ_t$. Finally, for an L_μ-Lipschitz prediction gradient define $\bar{\epsilon}_t^{\mathrm{lin}} = \frac{L_\mu}{2}(S + \|\boldsymbol{\theta}_t\|)^2$, $\bar{F}_t = \sum_{s<t}(\bar{\epsilon}_s^{\mathrm{lin}})^2$, and

$$\begin{aligned}\bar{\beta}_t &= \sqrt{\hat{\gamma}_{t-1} + 2\log(1/\delta)} + \sqrt{\lambda}S + \sigma^{-1}\sqrt{\bar{F}_t},\\ \hat{\gamma}_{t-1} &= \sum_{s<t}\log(1+q_s), \qquad q_s = \frac{u_s(a_s)\tilde{s}_s^2(a_s)}{\sigma^2(1-\bar{\varepsilon}_s)}.\end{aligned} \tag{64}$$

Corollary 16 (Operational CC-UCB)**.** *Under the structural hypotheses of Theorem 12, with its certificate inputs instantiated by valid smoothness constants, the optimizer residual bound ζ_t, uniform CG certificates, and a valid $\kappa_{+,t}$, (61)–(64) are predictable and satisfy every nonlinear input of the theorem. Thus Theorem 12 holds with E_T replaced by $\bar{E}_T = \sum_{t\le T}\bar{\epsilon}_t^{\mathrm{lin}}$ and the observable CG upper bound on $\Lambda_T^{\mathcal{C}}$ from Corollary 36.*

Proof. Lemma 24 proves $\bar{\chi}_t \ge \chi_t$, $\bar{\psi}_t \ge \psi_t$, $\bar{F}_t \ge F_t$, and $\bar{E}_T \ge E_T$. The one-sided operator relation and Corollary 42 give u_t and $\hat{\gamma}_{t-1} \ge \gamma_{t-1}$, hence $\bar{\beta}_t \ge \beta_t$. All quantities use the summary through round $t-1$ and the already formed $\boldsymbol{\theta}_t$; uniform candidate-action CG accuracy is checked before selection. Substitution in Theorem 12 and Corollary 36 proves the claim. □

The proof, filtration-order pseudocode, and a scalar-tanh specialization with $G \le B_\phi$ and $L_\mu = L_g \le 4B_\phi^2/(3\sqrt{3})$ are in Lemma 24 and Corollary 28.

For an abstract near-linearity scale W, suppose $\|\boldsymbol{\theta}_t\|, \|\boldsymbol{\theta}^*\| \le R$, $L_\mu \le c_\mu/\sqrt{W}$, $L_g \le c_g/\sqrt{W}$, $\zeta_t \le \zeta_0/\sqrt{W}$, and $R_t^{\mathrm{col}} \le C_R t P_T$ for a supplied $P_T \ge 1$. Then

$$\begin{aligned}\bar{E}_T &\le \frac{2c_\mu R^2 T}{\sqrt{W}}, \qquad \bar{F}_{T+1} \le \frac{4c_\mu^2R^4T}{W},\\ \bar{\chi}_t &\le \frac{2c_gR\sqrt{t-1}}{\sigma\sqrt{\lambda W}}.\end{aligned} \tag{65}$$

and Corollary 25 gives an explicit trust-region bound on $\bar{\psi}_t$. If a supplied fixed active subspace gives rank bound $r \ge 1$, define

$$\Gamma_{T,r} = r_T\log\left(1 + \frac{TG^2}{r_T\lambda\sigma^2}\right), \qquad r_T = \min\{r, T\}. \tag{66}$$

and set $\Gamma_{0,r} := 0$ for the round-one confidence schedule. The bound r must be known before selection; a terminal realized rank is only post-hoc. Then $\gamma_T \le \Gamma_{T,r}$ and Corollary 27 gives the fully explicit conditional projected-Gram bound

$$R_T \le 2\alpha_I\sqrt{\rho_+/\rho_-}\,\Omega_{T,r}\sqrt{(\sigma^2 + G^2/\lambda)T\Gamma_{T,r}} + 2\bar{E}_T, \tag{67}$$

where $x_T = 2c_gR\sqrt{T-1}/(\sigma\sqrt{\lambda W})$, $\rho_- = (1-x_T)^2$, $\rho_+ = (1+x_T)^2$, and $\Omega_{T,r}$ is displayed with every $T, W, r, P_T, \lambda, \sigma$ dependence in the appendix. At fixed r and problem constants, $W = \Omega(T^2P_T)$ makes the path inflation bounded and gives $R_T = O(r\sqrt{T}\log T) + O(T/\sqrt{W})$. This is a low-rank tangent-space or parameter-efficient adaptation regime, not generic full-network training.

Theorem 17 (Conditional growing-window bound under fixed-subspace excitation)**.** *Adopt the hypotheses, events, and valid predictable schedules of Theorem 12, let $m_t = \min\{t-1, \lceil t^q\rceil\}$ for $q \in (0,1]$, and let $\mathcal{C}_t$ be the unrescaled current-parameter Gram over the last m_t observations. Thus Proposition 44 gives $\kappa_{+,t} = 1$ and $u_t = (1+\bar{\chi}_t)^2$. Suppose candidate gradients lie in a fixed subspace U, and let P_U be its orthogonal projector. For a supplied burn-in index $\tau_w \in \{1,\ldots,T+1\}$, assume the predictable window-excitation condition*

$$\mathcal{C}_t \succeq \lambda\mathbf{I} + \kappa_w m_t P_U$$

holds for a known $\kappa_w > 0$ and every $t \geq \tau_w$. Then, on the same event of probability at least $1-\delta-\delta_{\mathrm{cert}}$ as Theorem 12,

$$\begin{aligned}
\Lambda_T^{\mathcal{C}} &\leq L_q(T;\tau_w), \\
L_q(T;\tau_w) &:= (\tau_w-1)\log\left(1+\frac{G^2}{\lambda\sigma^2}\right) \\
&\quad + \frac{G^2}{\sigma^2}\sum_{t=\tau_w}^{T}\frac{1}{\lambda+\kappa_w m_t}, \qquad (68)\\
L_q(T;\tau_w) &= \begin{cases} O(\tau_w+T^{1-q}), & q<1, \\ O(\tau_w+\log T), & q=1. \end{cases}
\end{aligned}$$

For $\tau_w = 1$, more explicitly, $L_q(T;1) \leq (G^2/\sigma^2)[\lambda^{-1} + \kappa_w^{-1} h_q(T)]$, where, for $T \geq 2$, $h_q(T) = 1 + ((T-1)^{1-q} - 1)/(1-q)$ for $q < 1$ and $h_1(T) = 1 + \log(T-1)$, while $h_q(1) := 0$ in both cases. Writing $r_U = \dim U$, the excitation premise can first hold nontrivially only once $m_t \geq r_U$, because $\operatorname{rank}(\mathcal{C}_t - \lambda I) \leq m_t$; thus τ_w includes the required rank-r_U burn-in. If $H_T = \max_{t\leq T}\alpha_t\sqrt{u_t(a_t)}\,\omega_t$, then

$$R_T \leq 2H_T\sqrt{(\sigma^2 + G^2/\lambda)TL_q(T;\tau_w)} + 2E_T. \tag{69}$$

If $H_T = \widetilde{O}(1)$, $E_T = \widetilde{O}(T^{1-q/2})$, and the burn-in obeys $\tau_w = \widetilde{O}(T^{1-q})$ for $q < 1$ (or $\tau_w = \widetilde{O}(1)$ for $q = 1$), then (69) *simplifies to $R_T = \widetilde{O}(T^{1-q/2})$; otherwise its displayed L_q and E_T terms remain. This is conditional on the fixed subspace, excitation, and all confidence, transfer, optimizer, and CG certificates above. Certified worst-case* width-solve *work is $O(KT^{1+3q/2}\log(1/\bar{\varepsilon}))$ sample-CVPs when a fixed uniform target $\bar{\varepsilon} \in (0,1)$ is enforced; this count excludes model optimization, candidate-gradient computation, and certificate construction.*

Proof. Before τ_w, $\mathcal{C}_t \succeq \lambda\mathbf{I}$ bounds each logarithm in $\Lambda_T^{\mathcal{C}}$ by $\log(1+G^2/(\lambda\sigma^2))$. Thereafter, for $g_t(a) \in U$, inversion of the excitation inequality gives $s^2_{\mathcal{C},t}(a) \leq G^2/(\lambda+\kappa_w m_t)$; summing $\log(1+x) \leq x$ proves (68). Since $S_T \leq TH_T^2$, substitution in Theorem 12 proves (69). The condition-number bound $\kappa(\mathcal{C}_t) \leq 1 + m_t G^2/(\lambda\sigma^2)$ makes CG use $O(\sqrt{m_t}\log(1/\bar{\varepsilon}))$ iterations, each replaying m_t samples; summing $Km_t^{3/2}$ proves the cost claim. □

Under the hypotheses of Corollary 27, if the window policy uses the predictable rank envelope $\Gamma_{t-1,r}$ instead of the realized $\hat{\gamma}_{t-1}$ in its confidence radius, then $H_T \leq \alpha_I(1+x_T)\Omega^{\mathrm{win}}_{T,r}$, where $\Omega^{\mathrm{win}}_{T,r}$ is the explicit $\Omega_{T,r}$ in Corollary 27 with $A_{\inf} = 1$. Hence (69) contains no uncontrolled information-gain placeholder. The excitation premise is restrictive and standard in persistently excited identification; contextual diversity alone does not imply it. For $\dim U = r > 1$ one must choose τ_w after a rank-r burn-in, as the formal early charge records. The predictable rank schedule in Corollary 27 makes H_T polylogarithmic under the same bounded-path conditions. These sufficient upper-bound exponents are not a Pareto frontier or a lower bound. When U is supplied, the projected Gram in (86) can instead be solved directly.

F Deferred main-text derivations

This appendix collects the formal statements, proofs, and derivations deferred from the compact main text; all cross-referenced results are stated here.

For the sequential frozen-feature Gram, the usual information gain is

$$\gamma_T := \log\frac{\det(\bar{\mathbf{C}}_{T+1})}{\det(\lambda\mathbf{I})} = \sum_{t=1}^{T}\log\left(1+\sigma^{-2}\bar{s}_t^2(a_t)\right). \tag{70}$$

Lemma 18 (Rank and statistical-dimension information closure)**.** *For $A \succeq 0$, $r \geq \operatorname{rank}(A)$, and $q_A = \|A\|_{\mathrm{op}}/\lambda$,*

$$\log\det(\mathbf{I} + A/\lambda) \leq \min\left\{ r\log\left(1+\frac{\operatorname{tr}A}{r\lambda}\right), k(q_A)d_{\mathrm{eff},\lambda}(A)\right\}, \tag{71}$$

where $d_{\mathrm{eff},\lambda}(A) = \operatorname{tr}[A(A+\lambda\mathbf{I})^{-1}]$, $k(0) = 1$, and $k(q) = (1+q)\log(1+q)/q$ for $q > 0$; the rank term is defined as zero when $r = 0$ (which forces $A = 0$). Consequently, if $\operatorname{rank}(\bar{\mathbf{C}}_{T+1} - \lambda\mathbf{I}) \leq r$, *then $\gamma_T \leq \Gamma_{T,r}$ from* (66).

Proof. Writing $x_i = \lambda_i(A)/\lambda$, concavity of $\log(1+x)$ after padding the spectrum with zeros gives the rank–trace term. Also

$$\log(1+x) = \frac{(1+x)\log(1+x)}{x}\frac{x}{1+x}.$$

The first factor is increasing because its derivative is $(x - \log(1+x))/x^2 \geq 0$; bounding it at q_A and summing proves the statistical-dimension term. For the frozen Gram, $\operatorname{tr}(\bar{\mathbf{C}}_{T+1} - \lambda\mathbf{I}) \leq TG^2/\sigma^2$, which gives (66) (with $r_T = \min\{r, T\}$). $\square$

The second term is useful only with a predictable or deterministic envelope on $d_{\mathrm{eff},\lambda}$; a realized value is post-hoc. It is not the empirical trace effective rank $\operatorname{tr}(A)/\|A\|_{\mathrm{op}}$, which does not in general control log determinant.

Lemma 19 (Spectral-tail information closure)**.** *Let*

$$A_T := \bar{\mathbf{C}}_{T+1} - \lambda\mathbf{I} = \sigma^{-2}\sum_{t=1}^{T}\mathbf{g}_t\mathbf{g}_t^\top \succeq 0$$

have eigenvalues $\nu_1 \geq \cdots \geq \nu_d \geq 0$, and suppose the frozen features in this sum satisfy $\|\mathbf{g}_t\|_2 \leq G$ for every $t \leq T$. For an integer $r \in \{0, \ldots, d\}$, let $\Delta_{T,r} := \sum_{i>r}\nu_i$, $\tau_T := \operatorname{tr}(A_T)$, $r_T := \min\{r, T\}$, and $q_{T,r} := \min\{d-r, \max\{T-r, 0\}\}$. Then

$$\begin{aligned}\gamma_T &\leq r_T\log\left(1 + \frac{\tau_T - \Delta_{T,r}}{r_T\lambda}\right) + q_{T,r}\log\left(1 + \frac{\Delta_{T,r}}{q_{T,r}\lambda}\right)\\ &=: \Gamma_{T,r}^{\mathrm{split}},\end{aligned} \tag{72}$$

where either term is defined as zero when its multiplier is zero. Moreover,

$$\Gamma_{T,r}^{\mathrm{split}} \leq r_T\log\left(1 + \frac{TG^2}{r_T\lambda\sigma^2}\right) + \frac{\Delta_{T,r}}{\lambda}, \tag{73}$$

where the first term is defined as zero when $r_T = 0$. Thus $r = 0$ gives $\gamma_T \leq \operatorname{tr}(A_T)/\lambda$, while $r \geq \operatorname{rank}(A_T)$ gives $\Delta_{T,r} = 0$ and recovers the rank closure. The statement also covers $T = 0$, when both sides vanish.

Proof. Because A_T is a sum of T rank-one matrices, $\operatorname{rank}(A_T) \leq T$, and the stated feature bound gives $\operatorname{tr}(A_T) \leq TG^2/\sigma^2$. If $r_T > 0$, concavity of $x \mapsto \log(1 + x/\lambda)$ on the leading eigenvalues gives

$$\sum_{i=1}^{r}\log(1+\nu_i/\lambda) = \sum_{i=1}^{r_T}\log(1+\nu_i/\lambda) \leq r_T\log\left(1 + \frac{\tau_T - \Delta_{T,r}}{r_T\lambda}\right).$$

The equality uses zero eigenvalues when $r > T$. For the tail, at most $q_{T,r}$ eigenvalues can be nonzero: there are at most $d - r$ tail coordinates and at most $T - r$ nonzero eigenvalues after the first r. Concavity, with zero padding when necessary, therefore gives

$$\sum_{i>r}\log(1+\nu_i/\lambda) \leq q_{T,r}\log\left(1 + \frac{\Delta_{T,r}}{q_{T,r}\lambda}\right).$$

If $q_{T,r} = 0$, the rank bound forces $\Delta_{T,r} = 0$ and the tail sum is empty. This proves (72); the $r_T = 0$ case similarly uses only the tail term.

For the second inequality, use $\tau_T - \Delta_{T,r} \leq \tau_T \leq TG^2/\sigma^2$ in the head term and $q\log(1 + x/q) \leq x$ with $x = \Delta_{T,r}/\lambda$ in the tail term. This proves (73) without evaluating a zero denominator. The stated edge cases follow immediately. $\square$

Proposition 20 (Tightness of the split spectral-tail expression)**.** *Fix integers $r, q \geq 1$, masses $H, \Delta \geq 0$, ambient dimension $d = r + q$, and $T \geq d$. Let a PSD $d \times d$ matrix have r head eigenvalues equal to H/r, q tail eigenvalues equal to Δ/q, and all remaining eigenvalues zero, with the head values ordered no smaller than the tail values. Splitting after r makes (72) an equality:*

$$\log\det(\mathbf{I}_d + A/\lambda) = r\log\left(1 + \frac{H}{r\lambda}\right) + q\log\left(1 + \frac{\Delta}{q\lambda}\right).$$

If $\Delta/(q\lambda) \le 1$, then the tail contribution obeys

$$q \log\left(1 + \frac{\Delta}{q\lambda}\right) \ge \frac{\Delta}{2\lambda}.$$

Thus dependence linear in the tail mass cannot in general be removed without additional information about the tail rank or spectrum. This is a tightness statement for the information inequality, not a regret lower bound for the nonlinear class. The example is algebraic; to embed it in the bounded-feature bandit class one additionally chooses factors satisfying $H + \Delta \le TG^2/\sigma^2$.

Proof. The equality follows by substituting the two constant spectral blocks. For $x \in [0,1]$, $\log(1+x) \ge x/2$; apply this with $x = \Delta/(q\lambda)$ and multiply by q. $\square$

A deterministic or history-measurable prefix envelope $\bar{\Delta}_{t,r} \ge \Delta_{t,r}$ may enter an online confidence schedule through the prefix form of (73). A second valid schedule is (59), which uses the observable full trace for the head and the tail envelope only for the tail. It does not subtract an upper tail envelope from the trace. In contrast, a realized terminal $\Delta_{T,r}$ is an a posteriori path-complexity bound only. Since the inequality is pathwise and simultaneous over the finite set of ranks, the terminal report may minimize over r. Neither the terminal tail nor that post-hoc rank may choose an eigenspace, a radius, an action, or any other action-time quantity. If valid predictable envelopes are available for several ranks, their pointwise minimum is itself a valid predictable information bound. The existing observable $\hat{\gamma}_t$ schedule remains valid when only realized terminal tails are reported.

The dynamic-potential and CG-observable consequences of Theorem 12 are

$$\begin{aligned} R_T &\le 2\sqrt{(\sigma^2 + G^2/\lambda)\Gamma_T^{\mathrm{dyn}} S_T} + 2E_T, \\ R_T &\le 2\sqrt{(\sigma^2 + G^2/\lambda)\widehat{\Lambda^{\mathcal{C}}}_T S_T} + 2E_T. \end{aligned} \tag{74}$$

Corollary 21 (Corrected-center variant, no centering certificate)**.** *Replace the predictive center in Algorithm 2 by*

$$\widehat{\mu}_t^{\mathrm{corr}}(a) := \mu_{\boldsymbol{\theta}_t}(\mathbf{x}_t, a) + \mathbf{g}_t(a)^\top(\hat{\boldsymbol{\theta}}_t^{\mathrm{lin}} - \boldsymbol{\theta}_t) \tag{75}$$

and use $\omega_t = \bar{\beta}_t$. Under the hypotheses of Theorem 12 except Assumption 5, the bounds (54)*–*(74) *hold.*

Proof. Local linearization gives $\mu^*(\mathbf{x}_t, a) - \widehat{\mu}_t^{\mathrm{corr}}(a) = \mathbf{g}_t(a)^\top(\boldsymbol{\theta}^* - \hat{\boldsymbol{\theta}}_t^{\mathrm{lin}}) + \rho_t$. Lemma 40 bounds the first term by $\beta_t \bar{s}_t(a)$ and $|\rho_t| \le \varepsilon_{\mathrm{lin}}(t)$, which is the first step of the regret proof with $\omega_t = \bar{\beta}_t$; all later steps are unchanged. $\square$

Remark (Cost of the corrected center)**.** *Exact evaluation requires a solve against $\bar{\mathbf{C}}_t$. A constant number of $O(d)$ CG vectors is insufficient by itself: one additionally needs stored historical frozen gradients, raw replay data with the historical parameter checkpoints needed to reconstruct them, a frozen-feature matrix–vector oracle, an explicit low-dimensional representation, or a certified sketch. Maintaining $\bar{\mathbf{C}}_t$ explicitly costs $O(d^2)$ memory. Approximate frozen-feature access introduces an error that must be returned to the centering certificate unless separately controlled.*

F.1 Assumptions

Assumption 2 (Sub-Gaussian noise)**.** *The noise η_t is conditionally σ-sub-Gaussian given the context and action: $\mathbb{E}[\exp(s\eta_t) \mid \mathcal{G}_t] \le \exp(s^2\sigma^2/2)$ for all $s \in \mathbb{R}$, where $\mathcal{G}_t := \mathcal{H}_t \vee \sigma(a_t)$ is the pre-reward sigma-algebra of* (144)*, available immediately after the action is selected and before the reward is revealed.*

Assumption 3 (Bounded gradient features)**.** *For a constant $G > 0$,*

$$\sup_{t \le T} \sup_{s \le t} \sup_a \|\nabla_{\boldsymbol{\theta}} \mu_{\boldsymbol{\theta}_t}(\mathbf{x}_s, a)\| \ \le\ G.$$

This bounds both the current-round features $\mathbf{g}_t(\mathbf{x}_t, a)$ and the replayed features $\mathbf{g}_t(\mathbf{x}_s, a_s)$ appearing in the curvature matrix (44)*, as well as the frozen features $\mathbf{g}_s(\mathbf{x}_s, a_s)$ in* (45)*.*

Assumption 4 (Local linearization). *There is a known convex parameter region Θ containing $\boldsymbol{\theta}^*$ and every realized $\boldsymbol{\theta}_t$, on which $\nabla_\theta \mu_\theta(\mathbf{x}, a)$ is L_μ-Lipschitz uniformly over candidate $(\mathbf{x}, a)$. Thus*

$$\mu_{\boldsymbol{\theta}}(\mathbf{x}, a) \;=\; \mu_{\boldsymbol{\theta}_t}(\mathbf{x}, a) + \mathbf{g}_t(\mathbf{x}, a)^\top(\boldsymbol{\theta} - \boldsymbol{\theta}_t) + \rho_t(\mathbf{x}, a, \boldsymbol{\theta}), \tag{76}$$

where the remainder satisfies $|\rho_t(\mathbf{x}, a, \boldsymbol{\theta})| \le \frac{L_\mu}{2}\|\boldsymbol{\theta} - \boldsymbol{\theta}_t\|^2$ for $\boldsymbol{\theta} \in \Theta$. Evaluating at $\boldsymbol{\theta} = \boldsymbol{\theta}^*$ gives a time-varying linearization error

$$\varepsilon_{\text{lin}}(t) \;:=\; \sup_{a \in \mathcal{A}_t} |\rho_t(\mathbf{x}_t, a, \boldsymbol{\theta}^*)| \;\le\; \frac{L_\mu}{2}\|\boldsymbol{\theta}^* - \boldsymbol{\theta}_t\|^2. \tag{77}$$

For Theorem 12 to give a sublinear guarantee, we require the cumulative linearization error $E_T := \sum_{t=1}^T \varepsilon_{\text{lin}}(t) = o(T)$; see Theorem 12 for the precise dependence. Concretely, $E_T = O(1)$ when $\|\boldsymbol{\theta}^* - \boldsymbol{\theta}_t\| = O(1/t)$ (since $\varepsilon_{\text{lin}}(t) = O(1/t^2)$ and $\sum t^{-2} < \infty$; achievable in convex or strongly convex regimes), and $E_T = O(\sqrt{T})$ when $\|\boldsymbol{\theta}^* - \boldsymbol{\theta}_t\| = O(t^{-1/4})$. In practice this is enforced by restricting fine-tuning depth, using trust-region updates, or freezing the backbone (which renders the last-layer model convex or near-linear with small L_μ). These rate examples are illustrative only. The present paper does not prove convergence of the nonconvex optimization path $\{\boldsymbol{\theta}_t\}$ to $\boldsymbol{\theta}^*$; a sublinear guarantee from this bound therefore requires this term to be controlled by an external stability or convergence argument.

Assumption 5 (Action-scaled centering certificate). *Lemma 22 below proves the geometric bound $\left|\mathbf{g}_t(\mathbf{x}_t, a)^\top(\boldsymbol{\theta}_t - \hat{\boldsymbol{\theta}}_t^{\text{lin}})\right| \le \bar{s}_t(a)\,\psi_t$ for every $a \in \mathcal{A}_t$, where*

$$\psi_t \;:=\; \frac{\zeta_t}{\sqrt{\lambda}} \;+\; \|M_t\|_{\bar{\mathbf{C}}_t^{-1}}, \tag{78}$$

$\zeta_t \ge 0$ is an $\mathcal{H}_t$-measurable optimizer gradient-residual bound ($\|\nabla \mathcal{L}_t(\boldsymbol{\theta}_t)\|_2 \le \zeta_t$) and M_t is the gradient mismatch vector defined there. We assume there is an $\mathcal{H}_t$-measurable upper bound $\bar{\psi}_t \ge \psi_t$ (the centering certificate*) such that, for every action $a \in \mathcal{A}_t$,*

$$\left|\mathbf{g}_t(\mathbf{x}_t, a)^\top\big(\boldsymbol{\theta}_t - \hat{\boldsymbol{\theta}}_t^{\text{lin}}\big)\right| \;\le\; \bar{\psi}_t\,\bar{s}_t(a). \tag{79}$$

The discrepancy is thus action-scaled*: it multiplies the confidence width $\bar{s}_t(a)$ rather than entering additively. The certificate absorbs the same three sources as before—optimizer residual ζ_t, replayed-Jacobian drift inside M_t, and higher-order nonlinear terms—but now inside a single per-round scalar $\bar{\psi}_t$. In the convex-in-parameters regime (frozen backbone with a linear reward head), under exact optimization $\hat{\boldsymbol{\theta}}_t^{\text{lin}} = \boldsymbol{\theta}_t$ and $\bar{\psi}_t = 0$. The corrected-center variant of Corollary 21 removes $\bar{\psi}_t$ entirely, at the cost of an explicit frozen-feature solve.*

Remark (Status of Assumption 5). *Lemma 24 supplies a predictable valid $\bar{\psi}_t$ from the optimizer residual, Welford path moments, and observed residual energy. We do not prove that this conservative quantity is small outside a restricted path. It enters the exploration term through $\omega_t = \bar{\beta}_t + \bar{\psi}_t$ (Equation (135)), not as an additive $O(T)$ term.*

The next lemma gives the primitive geometric bound underlying Assumption 5, expressed through the certifiable quantity ψ_t.

Lemma 22 (Primitive upper bound for prediction discrepancy). *Let $z_s = (\mathbf{x}_s, a_s)$, $\mathbf{g}_s = \nabla_{\boldsymbol{\theta}} \mu_{\boldsymbol{\theta}_s}(z_s)$, $\mathbf{g}_{s,t} = \nabla_{\boldsymbol{\theta}} \mu_{\boldsymbol{\theta}_t}(z_s)$, and $\delta_{s,t} = \mathbf{g}_{s,t} - \mathbf{g}_s$. Define*

$$\bar{\mu}_{s,t} := \mu_{\boldsymbol{\theta}_s}(z_s) + \mathbf{g}_s^\top(\boldsymbol{\theta}_t - \boldsymbol{\theta}_s), \qquad e_{s,t} := \mu_{\boldsymbol{\theta}_t}(z_s) - \bar{\mu}_{s,t}, \qquad \bar{r}_{s,t} := \bar{\mu}_{s,t} - r_s.$$

Let $\bar{\mathcal{L}}_t$ be the frozen-feature linearized ridge objective whose minimizer is $\hat{\boldsymbol{\theta}}_t^{\text{lin}}$, and define

$$M_t := \frac{1}{\sigma^2}\sum_{s<t}\left[\bar{r}_{s,t}\delta_{s,t} + e_{s,t}\mathbf{g}_s + e_{s,t}\delta_{s,t}\right].$$

If the nonlinear optimizer satisfies $\|\nabla \mathcal{L}_t(\boldsymbol{\theta}_t)\|_2 \le \zeta_t$, then for every candidate action $a \in \mathcal{A}_t$,

$$\left|\mathbf{g}_t(a)^\top(\boldsymbol{\theta}_t - \hat{\boldsymbol{\theta}}_t^{\text{lin}})\right| \;\le\; \bar{s}_t(a)\left(\frac{\zeta_t}{\sqrt{\lambda}} + \|M_t\|_{\bar{\mathbf{C}}_t^{-1}}\right), \qquad \bar{s}_t(a) := \sqrt{\mathbf{g}_t(a)^\top \bar{\mathbf{C}}_t^{-1}\mathbf{g}_t(a)}.$$

Assumption	Status	Use	Operational check
Conditional sub-Gaussian noise	Standard	Self-normalized confidence	Distributional premise
Realizability, $\|\theta^*\| \le S$	Restrictive	Bias and Taylor schedules	Not generally testable
Bounded gradients	Standard on compact sets	Width and condition bounds	Analytic bound or pre-reward monitor
L_μ, L_g on convex Θ	Restrictive	Linearization, transfer, centering	Analytic for a specified model class
Optimizer residual ζ_t	Algorithmic	Centering certificate	Pre-action full-objective gradient norm
Operator and CG certificates	Algorithmic	Optimism and solve error	Analytic Loewner and residual proof
Residual-energy envelope	Rate-only	Near-linear scaling	Observable after collection, not policy input
Fixed active rank r	Restrictive	Closes γ_T	Supplied subspace, not a terminal audit
Window excitation κ_w	Restrictive	Closes $\Lambda_T^{\mathcal{C}}$	Analytic construction or explicit premise

Table 2: Assumption ledger for the legacy one-sided theorem. Analytic and pre-action quantities are policy available; post-reward quantities are used only in the theorem statement or its audit.

The left-hand side is exactly the geometric bound $|\mathbf{g}_t(a)^\top(\boldsymbol{\theta}_t - \hat{\boldsymbol{\theta}}_t^{\text{lin}})| \le \bar{s}_t(a)\,\psi_t$ *of Assumption 5 with* $\psi_t = \zeta_t/\sqrt{\lambda} + \|M_t\|_{\bar{\mathbf{C}}_t^{-1}}$ *(Equation (78)). Consequently, since* $\|M_t\|_{\bar{\mathbf{C}}_t^{-1}} \le \|M_t\|_2/\sqrt{\lambda}$*, a valid centering certificate is*

$$\bar{\psi}_t \;:=\; \frac{\zeta_t + \|M_t\|_2}{\sqrt{\lambda}} \;\ge\; \psi_t.$$

If, in addition, on a fixed event, $\boldsymbol{\theta} \mapsto \nabla_{\boldsymbol{\theta}}\mu_{\boldsymbol{\theta}}(z)$ *is* L_g*-Lipschitz,* $|\bar{r}_{s,t}| \le B_{t,\delta}$ *for all* $s < t$*, and* $\Delta_{s,t} := \|\boldsymbol{\theta}_t - \boldsymbol{\theta}_s\|$*, then*

$$\|M_t\|_2 \;\le\; \frac{1}{\sigma^2}\sum_{s<t}\left[B_{t,\delta}L_g\Delta_{s,t} + \frac{GL_g}{2}\Delta_{s,t}^2 + \frac{L_g^2}{2}\Delta_{s,t}^3\right].$$

Proof. The linearized ridge objective satisfies $\nabla\bar{\mathcal{L}}_t(\hat{\boldsymbol{\theta}}_t^{\text{lin}}) = 0$ and, by its quadratic form,

$$\nabla\bar{\mathcal{L}}_t(\boldsymbol{\theta}_t) = \bar{\mathbf{C}}_t(\boldsymbol{\theta}_t - \hat{\boldsymbol{\theta}}_t^{\text{lin}}).$$

Thus $\boldsymbol{\theta}_t - \hat{\boldsymbol{\theta}}_t^{\text{lin}} = \bar{\mathbf{C}}_t^{-1}\nabla\bar{\mathcal{L}}_t(\boldsymbol{\theta}_t)$. Using $\mu_{\boldsymbol{\theta}_t}(z_s) - r_s = \bar{r}_{s,t} + e_{s,t}$ and $\mathbf{g}_{s,t} = \mathbf{g}_s + \delta_{s,t}$, the gradients of the nonlinear loss and the frozen-feature linearized loss satisfy

$$\nabla\mathcal{L}_t(\boldsymbol{\theta}_t) = \nabla\bar{\mathcal{L}}_t(\boldsymbol{\theta}_t) + M_t, \qquad \text{so} \qquad \nabla\bar{\mathcal{L}}_t(\boldsymbol{\theta}_t) = \nabla\mathcal{L}_t(\boldsymbol{\theta}_t) - M_t.$$

Therefore, by Cauchy–Schwarz in the $\bar{\mathbf{C}}_t$-norm,

$$|\mathbf{g}_t(a)^\top(\boldsymbol{\theta}_t - \hat{\boldsymbol{\theta}}_t^{\text{lin}})| \le \bar{s}_t(a)\|\nabla\bar{\mathcal{L}}_t(\boldsymbol{\theta}_t)\|_{\bar{\mathbf{C}}_t^{-1}} \le \bar{s}_t(a)\left(\|\nabla\mathcal{L}_t(\boldsymbol{\theta}_t)\|_{\bar{\mathbf{C}}_t^{-1}} + \|M_t\|_{\bar{\mathbf{C}}_t^{-1}}\right).$$

Since $\bar{\mathbf{C}}_t \succeq \lambda\mathbf{I}$, $\|\nabla\mathcal{L}_t(\boldsymbol{\theta}_t)\|_{\bar{\mathbf{C}}_t^{-1}} \le \zeta_t/\sqrt{\lambda}$, which gives the geometric inequality $|\mathbf{g}_t(a)^\top(\boldsymbol{\theta}_t - \hat{\boldsymbol{\theta}}_t^{\text{lin}})| \le \bar{s}_t(a)\big(\zeta_t/\sqrt{\lambda} + \|M_t\|_{\bar{\mathbf{C}}_t^{-1}}\big) = \bar{s}_t(a)\,\psi_t$, i.e. the bound of (78). For the certificate, $\bar{\mathbf{C}}_t \succeq \lambda\mathbf{I}$ also gives $\|M_t\|_{\bar{\mathbf{C}}_t^{-1}} \le \|M_t\|_2/\sqrt{\lambda}$, so $\psi_t \le (\zeta_t + \|M_t\|_2)/\sqrt{\lambda} =: \bar{\psi}_t \ge \psi_t$, the proposed certificate. For the final display, Lipschitzness gives $\|\delta_{s,t}\| \le L_g\Delta_{s,t}$ and the Taylor remainder bound $|e_{s,t}| \le (L_g/2)\Delta_{s,t}^2$; applying these together with $\|\mathbf{g}_s\| \le G$ (Assumption 3) and $|\bar{r}_{s,t}| \le B_{t,\delta}$ term by term in the definition of M_t gives the claimed bound on $\|M_t\|_2$. ☐

Lemma 23 (Bounded-output residual energy)**.** *Fix a horizon* $T \ge 1$ *and* $\delta_R \in (0,1)$*. Suppose* $|\mu_{\boldsymbol{\theta}}(x,a)| \le B_\mu$ *throughout the certified parameter region and* $|\mu^*(x,a)| \le B_\mu$*. Under Assumption 2, with probability at least* $1 - \delta_R$*, simultaneously for every* $t \le T + 1$*,*

$$R_t^{\text{col}} \le \left[8B_\mu^2 + 4\sigma^2\log\left(\frac{2T}{\delta_R}\right)\right](t-1). \tag{80}$$

This finite-horizon envelope is not claimed to be tight.

Proof. Conditional sub-Gaussianity gives

$$\Pr\Big(|\eta_s| > \sigma\sqrt{2\log(2T/\delta_R)} \mid \mathcal{G}_s\Big) \le \delta_R/T.$$

Taking expectations and a union bound over $s = 1, \dots, T$ yields an event of probability at least $1 - \delta_R$ on which this bound holds for every noise term. On that event, $c_s = \mu_{\boldsymbol{\theta}_s}(x_s, a_s) - \mu^*(x_s, a_s) - \eta_s$ and $(a+b)^2 \le 2a^2 + 2b^2$ give

$$c_s^2 \le 2(2B_\mu)^2 + 2\sigma^2\, 2\log(2T/\delta_R) = 8B_\mu^2 + 4\sigma^2 \log(2T/\delta_R).$$

Summing the first $t-1$ terms proves (80); at $t = 1$ both sides are zero. □

Lemma 24 ($O(d)$-state path certificates)**.** *Suppose $\nabla_\theta \mu_\theta(z)$ is L_g-Lipschitz and bounded by G on a convex region containing the realized iterates. Before round t, let $n_t = t-1$, let $\bar{\boldsymbol{\theta}}_{<t}$ and $J_{<t} = \sum_{s<t} \|\boldsymbol{\theta}_s - \bar{\boldsymbol{\theta}}_{<t}\|^2$ summarize past iterates, and let $R_t^{\mathrm{col}} = \sum_{s<t} c_s^2$ with $c_s = \mu_{\boldsymbol{\theta}_s}(z_s) - r_s$. At $t = 1$ use the initialization $\bar{\boldsymbol{\theta}}_{<1} = \mathbf{0}$, $J_{<1} = Q_1 = R_1^{\mathrm{col}} = 0$. Then, for all t,*

$$Q_t := J_{<t} + n_t\|\boldsymbol{\theta}_t - \bar{\boldsymbol{\theta}}_{<t}\|^2 = \sum_{s<t} \|\boldsymbol{\theta}_t - \boldsymbol{\theta}_s\|^2$$

and the following $\mathcal{H}_t$-measurable quantities are valid:

$$\bar{\chi}_t = \frac{L_g}{\sigma\sqrt{\lambda}}\sqrt{Q_t} \ge \chi_t, \tag{81}$$

$$\bar{M}_t = \frac{1}{\sigma^2}\left[L_g\sqrt{R_t^{\mathrm{col}} Q_t} + \frac{3GL_g}{2}Q_t + \frac{L_g^2}{2}Q_t^{3/2}\right] \ge \|M_t\|_2. \tag{82}$$

If $\|\boldsymbol{\theta}_s\|, \|\boldsymbol{\theta}_t\| \le R$, the final term in (82) may be replaced by $L_g^2 R Q_t$. If a nonnegative $\mathcal{H}_t$-measurable optimizer certificate satisfies $\|\nabla\mathcal{L}_t(\boldsymbol{\theta}_t)\| \le \zeta_t$, then

$$\bar{\psi}_t = (\zeta_t + \bar{M}_t)/\sqrt{\lambda} \ge \psi_t.$$

If prediction gradients are L_μ-Lipschitz on a convex region containing $\boldsymbol{\theta}_t$ and $\boldsymbol{\theta}^*$ and $\|\boldsymbol{\theta}^*\| \le S$, then

$$\bar{\epsilon}_t^{\mathrm{lin}} := \frac{L_\mu}{2}(S + \|\boldsymbol{\theta}_t\|)^2 \ge \varepsilon_{\mathrm{lin}}(t).$$

Consequently $\bar{F}_t = \sum_{s<t}(\bar{\epsilon}_s^{\mathrm{lin}})^2 \ge F_t$ and $\bar{E}_T = \sum_{t\le T} \bar{\epsilon}_t^{\mathrm{lin}} \ge E_T$.

Proof. The parallel-axis identity follows after writing $\boldsymbol{\theta}_t - \boldsymbol{\theta}_s = (\boldsymbol{\theta}_t - \bar{\boldsymbol{\theta}}_{<t}) + (\bar{\boldsymbol{\theta}}_{<t} - \boldsymbol{\theta}_s)$; the cross term sums to zero. Lemma 30 and $\|\nabla\mu_{\boldsymbol{\theta}_t}(z_s) - \nabla\mu_{\boldsymbol{\theta}_s}(z_s)\| \le L_g\|\boldsymbol{\theta}_t - \boldsymbol{\theta}_s\|$ give (81). Let $d_{s,t} = \|\boldsymbol{\theta}_t - \boldsymbol{\theta}_s\|$. In Lemma 22,

$$|\bar{r}_{s,t}| \le |c_s| + G d_{s,t}, \qquad \|\delta_{s,t}\| \le L_g d_{s,t}, \qquad |e_{s,t}| \le (L_g/2) d_{s,t}^2.$$

Thus each summand defining M_t has norm at most

$$L_g|c_s|d_{s,t} + \frac{3GL_g}{2}d_{s,t}^2 + \frac{L_g^2}{2}d_{s,t}^3.$$

Cauchy–Schwarz gives $\sum |c_s| d_{s,t} \le \sqrt{R_t^{\mathrm{col}} Q_t}$, while $\sum d_{s,t}^2 = Q_t$ and $\sum d_{s,t}^3 \le Q_t^{3/2}$, proving (82). Under the trust region, $d_{s,t} \le 2R$, so $\sum d_{s,t}^3 \le 2RQ_t$. The centering claim follows from Lemma 22 and $\bar{\mathbf{C}}_t \succeq \lambda\mathbf{I}$. Finally Taylor's theorem and $\|\boldsymbol{\theta}^* - \boldsymbol{\theta}_t\| \le S + \|\boldsymbol{\theta}_t\|$ prove the remainder, F_t, and E_T claims. □

Filtration-order implementation. At round t, the summaries through $t-1$, $\boldsymbol{\theta}_t$, and the certificate ζ_t are available before selection. The learner computes $Q_t, \bar{\chi}_t, u_t, \bar{M}_t, \bar{\psi}_t, \bar{\epsilon}_t^{\mathrm{lin}}$ and $\bar{\beta}_t$, then solves every candidate system to the uniform tolerance and selects a_t. After observing r_t, it adds c_t^2 to R_{t+1}^{col}, adds $\boldsymbol{\theta}_t$ with the Welford update

$$d = \boldsymbol{\theta}_t - \bar{\boldsymbol{\theta}}_{<t}, \quad \bar{\boldsymbol{\theta}}_{\le t} = \bar{\boldsymbol{\theta}}_{<t} + d/t, \quad J_{\le t} = J_{<t} + d^\top(\boldsymbol{\theta}_t - \bar{\boldsymbol{\theta}}_{\le t}),$$

and updates $\bar{F}_{t+1}$ and $\hat{\gamma}_t$. It then forms $\boldsymbol{\theta}_{t+1}$ and certifies the nonnegative bound ζ_{t+1} before the next action. No current reward enters the current score.

In frozen linear-head models Q_t may be nonzero while $L_g = L_\mu = 0$; then $\bar{\chi}_t = \bar{M}_t = \bar{\epsilon}_t^{\mathrm{lin}} = 0$, and exact optimization also gives $\bar{\psi}_t = 0$.

Corollary 25 (Bounded-path near-linear rate)**.** *Fix T and suppose for every $t \le T$*

$$\|\boldsymbol{\theta}_t\|, \|\boldsymbol{\theta}^*\| \le R, \quad L_\mu \le c_\mu/\sqrt{W}, \quad L_g \le c_g/\sqrt{W}, \quad \zeta_t \le \zeta_0/\sqrt{W}, \quad R_t^{\mathrm{col}} \le C_R t P_T,$$

where $P_T \ge 1$ is an explicitly supplied residual-energy factor. For exact current-parameter curvature and a uniform CG tolerance $\bar{\varepsilon}_t \le \bar{\varepsilon} < 1$, set

$$x_T = \frac{2c_g R\sqrt{T-1}}{\sigma\sqrt{\lambda W}}, \quad \rho_- = (1-x_T)^2, \qquad \rho_+ = (1+x_T)^2.$$

If $x_T < 1$, then $\rho_-\bar{\mathbf{C}}_t \preceq \mathbf{C}_t \preceq \rho_+\bar{\mathbf{C}}_t$ for all $t \le T$. Writing

$$\alpha_I = \sqrt{\frac{1+\bar{\varepsilon}}{1-\bar{\varepsilon}}}, \qquad A_{\inf} = \alpha_I^2\frac{\rho_+}{\rho_-},$$

the observable information schedule satisfies $\gamma_T \le \hat{\gamma}_T \le A_{\inf}\gamma_T$, and

$$R_T \le 2\alpha_I\sqrt{\rho_+/\rho_-}\,\omega^{\mathrm{op}}_{\max,T}\sqrt{(\sigma^2+G^2/\lambda)T\gamma_T} + 2\bar{E}_T \tag{83}$$

holds with

$$\omega^{\mathrm{op}}_{\max,T} := \max_{t\le T}(\bar{\beta}_t + \bar{\psi}_t)$$

and

$$\begin{aligned}
\omega^{\mathrm{op}}_{\max,T} &\le \sqrt{A_{\inf}\gamma_T + 2\log(1/\delta)} + \sqrt{\lambda}R + \frac{2c_\mu R^2\sqrt{T}}{\sigma\sqrt{W}} \\
&\quad + \frac{A_T}{\sqrt{\lambda W}} + \frac{B_T}{\sqrt{\lambda W}}, \\
A_T &= \zeta_0 + \sigma^{-2}\Big[2c_g R\sqrt{C_R T(T-1)P_T} + 6Gc_g R^2(T-1)\Big], \\
B_T &= 4c_g^2 R^3(T-1)/\sigma^2.
\end{aligned}$$

In addition, (65) *holds and*

$$\bar{\psi}_t \le \frac{1}{\sqrt{\lambda}}\left\{\frac{\zeta_0}{\sqrt{W}} + \frac{1}{\sigma^2}\left[\frac{2c_g R\sqrt{C_R t(t-1)P_T} + 6Gc_g R^2(t-1)}{\sqrt{W}} + \frac{4c_g^2 R^3(t-1)}{W}\right]\right\}.$$

Proof. The trust region gives $Q_t \le 4R^2(t-1)$; substitution in Lemma 24 proves all displayed path bounds. Corollary 32 gives the two-sided operator relation. The CG sandwich and $u_t \le (1+x_T)^2 = \rho_+$ imply that every logarithm in $\hat{\gamma}_T$ is at most $\log(1 + A_{\inf}\sigma^{-2}\bar{s}_t^2(a_t))$. For $A \ge 1$, $\log(1+Ax) \le A\log(1+x)$, proving the information comparison. Substitution into Corollary 50 gives the regret display. □

Corollary 26 (Corrected-center near-linear rate)**.** *Fix T and suppose, for every $t \le T$,*

$$\|\boldsymbol{\theta}_t\|, \|\boldsymbol{\theta}^*\| \le R, \qquad L_\mu \le c_\mu/\sqrt{W}, \qquad L_g \le c_g/\sqrt{W},$$

where W is an abstract near-linearity scale. Use the corrected center, exact current curvature, and a uniform CG tolerance $\bar{\varepsilon}_t \le \bar{\varepsilon} < 1$. Fix $r \in \{0, \dots, d\}$ and suppose a predictable prefix information envelope $\mathcal{G}_{n,r} \ge \gamma_n$ is supplied for $n = 0, \dots, T$. Valid choices are the rank envelope $\Gamma_{n,r}$ under a known rank bound, or the spectral-tail envelope in Corollary 14 with supplied prefix tails $\bar{\Delta}_{n,r}$. Define its predictable monotone closure $\mathcal{G}^{\uparrow}_{n,r} := \max_{0\le j\le n}\mathcal{G}_{j,r}$ and set

$$x_T := \frac{2c_g R\sqrt{T-1}}{\sigma\sqrt{\lambda W}}, \qquad \alpha_I := \sqrt{\frac{1+\bar{\varepsilon}}{1-\bar{\varepsilon}}},$$

$$\mathcal{B}^{\mathrm{corr}}_{T,r} := \sqrt{\mathcal{G}^{\uparrow}_{T,r} + 2\log(1/\delta)} + \sqrt{\lambda}R + \frac{2c_\mu R^2\sqrt{T}}{\sigma\sqrt{W}}.$$

If $x_T < 1$, the predictable radius

$$\bar{\beta}_t^{\mathrm{corr}} := \sqrt{\mathcal{G}_{t-1,r}^{\uparrow} + 2\log(1/\delta)} + \sqrt{\lambda}R + \frac{2c_\mu R^2\sqrt{t-1}}{\sigma\sqrt{W}} \tag{84}$$

is valid and the corrected-center policy satisfies

$$\begin{aligned} R_T \le 2\alpha_I \frac{1+x_T}{1-x_T}\mathcal{B}_{T,r}^{\mathrm{corr}}\sqrt{(\sigma^2+G^2/\lambda)T\mathcal{G}_{T,r}^{\uparrow}} \\ + \frac{4c_\mu R^2 T}{\sqrt{W}}. \end{aligned} \tag{85}$$

In particular, bounded transfer is ensured by $W > 4c_g^2R^2(T-1)/(\lambda\sigma^2) = \Omega(T)$ up to problem constants, and $E_T \le 2c_\mu R^2T/\sqrt{W}$. If $\mathcal{G}_{T,r}^{\uparrow} = O(r\log T)$—for example, if $\max_{n\le T}\bar{\Delta}_{n,r} = O(r\log T)$ at fixed λ—and $W = \Omega(T)$ with a sufficiently large constant to keep x_T bounded away from one, then

$$R_T = O(r\sqrt{T}\log T) + O(\sqrt{T})$$

up to confidence and CG constants.

This corollary assumes neither an optimizer-residual rate nor an envelope on R_t^{col}. Its cost is frozen-feature access: exact corrected centers require an explicit $d\times d$ frozen Gram, stored frozen gradients, replay with historical checkpoints, a supplied low-dimensional representation, or a separately certified sketch. The result therefore does not have $O(d)$ total memory merely because current-curvature products are matrix-free.

Proof. The running maximum $\mathcal{G}_{n,r}^{\uparrow}$ is predictable and dominates both $\mathcal{G}_{n,r}$ and every earlier prefix envelope. The trust region gives $Q_t \le 4R^2(t-1)$, so Lemma 24 yields $\bar{\chi}_t \le 2c_gR\sqrt{t-1}/(\sigma\sqrt{\lambda W}) \le x_T$, $\bar{\epsilon}_t^{\mathrm{lin}} \le 2c_\mu R^2/\sqrt{W}$, $\bar{F}_t \le 4c_\mu^2R^4(t-1)/W$, and $\bar{E}_T \le 2c_\mu R^2T/\sqrt{W}$. Hence (84) dominates (138). Apply the corrected-center form of Corollary 13, bound $\alpha_t \le \alpha_I$, $(1+\bar{\chi}_t)/(1-\bar{\chi}_t) \le (1+x_T)/(1-x_T)$, and $\bar{\beta}_t^{\mathrm{corr}} \le \mathcal{B}_{T,r}^{\mathrm{corr}}$, then use $\gamma_T \le \mathcal{G}_{T,r} \le \mathcal{G}_{T,r}^{\uparrow}$. Finally substitute $2\bar{E}_T \le 4c_\mu R^2T/\sqrt{W}$. The stated orders follow directly. □

Remark (Why W is an abstract near-linearity scale)**.** *The all-weights MLP used in the nonlinear diagnostics has action-a mean $v_a^\top\tanh(Ux+b)+c_a$ and no $W^{-1/2}$ forward normalization. With $\bar{x}=(x,1)$, hidden vector $q_i=(U_i,b_i)$, and $z_i = q_i^\top\bar{x}$, the Hessian block on $(q_i, v_{a,i})$ is*

$$\begin{bmatrix} v_{a,i}\tanh''(z_i)\bar{x}\bar{x}^\top & \tanh'(z_i)\bar{x} \\ \tanh'(z_i)\bar{x}^\top & 0 \end{bmatrix}.$$

At $z_i = 0$ and $v_{a,i}=0$ its operator norm is $\|\bar{x}\|$, independent of hidden width. Thus the executed all-weights parameterization does not imply $L_\mu, L_g = O(W^{-1/2})$. Such a conclusion would require a different explicitly normalized parameterization and coordinate constraints, which would also change the ridge penalty and GGN geometry. We therefore do not identify W with generic neural-network width or derive optimizer-residual scaling from it.

Corollary 27 (Conditional projected-Gram rate for a supplied subspace)**.** *Under Corollary 25, additionally suppose a subspace $U \subseteq \mathbb{R}^d$ with $\dim U \le r$ is fixed and supplied before interaction, and every candidate or replayed tangent feature through round T lies in U. Hence $\mathrm{rank}(\bar{\mathbf{C}}_t - \lambda\mathbf{I}) \le r$ for every $t \le T+1$. Let $r_U = \dim U$, let $Q_U \in \mathbb{R}^{d\times r_U}$ have orthonormal columns spanning U, and define*

$$\mathbf{z}_{t,s} = Q_U^\top\mathbf{g}_t(\mathbf{x}_s, a_s), \qquad \mathbf{z}_t(a) = Q_U^\top\mathbf{g}_t(\mathbf{x}_t, a).$$

Then the current-GGN width reduces exactly to

$$\mathbf{g}_t(a)^\top\mathbf{C}_t^{-1}\mathbf{g}_t(a) = \mathbf{z}_t(a)^\top\left(\lambda I_{r_U} + \sigma^{-2}\sum_{s<t}\mathbf{z}_{t,s}\mathbf{z}_{t,s}^\top\right)^{-1}\mathbf{z}_t(a), \tag{86}$$

with the analogous identity for $\bar{\mathbf{C}}_t$ using $Q_U^\top\mathbf{g}_s(\mathbf{x}_s,a_s)$. Thus a supplied basis permits an exact dense $r_U\times r_U$ Gram with $O(r_U^2)$ curvature state. Given tangent coordinates, forming the current Gram costs $O((t-1)r_U^2)$, factorization

$O(r_U^3)$*, and each width query* $O(r_U^2)$*; current-parameter replay, model training, and gradient generation can still be* d*-dimensional. Full-dimensional CG is not needed in this corollary's covered regime.*

Define $\Gamma_{T,r}$ *by* (66)*. Then* (67) *holds with*

$$\begin{aligned}\Omega_{T,r} &:= \sqrt{A_{\inf}\Gamma_{T,r} + 2\log(1/\delta)} + \sqrt{\lambda}R + \frac{2c_\mu R^2\sqrt{T}}{\sigma\sqrt{W}} + \frac{A_T}{\sqrt{\lambda W}} + \frac{B_T}{\sqrt{\lambda}W}, \\ A_T &= \zeta_0 + \sigma^{-2}\Big[2c_g R\sqrt{C_R T(T-1)P_T} + 6Gc_g R^2(T-1)\Big], \\ B_T &= 4c_g^2 R^3(T-1)/\sigma^2,\end{aligned}$$

and $A_{\inf}, \rho_\pm$ *are exactly those in Corollary 25. In particular, for fixed* r *and fixed problem constants, if* $W \geq CT^2 P_T$ *for a constant* C *satisfying the explicit thresholds in* (87)*, then*

$$R_T = O(r\sqrt{T}\log T) + O(T/\sqrt{W}).$$

This rate is conditional on the ex ante subspace U *and every bounded-path, near-linearity, residual, optimizer, transfer, and solver premise of Corollary 25. It neither learns* U *nor covers unrestricted representation learning or end-to-end neural-network training; it is an effective projected-feature closure, not a full-dimensional computational advantage for CC-UCB.*

Proof. Extend Q_U to an orthogonal basis of $\mathbb{R}^d$. In that basis $\mathbf{C}_t$ is block diagonal with the $r_U \times r_U$ matrix in (86) and λI_{d-r_U}; the query has zero coordinates in the second block. This proves (86), and the same argument proves the frozen-Gram identity. Lemma 18 gives $\gamma_T \leq \Gamma_{T,r}$. Every occurrence of γ_T in Corollary 25 is monotone, so direct substitution gives the display. Under the stated near-linearity scaling, x_T, the centering terms, and the linearization terms are bounded, while $\Gamma_{T,r} = O(r\log T)$. Hence $\Omega_{T,r} = O(\sqrt{r\log T} + 1)$ and the leading product is $O(r\sqrt{T}\log T)$. □

Remark (Linear-subclass lower bound)**.** *The exact-rank class contains a standard* r*-dimensional stochastic linear bandit. Take a constant context, a known orthonormal basis* Q_U*, the finite action set* $z(a) \in \{\pm 1/\sqrt{r}\}^r$*, linear mean* $\mu_\vartheta(a) = z(a)^\top\vartheta$*, and Gaussian noise. Choosing the usual hypercube parameter family with coordinates of order* $\sqrt{r/T}$ *keeps the parameter norm bounded when* $T \gtrsim r^2$*. Here* $L_\mu = L_g = E_T = 0$, $\mathbf{C}_t = \bar{\mathbf{C}}_t$*, and all features lie in* U*. The stochastic linear-bandit minimax expected-regret lower bound is* $\Omega(r\sqrt{T})$ *in this regime (Dani et al., 2008). Hence the leading* $r\sqrt{T}$ *dependence of the exact-rank upper bound is sharp up to logarithms on this linear subclass. This does not prove a lower bound for the nonlinear or spectral-tail classes, nor a computational lower bound.*

For explicit targets $\bar{x}_{\text{drift}} \in (0,1)$ and $\Psi, b_F, \eta_E > 0$, the sufficient conditions

$$\begin{aligned}&W \geq \frac{4c_g^2 R^2(T-1)}{\lambda\sigma^2\bar{x}_{\text{drift}}^2}, \qquad W \geq \frac{4c_\mu^2 R^4 T}{\sigma^2 b_F^2}, \quad W \geq \frac{4c_\mu^2 R^4}{\eta_E^2}, \\ &W \geq \max\left\{\frac{4A_T^2}{\lambda\Psi^2}, \frac{2B_T}{\sqrt{\lambda}\Psi}\right\}\end{aligned} \tag{87}$$

respectively ensure $x_T \leq \bar{x}_{\text{drift}} < 1$ (and therefore $\rho_- \geq (1-\bar{x}_{\text{drift}})^2$ and $\rho_+ \leq (1+\bar{x}_{\text{drift}})^2$), $\sqrt{\bar{F}_{T+1}}/\sigma \leq b_F$, $\bar{E}_T/T \leq \eta_E$, and $\bar{\psi}_T \leq \Psi$. Thus the centering requirement has leading sufficient scaling $W = \Omega(T^2 P_T)$; these conditions do not bound γ_T.

Corollary 28 (Nonlinear tanh-link Gaussian bandit)**.** *For* $\mu_\theta(x,a) = \tanh(\phi(x,a)^\top\theta)$ *with* $\|\phi(x,a)\| \leq B_\phi$*, one may take*

$$G = B_\phi, \qquad L_\mu = L_g = \frac{4}{3\sqrt{3}}B_\phi^2.$$

On any certified trust region, Corollary 16 therefore gives an executable nonlinear Gaussian-reward CC-UCB policy using only the online path statistics, a pre-action optimizer-residual certificate, and the stated operator/CG certificates. Under realizability, both the model and teacher outputs have $B_\mu = 1$*. Hence Lemma 23 removes the supplied residual-energy factor in Corollary 25: on its event one may take*

$$C_R = 1, \qquad P_T = 8 + 4\sigma^2\log(2T/\delta_R),$$

and allocate δ_R *to the certificate failure budget.*

Proof. $\nabla_\theta \mu = \operatorname{sech}^2(\phi^\top \theta)\phi$, so its norm is at most B_ϕ. The Hessian is $-2\tanh(z)\operatorname{sech}^2(z)\phi\phi^\top$. For $y = |\tanh z|$, $2y(1-y^2)$ is maximized at $y = 1/\sqrt{3}$ with value $4/(3\sqrt{3})$, proving both Lipschitz constants. Apply Lemma 24 and Corollary 16. □

Corollary 29 (Scaled-tanh relative-link rate)**.** *Consider the explicit zero-offset scalar-link family*

$$\mu_{\boldsymbol{\theta},W}(z) = \sqrt{W}\,\tanh\!\left(\frac{\phi(z)^\top \boldsymbol{\theta}}{\sqrt{W}}\right), \qquad \|\phi(z)\| \le B_\phi, \qquad \|\boldsymbol{\theta}_t\|, \|\boldsymbol{\theta}^*\| \le R. \tag{88}$$

This is a scaled generalized-linear family in the displacement coordinates, not a claim that generic neural-network width provides the scale W. Let

$$c_h := \frac{4B_\phi^2}{3\sqrt{3}}, \qquad \rho_W := \exp\!\left(\frac{4B_\phi R}{\sqrt{W}}\right) - 1.$$

Then valid global constants on the certified region are

$$G = B_\phi, \qquad L_\mu = L_g = \frac{c_h}{\sqrt{W}}, qquad|\mu_{\boldsymbol{\theta},W}(z)| \le B_\phi R, qquad \rho_t \le \rho_W. \tag{89}$$

The stated exponential relative-drift certificate obeys $\rho_W < 1$ exactly when

$$W > \left(\frac{4B_\phi R}{\log 2}\right)^2. \tag{90}$$

Fix a uniform CG certificate $\bar{\varepsilon}_t \le \bar{\varepsilon} < 1$, let $\alpha_I = \sqrt{(1+\bar{\varepsilon})/(1-\bar{\varepsilon})}$, and suppose $\zeta_t \le \zeta_0/\sqrt{W}$ and $R_t^{\mathrm{col}} \le C_R t P_T$ for $P_T \ge 1$. Fix a rank index r and supply a predictable prefix information envelope $\mathcal{G}_{n,r} \ge \gamma_n$; write $\mathcal{G}_{n,r}^{\uparrow} = \max_{j\le n} \mathcal{G}_{j,r}$. The envelope may be the exact-rank bound or either valid predictable spectral-tail schedule in Corollary 14. Define

$$\begin{aligned}\Psi_{T,W}^{\mathrm{rel}} := \frac{\zeta_0}{\sqrt{\lambda W}} + \frac{1}{\sigma}\Bigg\{ &\rho_W\left[\sqrt{C_R T P_T} + 2B_\phi R\sqrt{T-1}\right] \\ &+ \frac{2(1+\rho_W)c_h R^2\sqrt{T-1}}{\sqrt{W}}\Bigg\},\end{aligned} \tag{91}$$

$$\mathcal{B}_{T,r}^{\mathrm{rel}} := \sqrt{\mathcal{G}_{T,r}^{\uparrow} + 2\log(1/\delta)} + \sqrt{\lambda}R + \frac{2c_h R^2\sqrt{T}}{\sigma\sqrt{W}} + \Psi_{T,W}^{\mathrm{rel}}. \tag{92}$$

If $\rho_W < 1$, exact current curvature with the original nonlinear center and the relative certificate of Lemma 31 satisfies

$$R_T \le 2\alpha_I \frac{1+\rho_W}{1-\rho_W}\mathcal{B}_{T,r}^{\mathrm{rel}}\sqrt{\left(\sigma^2 + \frac{B_\phi^2}{\lambda}\right)T\mathcal{G}_{T,r}^{\uparrow}} + \frac{4c_h R^2 T}{\sqrt{W}}. \tag{93}$$

More explicitly, choose a fixed target $\rho_0 \in (0,1)$ and suppose

$$W \ge CTP_T, \qquad C \ge C_{\rho_0} := \left(\frac{4B_\phi R}{\log(1+\rho_0)}\right)^2. \tag{94}$$

Then $\rho_W \le \rho_0$, and the mean-value theorem gives the displayed uniform certificate

$$\begin{aligned}\Psi_{T,W}^{\mathrm{rel}} \le \frac{\zeta_0}{\sqrt{\lambda W}} + \frac{1}{\sigma}\Bigg[&4(1+\rho_0)B_\phi R\sqrt{\frac{C_R}{C}} \\ &+ \frac{8(1+\rho_0)B_\phi^2 R^2 + 2(1+\rho_0)c_h R^2}{\sqrt{CP_T}}\Bigg].\end{aligned} \tag{95}$$

Furthermore $\bar{F}_{T+1} \le 4c_h^2 R^4 T/W$ and $\bar{E}_T \le 2c_h R^2 T/\sqrt{W}$. Hence, if $\mathcal{G}_{T,r}^{\uparrow} = O(r\log T)$ with fixed problem, confidence, and CG constants,

$$R_T = O(r\sqrt{T}\log T) + O(T/\sqrt{W}).$$

At $W = \Theta(TP_T)$ this is $O(r\sqrt{T}\log T + \sqrt{T/P_T})$, and therefore $O(r\sqrt{T}\log T + \sqrt{T})$ up to the displayed residual factor.

Under realizability, Lemma 23 supplies the premise with probability at least $1 - \delta_R$ by taking

$$C_R = 1, \qquad P_T = \max\{1, 8B_\phi^2 R^2 + 4\sigma^2 \log(2T/\delta_R)\}\,. \tag{96}$$

Thus this fully analytic residual choice makes the sufficient width $W = \Omega(T\log(T/\delta_R))$ at fixed problem constants. Its event is added to the certificate failure allocation. No claim is made that the residual or width constants are tight.

Proof. For $h_W(q) = \sqrt{W}\tanh(q/\sqrt{W})$, $h'_W(q) = \operatorname{sech}^2(q/\sqrt{W})$ and

$$h''_W(q) = -\frac{2}{\sqrt{W}}\tanh(q/\sqrt{W})\operatorname{sech}^2(q/\sqrt{W}).$$

The maximum of $2y(1-y^2)$ on $y \in [0,1]$ is $4/(3\sqrt{3})$, proving the gradient and Hessian bounds in (89); also $\sqrt{W}|\tanh(q/\sqrt{W})| \le |q| \le B_\phi R$. Moreover

$$\left|\frac{d}{dq}\log h'_W(q)\right| = \frac{2}{\sqrt{W}}|\tanh(q/\sqrt{W})| \le \frac{2}{\sqrt{W}}.$$

Since $|q_{s,t} - q_s| \le B_\phi\|\boldsymbol{\theta}_t - \boldsymbol{\theta}_s\| \le 2B_\phi R$, $|\log c_{s,t}| \le 4B_\phi R/\sqrt{W}$. Hence $|c_{s,t} - 1| \le e^{4B_\phi R/\sqrt{W}} - 1 = \rho_W$, and solving $e^{4B_\phi R/\sqrt{W}} - 1 < 1$ gives (90) for this exponential certificate.

The trust region gives $Q_t \le 4R^2(t-1)$. Substitution in (114) proves $\bar{\psi}_t^{\mathrm{rel}} \le \Psi_{T,W}^{\mathrm{rel}}$. Taylor's theorem gives $\bar{\epsilon}_t^{\mathrm{lin}} \le 2c_h R^2/\sqrt{W}$, and hence the stated bounds on $\bar{F}_{T+1}, \bar{E}_T$ and the confidence radius (92). Apply Corollary 13, bound the time-varying factors by α_I, ρ_W, and $\mathcal{B}_{T,r}^{\mathrm{rel}}$, and use $\gamma_T \le \mathcal{G}_{T,r}^{\uparrow}$ to obtain (93).

Under (94), the exponential mean-value bound $e^x - 1 \le (1+\rho_0)x$ for $0 \le x \le \log(1+\rho_0)$ gives $\rho_W \le 4(1+\rho_0)B_\phi R/\sqrt{W}$. Substitute this and $W \ge CTP_T$ in (91) to prove (95); the rate follows. Finally Lemma 23 with $B_\mu = B_\phi R$ proves (96). □

Predictable algorithmic curvature under certified transfer. The analysis is conducted for a *generic predictable SPD algorithmic curvature sequence* $\{\mathcal{C}_t\}_{t\ge1}$ satisfying

$$\mathcal{C}_t \text{ is symmetric}, \qquad \mathcal{C}_t \succeq \lambda\mathbf{I} \succ 0, \qquad \mathcal{C}_t \text{ is } \mathcal{H}_t\text{-measurable}. \tag{97}$$

This is the operator whose inverse the algorithm actually applies (via CG) to form widths. The exact full-history result is the special case $\mathcal{C}_t = \mathbf{C}_t$ (relinearized curvature (44)); approximate, subsampled, windowed, refreshed, or sketched operators $\hat{\mathbf{C}}_t$ are also represented by $\mathcal{C}_t$, but are covered only when the required transfer certificate below is available. Write the algorithmic width

$$s_{\mathcal{C},t}^2(a) \;:=\; \mathbf{g}_t(a)^\top \mathcal{C}_t^{-1}\mathbf{g}_t(a). \tag{98}$$

Assumption 6 (One-sided optimism transfer)**.** *For every round $t \le T$ there is an $\mathcal{H}_t$-measurable factor $u_t(a) \ge 1$ such that, for every candidate action $a \in \mathcal{A}_t$,*

$$\bar{s}_t^2(a) \;\le\; u_t(a)\, s_{\mathcal{C},t}^2(a). \tag{99}$$

This is the *only* spectral relation the one-sided fallback theorem needs, and it is one-sided: it dominates the predictable frozen-feature width $\bar{s}_t(a)$ (the object of the self-normalized confidence bound) by an inflated algorithmic width. No *lower* Loewner factor on $\mathcal{C}_t$ is required for optimism. The factor may be action-dependent; when a uniform per-round u_t satisfies (99) for every action we write u_t and state the action-dependent theorem with $u_t(a)$. For the exact-curvature instantiation $\mathcal{C}_t = \mathbf{C}_t$, the whitened relative-drift lemma below certifies (99) with the uniform factor $u_t = (1+\bar{\chi}_t)^2$; for an approximate operator, the composition of Lemma 30 with a one-sided spectral bound $\hat{\mathbf{C}}_t \preceq \kappa_{+,t}\mathbf{C}_t$ gives $u_t = \kappa_{+,t}(1+\bar{\chi}_t)^2$ (Theorem 45).

The following lemma supplies the exact-curvature transfer factor through a *whitened, relative* feature-drift bound: the drift is measured in the $\bar{\mathbf{C}}_t^{-1}$ metric and accumulated in ℓ_2, and the resulting one-sided bound needs *no* smallness condition.

Lemma 30 (Whitened relative feature-drift bound)**.** *Let* $\mathbf{g}_s := \nabla_{\boldsymbol{\theta}}\mu_{\boldsymbol{\theta}_s}(\mathbf{x}_s, a_s)$*,* $\mathbf{g}_{s,t} := \nabla_{\boldsymbol{\theta}}\mu_{\boldsymbol{\theta}_t}(\mathbf{x}_s, a_s)$*,* $\delta_{s,t} := \mathbf{g}_{s,t} - \mathbf{g}_s$*, and* $\Delta_{s,t} := \|\boldsymbol{\theta}_t - \boldsymbol{\theta}_s\|$*. Using the symmetric (principal) square root* $\bar{\mathbf{C}}_t^{1/2}$*, define the* $d \times (t-1)$ *whitened column matrices*

$$\mathsf{G}_t := \Big[\bar{\mathbf{C}}_t^{-1/2}\mathbf{g}_s/\sigma\Big]_{s<t}, \qquad \mathsf{D}_t := \Big[\bar{\mathbf{C}}_t^{-1/2}\delta_{s,t}/\sigma\Big]_{s<t}, \qquad \chi_t := \|\mathsf{D}_t\|_{\mathrm{op}}. \tag{100}$$

Writing $n_t = t-1$*, also define the* $(d+n_t)\times d$ *stacked matrices*

$$\mathbf{A}_t := \begin{bmatrix} \sqrt{\lambda}\,\bar{\mathbf{C}}_t^{-1/2} \\ \mathsf{G}_t^\top \end{bmatrix}, \qquad \mathbf{B}_t := \begin{bmatrix} \sqrt{\lambda}\,\bar{\mathbf{C}}_t^{-1/2} \\ (\mathsf{G}_t + \mathsf{D}_t)^\top \end{bmatrix}. \tag{101}$$

At $t = 1$ *the lower blocks have zero rows. All square roots and inverse square roots in this statement are principal. Then:*

(a) $\mathsf{G}_t\mathsf{G}_t^\top = \mathbf{I} - \lambda\,\bar{\mathbf{C}}_t^{-1} \preceq \mathbf{I}$*, hence* $\|\mathsf{G}_t\|_{\mathrm{op}} \le 1$*.*

(b) $\bar{\mathbf{C}}_t^{-1/2}(\mathbf{C}_t - \bar{\mathbf{C}}_t)\bar{\mathbf{C}}_t^{-1/2} = \mathsf{G}_t\mathsf{D}_t^\top + \mathsf{D}_t\mathsf{G}_t^\top + \mathsf{D}_t\mathsf{D}_t^\top$*, so* $\big\|\bar{\mathbf{C}}_t^{-1/2}(\mathbf{C}_t - \bar{\mathbf{C}}_t)\bar{\mathbf{C}}_t^{-1/2}\big\|_{\mathrm{op}} \le 2\chi_t + \chi_t^2$*.*

(c) The stacked designs obey

$$\mathbf{A}_t^\top\mathbf{A}_t = \mathbf{I}_d, \qquad \mathbf{B}_t^\top\mathbf{B}_t = \bar{\mathbf{C}}_t^{-1/2}\mathbf{C}_t\bar{\mathbf{C}}_t^{-1/2}, \qquad \|\mathbf{B}_t - \mathbf{A}_t\|_{\mathrm{op}} = \chi_t. \tag{102}$$

(d) ***(One-sided, no smallness condition.)*** *For every* $\chi_t \ge 0$*,*

$$\mathbf{C}_t \;\preceq\; (1+\chi_t)^2\,\bar{\mathbf{C}}_t, \qquad \text{hence} \qquad \bar{s}_t^2(a) \;\le\; (1+\chi_t)^2\, s_t^2(a) \quad \forall a. \tag{103}$$

A predictable certificate $\bar{\chi}_t \ge \chi_t$ *therefore satisfies Assumption 6 in the exact case* $\mathcal{C}_t = \mathbf{C}_t$ *with the uniform factor* $u_t = (1+\bar{\chi}_t)^2$*.*

(e) ***(Whitened*** ℓ_2 ***primitive bound.)***

$$\chi_t \;\le\; \|\mathsf{D}_t\|_F = \left[\sigma^{-2}\sum_{s<t}\|\delta_{s,t}\|^2_{\bar{\mathbf{C}}_t^{-1}}\right]^{1/2} \;\le\; \frac{L_g}{\sigma\sqrt{\lambda}}\left[\sum_{s<t}\|\boldsymbol{\theta}_t - \boldsymbol{\theta}_s\|^2\right]^{1/2}, \tag{104}$$

the last step under the L_g*-Lipschitz-Jacobian assumption* $\|\delta_{s,t}\| \le L_g\Delta_{s,t}$*.*

Proof. (a) $\mathsf{G}_t\mathsf{G}_t^\top = \sigma^{-2}\sum_{s<t}\bar{\mathbf{C}}_t^{-1/2}\mathbf{g}_s\mathbf{g}_s^\top\bar{\mathbf{C}}_t^{-1/2} = \bar{\mathbf{C}}_t^{-1/2}\big(\sigma^{-2}\sum_{s<t}\mathbf{g}_s\mathbf{g}_s^\top\big)\bar{\mathbf{C}}_t^{-1/2} = \bar{\mathbf{C}}_t^{-1/2}(\bar{\mathbf{C}}_t - \lambda\mathbf{I})\bar{\mathbf{C}}_t^{-1/2} = \mathbf{I} - \lambda\bar{\mathbf{C}}_t^{-1}$. Since $\lambda\bar{\mathbf{C}}_t^{-1} \succeq 0$, $\mathsf{G}_t\mathsf{G}_t^\top \preceq \mathbf{I}$, and $\|\mathsf{G}_t\|_{\mathrm{op}}^2 = \|\mathsf{G}_t\mathsf{G}_t^\top\|_{\mathrm{op}} \le 1$.

(b) From $\mathbf{g}_{s,t} = \mathbf{g}_s + \delta_{s,t}$, $\mathbf{g}_{s,t}\mathbf{g}_{s,t}^\top - \mathbf{g}_s\mathbf{g}_s^\top = \mathbf{g}_s\delta_{s,t}^\top + \delta_{s,t}\mathbf{g}_s^\top + \delta_{s,t}\delta_{s,t}^\top$. Conjugating each summand by $\bar{\mathbf{C}}_t^{-1/2}$ and weighting by σ^{-2}, the three groups sum exactly to $\mathsf{G}_t\mathsf{D}_t^\top$, $\mathsf{D}_t\mathsf{G}_t^\top$, and $\mathsf{D}_t\mathsf{D}_t^\top$. By the triangle inequality and $\|\mathsf{G}_t\|_{\mathrm{op}} \le 1$, $\|\mathsf{G}_t\mathsf{D}_t^\top + \mathsf{D}_t\mathsf{G}_t^\top + \mathsf{D}_t\mathsf{D}_t^\top\|_{\mathrm{op}} \le 2\|\mathsf{G}_t\|_{\mathrm{op}}\|\mathsf{D}_t\|_{\mathrm{op}} + \|\mathsf{D}_t\|_{\mathrm{op}}^2 \le 2\chi_t + \chi_t^2$.

(c) By part (a), $\mathbf{A}_t^\top\mathbf{A}_t = \lambda\bar{\mathbf{C}}_t^{-1} + \mathsf{G}_t\mathsf{G}_t^\top = \mathbf{I}_d$. Expanding the current replay Gram gives

$$\mathbf{B}_t^\top\mathbf{B}_t = \lambda\bar{\mathbf{C}}_t^{-1} + (\mathsf{G}_t + \mathsf{D}_t)(\mathsf{G}_t + \mathsf{D}_t)^\top = \bar{\mathbf{C}}_t^{-1/2}\mathbf{C}_t\bar{\mathbf{C}}_t^{-1/2}.$$

The two stacks differ only in their bottom block, hence $\|\mathbf{B}_t - \mathbf{A}_t\|_{\mathrm{op}} = \|\mathsf{D}_t^\top\|_{\mathrm{op}} = \chi_t$. These identities also hold at $t = 1$, with empty bottom blocks and $\chi_1 = 0$.

(d) The matrix $W_t := \bar{\mathbf{C}}_t^{-1/2}(\mathbf{C}_t - \bar{\mathbf{C}}_t)\bar{\mathbf{C}}_t^{-1/2}$ is symmetric with $\|W_t\|_{\mathrm{op}} \le 2\chi_t + \chi_t^2$, hence $W_t \preceq (2\chi_t + \chi_t^2)\mathbf{I}$. Congruence by $\bar{\mathbf{C}}_t^{1/2}$ gives $\mathbf{C}_t - \bar{\mathbf{C}}_t \preceq (2\chi_t + \chi_t^2)\bar{\mathbf{C}}_t$, i.e. $\mathbf{C}_t \preceq (1+\chi_t)^2\bar{\mathbf{C}}_t$; no bound on χ_t is used. Both $\mathbf{C}_t, \bar{\mathbf{C}}_t \succ 0$, so inversion reverses the order: $\mathbf{C}_t^{-1} \succeq (1+\chi_t)^{-2}\bar{\mathbf{C}}_t^{-1}$. The quadratic form at $\mathbf{g}_t(a)$ gives $s_t^2(a) \ge (1+\chi_t)^{-2}\bar{s}_t^2(a)$, i.e. $\bar{s}_t^2(a) \le (1+\chi_t)^2 s_t^2(a)$. Since $x \mapsto (1+x)^2$ is increasing on $[0,\infty)$, replacing χ_t by any $\bar{\chi}_t \ge \chi_t$ preserves the inequality, giving $u_t = (1+\bar{\chi}_t)^2$.

(e) $\chi_t = \|\mathsf{D}_t\|_{\mathrm{op}} \le \|\mathsf{D}_t\|_F$, and $\|\mathsf{D}_t\|_F^2 = \sum_{s<t}\|\bar{\mathbf{C}}_t^{-1/2}\delta_{s,t}/\sigma\|^2 = \sigma^{-2}\sum_{s<t}\delta_{s,t}^\top\bar{\mathbf{C}}_t^{-1}\delta_{s,t} = \sigma^{-2}\sum_{s<t}\|\delta_{s,t}\|^2_{\bar{\mathbf{C}}_t^{-1}}$. Since $\bar{\mathbf{C}}_t^{-1} \preceq \lambda^{-1}\mathbf{I}$, $\|\delta_{s,t}\|^2_{\bar{\mathbf{C}}_t^{-1}} \le \|\delta_{s,t}\|^2/\lambda \le L_g^2\Delta_{s,t}^2/\lambda$; summing and taking square roots gives (104). ∎

Lemma 31 (Relative scalar-link drift and centering)**.** *Suppose the prediction model has the fixed-feature scalar-link form $\mu_{\boldsymbol{\theta}}(z) = h(\phi(z)^\top \boldsymbol{\theta})$, where $\phi(z) \in \mathbb{R}^d$, h is differentiable, and $h'(q)$ is nonzero throughout the certified scalar region. For $s < t$, write $\phi_s = \phi(z_s)$ and define*

$$q_s = \phi_s^\top \boldsymbol{\theta}_s, \qquad q_{s,t} = \phi_s^\top \boldsymbol{\theta}_t, \qquad c_{s,t} = \frac{h'(q_{s,t})}{h'(q_s)}, \qquad \rho_t = \max_{s<t} |c_{s,t} - 1|,$$

with $\rho_1 = 0$. Here ϕ_s, q_s, and $h'(q_s)$ are fixed before reward r_s, while $\boldsymbol{\theta}_t$ and every ratio above are $\mathcal{H}_t$-measurable before action a_t is selected. Then:

(a) The scalar-link gradients obey

$$\mathbf{g}_s = h'(q_s)\phi_s, \qquad \mathbf{g}_{s,t} = h'(q_{s,t})\phi_s = c_{s,t}\mathbf{g}_s, \qquad \delta_{s,t} = (c_{s,t} - 1)\mathbf{g}_s. \tag{105}$$

Writing $\mathbf{c}_t = (c_{1,t}, \ldots, c_{t-1,t})^\top \in \mathbb{R}^{t-1}$, the $d \times (t-1)$ matrices of Lemma 30 satisfy

$$\mathsf{D}_t = \mathsf{G}_t \operatorname{Diag}(\mathbf{c}_t - \mathbf{1}), \qquad \chi_t \le \|\mathsf{G}_t\|_{\mathrm{op}} \rho_t \le \rho_t. \tag{106}$$

At $t = 1$ the matrices have zero columns and the same identities hold under the empty-matrix convention.

(b) Exact current curvature therefore satisfies, without a smallness condition,

$$\mathbf{C}_t \preceq (1 + \rho_t)^2 \bar{\mathbf{C}}_t. \tag{107}$$

If $\rho_t < 1$, it also satisfies

$$(1 - \rho_t)^2 \bar{\mathbf{C}}_t \preceq \mathbf{C}_t \preceq (1 + \rho_t)^2 \bar{\mathbf{C}}_t. \tag{108}$$

Any predictable upper certificate $\bar{\rho}_t \ge \rho_t$ may replace ρ_t in the one-sided result, and may replace it in both sides when $\bar{\rho}_t < 1$.

(c) Reuse $\bar{r}_{s,t}, e_{s,t}$, and M_t from Lemma 22, and define

$$v_{s,t} := \bar{r}_{s,t}(c_{s,t} - 1) + c_{s,t} e_{s,t}, \qquad \mathbf{v}_t = (v_{1,t}, \ldots, v_{t-1,t})^\top \in \mathbb{R}^{t-1}. \tag{109}$$

Then the centering mismatch has the exact structured form

$$M_t = \frac{1}{\sigma^2} \sum_{s<t} v_{s,t} \mathbf{g}_s, \qquad \|M_t\|_{\bar{\mathbf{C}}_t^{-1}} \le \frac{\|\mathbf{v}_t\|_2}{\sigma}. \tag{110}$$

(d) If $\|\boldsymbol{\theta}_s\|, \|\boldsymbol{\theta}_t\| \le R$, the gradients are bounded by G and are L_g-Lipschitz on the radius-R region, then

$$\|\bar{\mathbf{r}}_t\|_2 \le \sqrt{R_t^{\mathrm{col}}} + G\sqrt{Q_t}, \qquad \bar{\mathbf{r}}_t = (\bar{r}_{1,t}, \ldots, \bar{r}_{t-1,t})^\top, \tag{111}$$

$$\|\mathbf{e}_t\|_2 \le L_g R \sqrt{Q_t}, \qquad \mathbf{e}_t = (e_{1,t}, \ldots, e_{t-1,t})^\top, \tag{112}$$

$$\|\mathbf{v}_t\|_2 \le \rho_t \left(\sqrt{R_t^{\mathrm{col}}} + G\sqrt{Q_t} \right) + (1 + \rho_t) L_g R \sqrt{Q_t}. \tag{113}$$

Consequently a valid original-center certificate is

$$\bar{\psi}_t^{\mathrm{rel}} := \frac{\zeta_t}{\sqrt{\lambda}} + \frac{\rho_t(\sqrt{R_t^{\mathrm{col}}} + G\sqrt{Q_t}) + (1 + \rho_t) L_g R \sqrt{Q_t}}{\sigma}. \tag{114}$$

The right side remains valid after replacing ρ_t by a predictable upper bound $\bar{\rho}_t$.

The exact ρ_t requires replaying ϕ_s and retaining at least the collection scalar q_s (or $h'(q_s)$) for every sample. Computing the exact $v_{s,t}$ additionally uses the stored reward and the same scalar-link replay; q_s suffices because both $\bar{r}_{s,t}$ and $e_{s,t}$ can be reconstructed from $q_s, q_{s,t}, h(q_s)$, and $h'(q_s)$. Thus the structured replay uses $O(t)$ scalar metadata in addition to raw-example replay. The Welford quantities Q_t, R_t^{col} remain $O(d)$ path-summary state, but neither implementation is $O(d)$ total memory.

Proof. Equation (105) is the chain rule. Applying it to each column of (100) gives (106). Since the operator norm of the diagonal matrix is ρ_t and $\|\mathsf{G}_t\|_{\mathrm{op}} \le 1$ by Lemma 30, $\chi_t \le \rho_t$. Equations (107) and (108) then follow respectively from the one-sided part of Lemma 30 and Corollary 32.

Substitute $\delta_{s,t} = (c_{s,t} - 1)\mathbf{g}_s$ in the definition of M_t:

$$\bar{r}_{s,t}\delta_{s,t} + e_{s,t}\mathbf{g}_s + e_{s,t}\delta_{s,t} = \{\bar{r}_{s,t}(c_{s,t} - 1) + c_{s,t}e_{s,t}\}\mathbf{g}_s.$$

This proves the first identity in (110). Left multiplication by $\bar{\mathbf{C}}_t^{-1/2}$ gives the transformed identity $\bar{\mathbf{C}}_t^{-1/2}M_t = \sigma^{-1}\mathsf{G}_t\mathbf{v}_t$, so

$$\|M_t\|_{\bar{\mathbf{C}}_t^{-1}} = \sigma^{-1}\|\mathsf{G}_t\mathbf{v}_t\|_2 \le \sigma^{-1}\|\mathbf{v}_t\|_2,$$

which proves the second identity and uses equivalently $\mathsf{G}_t^\top \mathsf{G}_t \preceq I_{t-1}$.

Finally let $d_{s,t} = \|\boldsymbol{\theta}_t - \boldsymbol{\theta}_s\|$. The collection residual $c_s = \mu_{\boldsymbol{\theta}_s}(z_s) - r_s$ gives $\bar{r}_{s,t} = c_s + \mathbf{g}_s^\top(\boldsymbol{\theta}_t - \boldsymbol{\theta}_s)$. The triangle and Cauchy–Schwarz inequalities therefore yield $\|\bar{\mathbf{r}}_t\|_2 \le \sqrt{R_t^{\mathrm{col}}} + G\sqrt{Q_t}$. Taylor's theorem gives $|e_{s,t}| \le (L_g/2)d_{s,t}^2$; because both parameters lie in the radius-R region, $d_{s,t} \le 2R$, hence $|e_{s,t}| \le L_g R d_{s,t}$ and $\|\mathbf{e}_t\|_2 \le L_g R\sqrt{Q_t}$. Applying the triangle inequality to $\mathbf{v}_t = \mathrm{Diag}(\mathbf{c}_t - \mathbf{1})\bar{\mathbf{r}}_t + \mathrm{Diag}(\mathbf{c}_t)\mathbf{e}_t$, using $\|\mathbf{c}_t - \mathbf{1}\|_\infty = \rho_t$ and $\|\mathbf{c}_t\|_\infty \le 1 + \rho_t$, proves (113). Combine this with Lemma 22 and (110) to obtain (114). □

Corollary 32 (Sharp two-sided feature-drift sandwich)**.** *If $\chi_t < 1$, then*

$$(1 - \chi_t)^2\, \bar{\mathbf{C}}_t \;\preceq\; \mathbf{C}_t \;\preceq\; (1 + \chi_t)^2\, \bar{\mathbf{C}}_t. \tag{115}$$

Consequently, if a predictable certificate satisfies $\chi_t \le \bar{\chi}_t < 1$, then $(1 - \bar{\chi}_t)^2\bar{\mathbf{C}}_t \preceq \mathbf{C}_t \preceq (1 + \bar{\chi}_t)^2\bar{\mathbf{C}}_t$.

Proof. Because $\mathbf{A}_t^\top \mathbf{A}_t = \mathbf{I}_d$, all d singular values of the $(d + n_t) \times d$ matrix $\mathbf{A}_t$ equal one. Singular-value perturbation and (102) give, for $j = 1, \dots, d$,

$$|\sigma_j(\mathbf{B}_t) - 1| \le \|\mathbf{B}_t - \mathbf{A}_t\|_{\mathrm{op}} = \chi_t.$$

For $\chi_t < 1$, every singular value of $\mathbf{B}_t$ therefore lies in $[1 - \chi_t, 1 + \chi_t]$. Squaring the singular values and using $\mathbf{B}_t^\top\mathbf{B}_t = \bar{\mathbf{C}}_t^{-1/2}\mathbf{C}_t\bar{\mathbf{C}}_t^{-1/2}$ yields $(1 - \chi_t)^2\mathbf{I} \preceq \bar{\mathbf{C}}_t^{-1/2}\mathbf{C}_t\bar{\mathbf{C}}_t^{-1/2} \preceq (1 + \chi_t)^2\mathbf{I}$. Congruence by the principal $\bar{\mathbf{C}}_t^{1/2}$ proves (115). If $0 \le \chi_t \le \bar{\chi}_t < 1$, then $(1 - \bar{\chi}_t)^2 \le (1 - \chi_t)^2$ and $(1 + \chi_t)^2 \le (1 + \bar{\chi}_t)^2$, proving the certificate version. □

The one-sided fallback theorem uses only the bound (103), which holds for *any* $\chi_t \ge 0$; the smallness condition $\chi_t < 1$ is needed only for the lower half of the two-sided sandwich (115), which in turn is used by the explicit near-linear corollary and by any optional analysis that explicitly assumes a two-sided sandwich. An unrescaled current-parameter window is handled by Proposition 44 through a one-sided relation and does not require this condition. The binding quantity is the *whitened ℓ_2 accumulation* $\chi_t \le \|\mathsf{D}_t\|_F$ of replayed-Jacobian drift, measured in the $\bar{\mathbf{C}}_t^{-1}$ metric—a relative, whitened ℓ_2 bound that can be sharper than an ℓ_1 sum normalized only by λ when the drift occurs in well-observed directions (it credits directions the design has already explored), though it does not dominate uniformly. Together, Lemmas 22 and 30 identify exact whitened diagnostic targets. The operational upper bounds in Lemma 24 replace those targets by the predictable path statistics Q_t and R_t, the certified optimizer residual ζ_t, and analytic smoothness constants. Thus $\bar{\psi}_t$ and $u_t = (1 + \bar{\chi}_t)^2$ require only $O(d)$ additional state; the exact whitened quantities remain post-hoc diagnostics used to measure the conservatism of these certificates.

Positive definiteness of both $\mathbf{C}_t$ and $\bar{\mathbf{C}}_t$ follows from $\lambda > 0$ and the outer-product form; it is not an independent assumption. All theoretical results hold under these explicit assumptions; we do *not* claim a regret guarantee for the full nonlinear model outside the local-linearization regime.

F.2 Novelty Relative to Prior Neural Bandits

NeuralUCB/NeuralTS use historical-time full-gradient Grams in their stated algorithms, while NeuralLinear and last-layer Laplace restrict uncertainty to a learned representation or final layer (Zhou et al., 2020; Zhang et al., 2021; Riquelme et al., 2018; Kristiadi et al., 2020). Matrix-free full-network predictive-variance solves also predate this work (Maddox et al., 2021). CC-UCB's narrower adaptation is to replay every historical Jacobian at the current parameter, separate the predictable frozen confidence Gram from that algorithmic operator, and expose centering, drift, transfer, and residual-checked CG terms. We claim neither the CG/CVP primitive nor a prior assertion of uniform full-metric superiority as new.

The one-sided whitened result in Lemma 30 is the main technical step: optimism transfer needs no lower spectral comparison. The determinant formula is bookkeeping rather than a standalone rate, and the supplied-U closure is exactly the projected Gram in (86), not a concrete neural-training theorem. Appendix G compares NTK, regression, EKF/LO-FI, structured Laplace, and sketching alternatives.

F.3 Computable Curvature Operator

Applying $\mathbf{C}_t\mathbf{v} = \lambda\mathbf{v} + \sigma^{-2}\sum_{s<t}\mathbf{g}_t(\mathbf{x}_s, a_s)\big[\mathbf{g}_t(\mathbf{x}_s, a_s)^\top\mathbf{v}\big]$ *naively* requires one curvature–vector product (CVP) per historical sample per CG step—cost $O(It)$ CVPs per round. Each CVP evaluates one Jacobian–vector product (forward mode) followed by one vector–Jacobian product (reverse mode) through the network, at roughly the cost of one gradient evaluation (one JVP plus one VJP)[1]; we do not store per-sample gradient features $\mathbf{g}_s$, only a constant number of $O(d)$ CG vectors for CG, with CVPs computed on the fly. We describe three strategies that limit the per-step cost to $|\mathcal{S}_t|$ CVPs:

(i) Subsampled GGN (worst-case spectral baseline). At each round draw a uniform subsample $\mathcal{S}_t \subset \{1, \dots, t-1\}$ of size n (without replacement) and set $w_s = (t-1)/n$ to obtain the rescaled operator. The following lemma is a worst-case uniform spectral guarantee.

Lemma 33 (Worst-case uniform spectral guarantee for subsampled GGN)**.** *Fix round t and set $m = t-1$. Let $Y_s = \sigma^{-2}\mathbf{g}_t(\mathbf{x}_s, a_s)\mathbf{g}_t(\mathbf{x}_s, a_s)^\top$ for $s = 1, \dots, m$, so that $\mathbf{C}_t = \lambda\mathbf{I} + V$ with $V = \sum_{s=1}^m Y_s$. Each Y_s is PSD with $\|Y_s\|_{\text{op}} \le G^2/\sigma^2 =: B$. For $\delta \in (0,1)$ and $1 \le n \le m$, draw $\mathcal{S}_t$ of size n uniformly without replacement from $\{1, \dots, m\}$ and define $\hat{V} = \frac{m}{n}\sum_{s\in\mathcal{S}_t} Y_s$ and $\hat{\mathbf{C}}_t = \lambda\mathbf{I} + \hat{V}$. Write $L := \log(2d/\delta)$.*

(i) Deviation bound. *With probability $\ge 1 - \delta$,*

$$\|\hat{V} - V\|_{\text{op}} \;\le\; mB\sqrt{\frac{2L}{n}} \;+\; \frac{4\,mB\,L}{3\,n}. \tag{116}$$

(ii) Sufficient sample size. *For any $\varepsilon \in (0,1)$, if*

$$n \;\ge\; 8L\left[\frac{m^2\,B^2}{\varepsilon^2\,\lambda^2} \;+\; \frac{m\,B}{3\,\varepsilon\,\lambda}\right], \tag{117}$$

then $\Pr[\|\hat{V} - V\|_{\text{op}} > \varepsilon\lambda] \le \delta$ *and consequently* $(1-\varepsilon)\mathbf{C}_t \preceq \hat{\mathbf{C}}_t \preceq (1+\varepsilon)\mathbf{C}_t$*.*

Proof. The $\{Y_s\}_{s=1}^m$ are fixed PSD self-adjoint matrices (deterministic conditional on the history $\mathcal{F}_{t-1}$ and the current parameter $\boldsymbol{\theta}_t$). The displayed proof is for *with-replacement* (i.i.d.) sampling. The without-replacement version follows from the domination argument of Gross and Nesme (2010), building on Hoeffding's reduction (Hoeffding, 1963): the matrix moment-generating function of a sum sampled without replacement from a fixed finite collection of self-adjoint matrices is dominated by that of the corresponding i.i.d. sum, so every tail bound derived from the matrix Laplace-transform method, including the matrix Bernstein inequality of Tropp (2015, Thm. 6.1.1), carries over with the same constants. In our setting the $\{Y_s\}$ are *fixed* PSD matrices (deterministic conditional on $\mathcal{F}_{t-1}$ and $\boldsymbol{\theta}_t$) and only the index set $\mathcal{S}_t$ is random, so the domination argument applies directly. We therefore state and prove the bound for i.i.d. draws $s_1, \dots, s_n$ uniform on $\{1, \dots, m\}$; the without-replacement guarantee is at least as strong. Define the centered random matrices $Z_k = (m/n)Y_{s_k} - V/n$ (mean zero, independent). The range satisfies $\|Z_k\|_{\text{op}} \le (m/n)B + (1/n)\|V\| \le 2mB/n =: R$ (using $\|V\| \le mB$). For the variance parameter, since $Y_s \preceq B\mathbf{I}$,

$$\mathbb{E}[Z_k Z_k^\top] \;\preceq\; \tfrac{m^2}{n^2}\mathbb{E}[Y_{s_1}^2] \;=\; \tfrac{m}{n^2}\sum_{s=1}^m Y_s^2 \;\preceq\; \tfrac{m}{n^2}B\,V \;\preceq\; \tfrac{m^2B^2}{n^2}\mathbf{I},$$

[1]Implementations based on double-backward Hessian–vector products cost closer to two gradient-equivalents per CVP; the constant does not affect the $O(\cdot)$ accounting.

so $v := \left\|\sum_{k=1}^{n} \mathbb{E}[Z_k Z_k^\top]\right\| \le m^2B^2/n$. By the matrix Bernstein inequality (Tropp, 2015, Thm. 6.1.1), $\Pr[\|\hat{V}-V\| \ge t] \le 2d\exp\bigl(-t^2/(2v+2Rt/3)\bigr)$. Choosing $t_0 = \sqrt{2vL} + \frac{2}{3}RL$ makes the matrix-Bernstein tail probability at most δ. Substituting $v \le m^2B^2/n$ and $R = 2mB/n$ gives $t_0 \le mB\sqrt{2L/n} + \frac{4mBL}{3n}$, yielding (116). For (117), require each summand in (116) to be $\le \varepsilon\lambda/2$: $mB\sqrt{2L/n} \le \varepsilon\lambda/2$ gives $n \ge 8m^2B^2L/(\varepsilon^2\lambda^2)$, and $4mBL/(3n) \le \varepsilon\lambda/2$ gives $n \ge 8mBL/(3\varepsilon\lambda)$. Summing the two conditions yields (117); this is a conservative sufficient condition (using $a+b$ rather than $\max\{a,b\}$), not a sharp threshold. The multiplicative sandwich follows: $\|\hat{V}-V\| \le \varepsilon\lambda$ gives $\mathbf{C}_t - \varepsilon\lambda\mathbf{I} \preceq \hat{\mathbf{C}}_t \preceq \mathbf{C}_t + \varepsilon\lambda\mathbf{I}$. Since $\mathbf{C}_t \succeq \lambda\mathbf{I}$, we have $\varepsilon\lambda\mathbf{I} \preceq \varepsilon\mathbf{C}_t$, so $\mathbf{C}_t - \varepsilon\lambda\mathbf{I} \succeq (1-\varepsilon)\mathbf{C}_t$ and $\mathbf{C}_t + \varepsilon\lambda\mathbf{I} \preceq (1+\varepsilon)\mathbf{C}_t$. □

With $B = G^2/\sigma^2$, the right-hand side of the sufficient condition (117) is $O\bigl(L \cdot [m^2G^4/(\varepsilon^2\lambda^2\sigma^4) + mG^2/(\varepsilon\lambda\sigma^2)]\bigr)$. The first (variance) term dominates unless $\varepsilon\lambda\sigma^2 \gtrsim mG^2$, so this worst-case sufficient threshold has a *quadratic leading dependence* on $m = t-1$ (it is a sufficient upper bound on the required sample size, not an exact or necessary sample complexity). The condition (117) is conservative and sufficient, not necessary, with worst-case leading dependence quadratic in m. If its threshold exceeds the population size $m = t-1$, it provides no nontrivial subsampling guarantee for $n < m$. Taking $n = m$ recovers the full population exactly for the separate reason that every summand is included.

This is a worst-case baseline guarantee, too conservative for practical use at large t. In particular, the bound above does not by itself justify computational savings at large t; it should be read as a worst-case spectral guarantee rather than as a practical scaling result. Subsampling reduces computation ($n \ll m$) only when m is moderate or $\varepsilon\lambda$ is large relative to mB; for large t the full history may be needed under the uniform Bernstein bound. Under additional spectral structure, leverage-score sampling or Frequent Directions sketches may yield useful approximations with smaller state. For linear bandits, Kuzborskij et al. (2019) give regret bounds in terms of the Frequent Directions spectral-tail error. Translating such structure into a uniform GGN sandwich and a concrete buffer or sketch size is left open. The Bernstein guarantee is not the mechanism by which we expect practical savings at scale. Practical savings require small buffers treated heuristically, spectral decay/effective dimension, ridge-leverage sampling, Frequent Directions, or empirical verification of the spectral sandwich. We leave structure-aware sampling and empirical spectral certification as future work. To obtain a spectral sandwich that holds simultaneously for all $t \le T$, union-bound over t (replacing δ by δ/T). Theorem 45 then exposes the one-sided transfer factor and realized width complexity as operator-dependent terms along the realized path; it does not identify a standalone causal regret inflation.

(ii) Windowed curvature (unrescaled, current parameter). Maintain a sliding window $\mathcal{S}_t \subseteq \{1,\ldots,t-1\}$ of the most recent n interactions with unit weights (w_s=1), *re-evaluating the Jacobians at the current parameter* $\boldsymbol{\theta}_t$. This is a nonnegative-weighted *subset* of the same current-parameter outer products, so $\hat{\mathbf{C}}_t \preceq \mathbf{C}_t$ (Proposition 44); Theorem 45 then applies with the *deterministic* approximate-to-full factor $\kappa_{+,t}$=1 and the full-to-frozen factor $u_t = (1+\bar{\chi}_t)^2$—no two-sided sandwich is required, though the overall guarantee remains conditional on that transfer certificate. The cost of the smaller operator is not hidden: it surfaces as larger realized widths $s^2_{\mathcal{C},t}(a_t)$ and hence a larger dynamic width complexity $\Lambda_T^{\mathcal{C}}$. The same holds for any current-parameter downweighted subset with $w_s \in [0,1]$ (including an unrescaled subsample). A *stale* window that reuses Jacobians from an older parameter is *not* covered; see (iii).

(iii) Periodic refresh (heuristic; stale operator). Recompute the curvature operator every M rounds on a cached buffer and reuse it for M consecutive actions. Because the reused operator evaluates Jacobians at an *older* parameter, it is no longer a subset of the current-parameter outer products, so the $\hat{\mathbf{C}}_t \preceq \mathbf{C}_t$ argument of Proposition 44 does not apply and the stale operator can both over- and under-estimate $\mathbf{C}_t$. This remains a heuristic: Theorem 45 covers it only when a one-sided factor $\kappa_{+,t}$ (an explicit bound $\hat{\mathbf{C}}_t \preceq \kappa_{+,t}\mathbf{C}_t$) is separately certified. The rescaled subsample of (i), with weights $w_s = (t-1)/n > 1$, is likewise not covered by Proposition 44; it instead uses the two-sided matrix Bernstein bound of Lemma 33.

F.4 Computational Cost Model

Primitive assumptions. CC-UCB requires a single computational primitive: the *curvature–vector product* (CVP) $\mathbf{v} \mapsto \mathbf{C}_t\mathbf{v}$, which decomposes into per-sample operations $\mathbf{v} \mapsto \mathbf{g}_t(\mathbf{x}_s, a_s)\bigl[\mathbf{g}_t(\mathbf{x}_s, a_s)^\top\mathbf{v}\bigr]$. Each such operation consists of:

1. a Jacobian–vector product (JVP) $\mathbf{v} \mapsto \mathbf{g}_t^\top\mathbf{v}$ (forward-mode AD, cost $\approx 1\times$ forward pass);

2. a vector–Jacobian product (VJP) $\alpha \mapsto \alpha\,\mathbf{g}_t$ (reverse-mode AD, cost $\approx 1\times$ backward pass).

Sharded autodiff can distribute JVP/VJP evaluation, but its communication pattern and sample parallelism are implementation-dependent and are not benchmarked here. For $m_t := |\mathcal{S}_t|$, one matrix-free oracle application consists of m_t sample-CVPs plus $O(d)$ damping and accumulation work. Thus $I_{t,a}$ CG recurrences for action a use $I_{t,a}m_t$ sample-CVPs; an explicit final check of the original-system residual $\mathbf{g}_t(a) - \mathcal{C}_t\tilde{\mathbf{u}}$ uses another m_t. The sequential Krylov state is $O(d)$ ($O(Kd)$ if right-hand sides are batched), excluding model and optimizer state, activations, and raw replay. Caching an explicit $m_t \times d$ Jacobian matrix instead uses $O(m_t d)$ additional state and is not the on-the-fly JVP/VJP implementation analyzed by this operation count.

Krylov methods for Hessian spectral analysis have been run on networks with $> 10^8$ parameters under distributed training (Granziol and Juarev, 2026), which shows that related curvature–vector primitives can be implemented at that scale. This does not establish feasibility for the exact per-example GGN JVP/VJP path analyzed here, nor does it measure end-to-end bandit training. Warm-starting CG from the previous round's solution can reduce the iteration count in practice.

Cost accounting. Let $C_F(W_{\text{net}}), C_B(W_{\text{net}}), C_J(W_{\text{net}})$, and $C_V(W_{\text{net}})$ denote the one-example forward, reverse, JVP, and VJP costs for model-compute scale W_{net}, and define

$$C_{\text{op},t} := m_t[C_J(W_{\text{net}}) + C_V(W_{\text{net}})] + O(d).$$

Parameter count d alone does not determine these costs when architecture, input length, or weight sharing changes. Let J^{opt}_{t+1} be the number of full-batch first-order steps after round t; $q^{\text{opt}}_{t+1} \in \{0,1\}$ records a separate full-gradient optimizer certificate, and $q^{\text{CG}}_{t,a} \in \{0,1\}$ records an explicit final CG residual application. Table 3 separates the training history N_t from the curvature buffer m_t.

Table 3: Per-round arithmetic cost. $N_t = t-1$ is the training-history size; $m_t = |\mathcal{S}_t|$ is the curvature-buffer size.

Component	Cost
Candidate means/features	$O(K[C_F + C_B])$
CG recurrences/checks	$C_{\text{op},t}\sum_{a=1}^{K}(I_{t,a} + q^{\text{CG}}_{t,a})$
Parameter update/certificate	$(J^{\text{opt}}_{t+1} + q^{\text{opt}}_{t+1})\{(N_t+1)[C_F + C_B] + O(d)\}$
Path statistics/scoring	$O(d+K)$

This is an operation count, not a wall-clock benchmark. For illustration, $K = 10$, $I = 25$, and $m_t = 256$ give 64,000 sample-CVPs for the recurrences and 66,560 when every solve receives one explicit final residual application. Both numbers exclude candidate gradients, full-history model training, and optimizer certification. Backtracking adds full-history objective passes, and second-order optimization has additional Hessian costs. A linear sufficient-statistic update may substitute a lower special-case cost, but must still supply the theorem's optimizer certificate.

Cost of the analyzed configuration. The full-history instantiation $\mathcal{C}_t = \mathbf{C}_t$, $\mathcal{S}_t = \{1,\dots,t-1\}$ with adaptive CG stopping has the certificate $\bar\kappa_t = 1 + (t-1)G^2/(\lambda\sigma^2)$, so maintaining $\max_a \varepsilon_{t,a} \le \bar\varepsilon_t$ is certified by $I_t = O\big(\sqrt{t}\,G/(\sigma\sqrt{\lambda})\,\log(1/\bar\varepsilon_t)\big)$ iterations per solve (Eq. (151)), so the per-round CG cost is $K\,I_t\,(t-1) = O(K\,t^{3/2})$ sample-CVPs, i.e. $O(K\,T^{5/2})$ cumulatively—the worst-case width-solve sample-CVP count under the stated CG certificate. End-to-end work additionally contains $\sum_{t\le T}(J^{\text{opt}}_t + q^{\text{opt}}_t)N_t$ full-data sample-gradient evaluations. This is $O((J+q)T^2)$ only when those counts are uniformly bounded, and curvature windowing does not reduce it. Table 3 and the numerical example describe fixed-buffer operators ($m_t \ll N_t$), which reduce width-solve work but are covered by Theorem 12 only under the corrected window/subsample conditions of §F.3: an unrescaled current-parameter buffer has the deterministic approximate-to-full factor $\kappa_{+,t}{=}1$ (Proposition 44)—the overall guarantee remaining conditional on the full-to-frozen transfer and the other certificates of Theorem 12—while a rescaled or stale buffer additionally requires the verified transfer factor of Theorem 45.

For the growing window of Theorem 17, one solve costs $O(m_t^{3/2}\log(1/\bar\varepsilon))$ sample-CVPs under the analytic condition-number bound. The following table uses the sufficient fixed-burn-in setting $\tau_w = O(1)$, $H_T = \widetilde{O}(1)$, and $E_T = \widetilde{O}(T^{1-q/2})$; $E_T = o(T)$ alone is insufficient. The sample-CVP column counts only width solves. With uniformly bounded full-batch optimizer/certificate passes, the final column adds the T^2 full-history training term. These are achieved upper bounds for this implementation, not a Pareto frontier, lower bound, or wall-clock measurement (logarithmic factors suppressed):

q	conditional regret	width-solve CVPs	full-policy sample work
$\frac{1}{2}$	$T^{\frac{3}{4}}$	$KT^{\frac{7}{4}}$	T^2
$\frac{2}{3}$	$T^{\frac{2}{3}}$	KT^2	T^2
1	$T^{\frac{1}{2}}$	$KT^{\frac{5}{2}}$	$T^{\frac{5}{2}}$

F.5 An Exact Log-Determinant Bookkeeping Identity

The regret bound is driven not by γ_T but by the realized *width complexity* of the widths the algorithm actually uses,

$$\Lambda_T^{\mathcal{C}} \;:=\; \sum_{t=1}^{T} \log\!\big(1+\sigma^{-2}\, s_{\mathcal{C},t}^2(a_t)\big). \tag{118}$$

Unlike $\bar{\mathbf{C}}_t$, the algorithmic curvature $\mathcal{C}_t$ need *not* be monotone in t: relinearization, subsampling, windowing, and periodic refresh can all *shrink* it, so the elementary determinant telescoping of (70) no longer applies. The following lemma is an exact pathwise bookkeeping identity for a general predictable SPD sequence. It is not a standalone rate result: its upper bound is informative only when the terminal log determinant and determinant-decreasing refresh charge are independently controlled, as in Lemma 35.

Lemma 34 (Dynamic log-determinant potential)**.** *Let $\{\mathcal{C}_t\}_{t\ge 1}$ satisfy (97) with $\mathcal{C}_1=\lambda\mathbf{I}$. For the played feature $\mathbf{h}_t := \mathbf{g}_t(\mathbf{x}_t, a_t)$ define the rank-one update $\mathcal{C}_t^+ := \mathcal{C}_t + \sigma^{-2}\mathbf{h}_t\mathbf{h}_t^\top$ and, using the symmetric (principal) root $(\mathcal{C}_t^+)^{-1/2}$, the normalized perturbation*

$$\Xi_t \;:=\; (\mathcal{C}_t^+)^{-1/2}\,(\mathcal{C}_{t+1}-\mathcal{C}_t^+)\,(\mathcal{C}_t^+)^{-1/2}, \tag{119}$$

so that $\mathbf{I}+\Xi_t = (\mathcal{C}_t^+)^{-1/2}\mathcal{C}_{t+1}(\mathcal{C}_t^+)^{-1/2}$ is SPD; equivalently, every eigenvalue of Ξ_t exceeds -1 and every eigenvalue of $\mathbf{I}+\Xi_t$ is positive (with no smallness condition on the jump $\mathcal{C}_{t+1}-\mathcal{C}_t^+$). Fix any end-of-round-$T$ *terminal operator $\mathcal{C}_{T+1}$ constructed only from data and randomness available through round T ($\mathcal{F}_T$-measurable, hence not a function of $\mathbf{x}_{T+1}$, $\mathcal{A}_{T+1}$, or future rewards; e.g. the canonical rank-one update $\mathcal{C}_{T+1}=\mathcal{C}_T^+$, or the operator the algorithm commits for round $T+1$). The identity below is then an a posteriori statement about the completed sequence $\mathcal{C}_1,\dots,\mathcal{C}_{T+1}$; the quantities Ξ_T, $V_T^{\mathcal{C}}$, and Γ_T^{dyn} are defined relative to this choice, and the bound $\Lambda_T^{\mathcal{C}} \le \Gamma_T^{\mathrm{dyn}}$ holds for* every *such valid choice. Then*

$$\Lambda_T^{\mathcal{C}} \;=\; \log\frac{\det(\mathcal{C}_{T+1})}{\det(\lambda\mathbf{I})} \;-\; \sum_{t=1}^{T}\log\det(\mathbf{I}+\Xi_t). \tag{120}$$

With the determinant-destroying variation charge and the dynamic potential

$$V_T^{\mathcal{C}} := \sum_{t=1}^{T}\big[-\log\det(\mathbf{I}+\Xi_t)\big]_+, \qquad \Gamma_T^{\mathrm{dyn}} := \log\frac{\det(\mathcal{C}_{T+1})}{\det(\lambda\mathbf{I})} + V_T^{\mathcal{C}}, \tag{121}$$

we have $\Lambda_T^{\mathcal{C}} \le \Gamma_T^{\mathrm{dyn}}$, and, using $s_{\mathcal{C},t}^2(a_t)\le G^2/\lambda$ and $\log(1+x)\ge x/(1+x)$,

$$\sum_{t=1}^{T} s_{\mathcal{C},t}^2(a_t) \;\le\; (\sigma^2+G^2/\lambda)\,\Lambda_T^{\mathcal{C}} \;\le\; (\sigma^2+G^2/\lambda)\,\Gamma_T^{\mathrm{dyn}}. \tag{122}$$

Proof. By the matrix determinant lemma applied to $\mathcal{C}_t^+ = \mathcal{C}_t + \sigma^{-2}\mathbf{h}_t\mathbf{h}_t^\top$,

$$\frac{\det(\mathcal{C}_t^+)}{\det(\mathcal{C}_t)} = 1+\sigma^{-2}\mathbf{h}_t^\top\mathcal{C}_t^{-1}\mathbf{h}_t = 1+\sigma^{-2}s_{\mathcal{C},t}^2(a_t). \tag{123}$$

Writing $B_t := (\mathcal{C}_t^+)^{-1/2}$, we have $\mathbf{I}+\Xi_t = B_t\mathcal{C}_t^+B_t + B_t(\mathcal{C}_{t+1}-\mathcal{C}_t^+)B_t = B_t\mathcal{C}_{t+1}B_t$, a congruence of the SPD matrix $\mathcal{C}_{t+1}$ by the invertible symmetric B_t, hence SPD (in particular its eigenvalues exceed -1 with no smallness condition on the jump $\mathcal{C}_{t+1} - \mathcal{C}_t^+$), and

$$\frac{\det(\mathcal{C}_{t+1})}{\det(\mathcal{C}_t^+)} = \det\big(B_t^{-1}(\mathbf{I} + \Xi_t)B_t^{-1}\big) \big/ \det(\mathcal{C}_t^+) = \det(\mathbf{I} + \Xi_t), \tag{124}$$

using $\det(\mathcal{C}_{t+1}) = \det(\mathcal{C}_t^+)\det(\mathbf{I} + \Xi_t)$. Taking logs of (123)–(124) and adding,

$$\log\big(1 + \sigma^{-2} s^2_{\mathcal{C},t}(a_t)\big) = \log\det(\mathcal{C}_{t+1}) - \log\det(\mathcal{C}_t) - \log\det(\mathbf{I} + \Xi_t).$$

Summing over $t = 1, \ldots, T$ telescopes the first difference to $\log\det(\mathcal{C}_{T+1}) - \log\det(\mathcal{C}_1)$; with $\mathcal{C}_1 = \lambda\mathbf{I}$ this is $\log[\det(\mathcal{C}_{T+1})/\det(\lambda\mathbf{I})]$, proving (120). Since $-\log\det(\mathbf{I}+\Xi_t) \le [-\log\det(\mathbf{I}+\Xi_t)]_+$ term by term, $-\sum_t \log\det(\mathbf{I}+\Xi_t) \le V_T^{\mathcal{C}}$, so $\Lambda_T^{\mathcal{C}} \le \Gamma_T^{\mathrm{dyn}}$. For (122): $\mathcal{C}_t \succeq \lambda\mathbf{I}$ gives $s^2_{\mathcal{C},t}(a_t) = \mathbf{h}_t^\top \mathcal{C}_t^{-1}\mathbf{h}_t \le \|\mathbf{h}_t\|^2/\lambda \le G^2/\lambda =: c_{\max}\sigma^2$. With $x_t := \sigma^{-2}s^2_{\mathcal{C},t}(a_t) \in [0, c_{\max}]$ and $\log(1+x) \ge x/(1+x) \ge x/(1+c_{\max})$, $\Lambda_T^{\mathcal{C}} \ge (1+c_{\max})^{-1}\sum_t x_t$, i.e. $\sum_t s^2_{\mathcal{C},t}(a_t) \le (\sigma^2 + G^2/\lambda)\Lambda_T^{\mathcal{C}}$; the second inequality is $\Lambda_T^{\mathcal{C}} \le \Gamma_T^{\mathrm{dyn}}$. □

Remark (Interpretation and scope of the dynamic potential)**.** *Identity* (120) *is pathwise. $V_T^{\mathcal{C}}$ charges only changes that reduce determinant relative to the rank-one update; determinant-increasing changes contribute zero. If $\mathcal{C}_1 \ne \lambda\mathbf{I}$, the denominator becomes $\det(\mathcal{C}_1)$. Neither $V_T^{\mathcal{C}}$ nor the endpoint term is generally small. Bounding both is sufficient, but not necessary because the exact identity may cancel them. Thus the identity does not imply sublinear regret; outside Corollary 27 and Theorem 17, controlling $\Lambda_T^{\mathcal{C}}$ for an adapting backbone remains open.*

Lemma 35 (Rank-sensitive refresh bound)**.** *Let $\mathcal{R} = \{t : \mathcal{C}_{t+1} \ne \mathcal{C}_t^+\}$. Suppose that for every $t \in \mathcal{R}$, the negative part $(\Xi_t)_- = (-\Xi_t)_+$ has rank at most r_t and $\lambda_{\min}(\mathbf{I} + \Xi_t) \ge 1 - \nu_t$ for $0 \le \nu_t < 1$. Then*

$$V_T^{\mathcal{C}} \le \sum_{t\in\mathcal{R}} r_t \log\frac{1}{1-\nu_t}. \tag{125}$$

If additionally $\mathcal{C}_{T+1} = \lambda\mathbf{I} + A$, where $A \succeq 0$, $\mathrm{rank}(A) \le r_{\mathrm{end}}$, and $\mathrm{tr}(A) \le W_{\mathrm{end}}G^2/\sigma^2$ for a specified total endpoint weight $W_{\mathrm{end}} \ge 0$, then, for $r_{\mathrm{end}} > 0$,

$$\Lambda_T^{\mathcal{C}} \le r_{\mathrm{end}} \log\left(1 + \frac{W_{\mathrm{end}}G^2}{r_{\mathrm{end}}\lambda\sigma^2}\right) + \sum_{t\in\mathcal{R}} r_t \log\frac{1}{1-\nu_t}. \tag{126}$$

For $r_{\mathrm{end}} = 0$, the first term is defined as zero.

Proof. Let $\xi_{t,i}$ be the eigenvalues of Ξ_t. At most r_t are negative, each is at least $-\nu_t$, and every nonnegative eigenvalue makes a nonpositive contribution to $-\sum_i \log(1+\xi_{t,i})$. Hence $[-\log\det(\mathbf{I}+\Xi_t)]_+ \le r_t\log(1/(1-\nu_t))$, proving (125). List the nonzero eigenvalues of A and pad with zeros to obtain $a_1, \ldots, a_{r_{\mathrm{end}}}$. Concavity gives

$$\sum_i \log(1 + a_i/\lambda) \le r_{\mathrm{end}} \log\left(1 + \frac{\sum_i a_i}{r_{\mathrm{end}}\lambda}\right),$$

and the trace premise plus Lemma 34 proves (126). □

Consequences. Sequential Gram updates have $\Xi_t = 0$. If canonical rank-one updates are used outside J_T refresh rounds, and each refresh has $r_t \le r$, $\nu_t \le \nu < 1$, then $V_T^{\mathcal{C}} \le J_T r\log(1/(1-\nu))$; a geometric schedule is therefore polylogarithmic only when these rank and spectral-floor bounds hold. For a replacement $\mathcal{C}_{t+1} - \mathcal{C}_t^+ = A_t^{\mathrm{new}} - A_t^{\mathrm{old}}$ with both terms PSD and $\mathrm{rank}(A_t^{\mathrm{old}}) \le r$, Sylvester inertia gives $r_t \le r$, but a spectral floor is still required.

For a fixed-feature window of length m, after warm-up each deletion has rank one. Writing $N_{\mathrm{drop}} = \max\{0, T-m\}$, Sherman–Morrison gives the following exact charge for each $t > m$. Let $v_t^{\mathrm{drop}} = \sigma^{-1}\mathbf{g}_{t-m}(\mathbf{x}_{t-m}, a_{t-m})$ and let $D_t = \mathcal{C}_t^+ - v_t^{\mathrm{drop}}(v_t^{\mathrm{drop}})^\top = \mathcal{C}_{t+1}$ be the remaining-window matrix. Then $-\log\det(\mathbf{I}+\Xi_t) = \log[1 + (v_t^{\mathrm{drop}})^\top D_t^{-1} v_t^{\mathrm{drop}}]$, and therefore

$$V_T^{\mathcal{C}} \le N_{\mathrm{drop}} \log\left(1 + \frac{G^2}{\lambda\sigma^2}\right).$$

This can be linear in T. A relinearized window need not be a rank-one replacement. Likewise, merely holding a stale operator fixed does not make non-refresh transitions canonical rank-one updates, so geometric refresh timing alone does not imply a logarithmic charge.

Corollary 36 (Observable CG upper bound on the width complexity)**.** *If the played-action CG solve satisfies $\varepsilon_{t,a_t} \le \bar{\varepsilon}_t < 1$ so that $\tilde{s}_t^2(a_t) \ge (1-\bar{\varepsilon}_t)\, s^2_{\mathcal{C},t}(a_t)$ (Lemma 37), then*

$$\Lambda_T^{\mathcal{C}} \;\le\; \widehat{\Lambda^{\mathcal{C}}}_T \;:=\; \sum_{t=1}^{T} \log\!\Big(1 + \frac{\tilde{s}_t^2(a_t)}{\sigma^2\,(1-\bar{\varepsilon}_t)}\Big). \tag{127}$$

Proof. $s^2_{\mathcal{C},t}(a_t) \le \tilde{s}_t^2(a_t)/(1-\bar{\varepsilon}_t)$ and $x \mapsto \log(1+\sigma^{-2}x)$ is increasing, so each summand of $\Lambda_T^{\mathcal{C}}$ is dominated term by term. $\square$

This gives an observable upper bound on the dynamic width-complexity term, conditional on the certified CG tolerances; $V_T^{\mathcal{C}}$ itself is *not* observable in general (it requires determinant estimation of the curvature sequence).

F.6 CG Approximation Lemma

Lemma 37 (CG multiplicative accuracy)**.** *Fix a round t and an action a with right-hand side $\mathbf{g} := \mathbf{g}_t(a) \ne \mathbf{0}$. Let $\mathbf{u} = \mathcal{C}_t^{-1}\mathbf{g}$ and let $\tilde{\mathbf{u}}$ be the iterate after $I_{t,a}$ steps of CG on the SPD system $\mathcal{C}_t\mathbf{u} = \mathbf{g}$ starting from $\tilde{\mathbf{u}}_0 = \mathbf{0}$, with $\mathcal{C}_t$ held* fixed *across all iterations. The relative energy-norm error is the per-action quantity $\varepsilon_{t,a}$ of* (48)*, and*

$$\varepsilon_{t,a} \;=\; \frac{\|\mathbf{u} - \tilde{\mathbf{u}}\|_{\mathcal{C}_t}}{\|\mathbf{u}\|_{\mathcal{C}_t}} \;\le\; 2\bigg(\frac{\sqrt{\bar{\kappa}_t} - 1}{\sqrt{\bar{\kappa}_t} + 1}\bigg)^{I_{t,a}}, \tag{128}$$

where $\kappa_t := \lambda_{\max}(\mathcal{C}_t)/\lambda_{\min}(\mathcal{C}_t)$ is the true condition number and $\bar{\kappa}_t \ge \kappa_t$ is a certified $\mathcal{H}_t$-measurable upper bound. The convergence rate in (128) *is right-hand-side independent, so it holds for every action with the same $\bar{\kappa}_t$. For a* nonnegative weighted outer-product *operator $\mathcal{C}_t = \lambda\mathbf{I} + \sigma^{-2}\sum_{s\in\mathcal{S}_t} w_s\,\mathbf{g}_t(\mathbf{x}_s, a_s)\mathbf{g}_t(\mathbf{x}_s, a_s)^\top$ ($w_s \ge 0$, so $\mathcal{C}_t \succeq \lambda\mathbf{I} \succ 0$ and CG is well-posed),*

$$\kappa_t \le \frac{\lambda_{\max}(\mathcal{C}_t)}{\lambda} \le 1 + \frac{W_t G^2}{\lambda\sigma^2} =: \bar{\kappa}_t,$$

with $W_t = \sum_{s\in\mathcal{S}_t} w_s$ and $\|\mathbf{g}_t(\mathbf{x}_s, a_s)\| \le G$ from Assumption 3; here $W_t = n$ for unrescaled windowed/refresh buffers, giving $\bar{\kappa}_t$ and α_t independent of T, and $W_t = t-1$ for full history and for rescaled uniform subsampling with $w_s = (t-1)/n$. For a general SPD operator that is not such an outer product, $\bar{\kappa}_t$ must be certified separately. Write $\tilde{s}_t^2(a) = \mathbf{g}^\top\tilde{\mathbf{u}}$ and $s^2_{\mathcal{C},t}(a) = \mathbf{g}^\top\mathcal{C}_t^{-1}\mathbf{g}$.

Assume *$\varepsilon_{t,a} < 1$ (see Appendix I for the minimum $I_{t,a}$ ensuring this as a function of $\bar{\kappa}_t$; Algorithm 2 enforces a known* per-round *tolerance $\bar{\varepsilon}_t$ uniformly over the scored actions,* $\max_a \varepsilon_{t,a} \le \bar{\varepsilon}_t < 1$*, via adaptive residual stopping or a fixed budget under a certified condition-number bound; see §C). Then*

$$(1-\varepsilon_{t,a})\, s^2_{\mathcal{C},t}(a) \;\le\; \tilde{s}_t^2(a) \;\le\; (1+\varepsilon_{t,a})\, s^2_{\mathcal{C},t}(a). \tag{129}$$

For $\mathbf{g}_t(a) = \mathbf{0}$ the algorithm sets $\tilde{s}_t^2(a) = 0 = s^2_{\mathcal{C},t}(a)$ and (129) *holds trivially.*

Proof. The CG minimax property gives $\|e_I\|_{\mathcal{C}_t}/\|e_0\|_{\mathcal{C}_t} \le \min_{p\in\mathcal{P}_I,\,p(0)=1} \max_{x\in[\lambda_{\min},\lambda_{\max}]} |p(x)|$. Choosing the scaled Chebyshev polynomial bounds this maximum by $2[(\sqrt{\kappa_t}-1)/(\sqrt{\kappa_t}+1)]^I$; this ratio is increasing in κ_t, so $\bar{\kappa}_t \ge \kappa_t$ proves (128). $\tilde{s}_t^2(a) - s^2_{\mathcal{C},t}(a) = \mathbf{g}^\top(\tilde{\mathbf{u}} - \mathbf{u}) = (\mathcal{C}_t\mathbf{u})^\top(\tilde{\mathbf{u}} - \mathbf{u}) = \mathbf{u}^\top\mathcal{C}_t(\tilde{\mathbf{u}} - \mathbf{u})$. By Cauchy–Schwarz in the $\mathcal{C}_t$-norm, $|\mathbf{u}^\top\mathcal{C}_t(\tilde{\mathbf{u}} - \mathbf{u})| \le \|\mathbf{u}\|_{\mathcal{C}_t}\|\tilde{\mathbf{u}} - \mathbf{u}\|_{\mathcal{C}_t} \le \varepsilon_{t,a}\|\mathbf{u}\|^2_{\mathcal{C}_t} = \varepsilon_{t,a}\, s^2_{\mathcal{C},t}(a)$. For the outer-product special case the condition-number bound follows from $\lambda_{\max}(\mathcal{C}_t) \le \lambda + \sigma^{-2}\sum_{s\in\mathcal{S}_t} w_s\|\mathbf{g}_t(\mathbf{x}_s, a_s)\|^2 \le \lambda + W_t G^2/\sigma^2$. $\square$

Lemma 38 (Fixed-preconditioner CG accuracy)**.** *Fix round t and let $P_t \in \mathbb{R}^{d\times d}$ be an $\mathcal{H}_t$-measurable SPD preconditioner, held fixed and applied as the same exact linear map throughout every iteration of a solve. For $\mathbf{g} = \mathbf{g}_t(a) \ne 0$, define the symmetric transformed system*

$$\widehat{C}_t\mathbf{y} = \widehat{\mathbf{g}}, \qquad \widehat{C}_t = P_t^{-1/2}\mathcal{C}_t P_t^{-1/2}, \qquad \widehat{\mathbf{g}} = P_t^{-1/2}\mathbf{g}, \tag{130}$$

using principal square roots. If CG starts from zero on this system and $\tilde{\mathbf{y}}_I$ is its Ith iterate for $I \ge 1$, set $\mathbf{u} = \mathcal{C}_t^{-1}\mathbf{g}$, $\mathbf{y} = P_t^{1/2}\mathbf{u}$, and $\tilde{\mathbf{u}}_I = P_t^{-1/2}\tilde{\mathbf{y}}_I$. Then

$$\frac{\|\mathbf{u} - \tilde{\mathbf{u}}_I\|_{\mathcal{C}_t}}{\|\mathbf{u}\|_{\mathcal{C}_t}} = \frac{\|\mathbf{y} - \tilde{\mathbf{y}}_I\|_{\widehat{C}_t}}{\|\mathbf{y}\|_{\widehat{C}_t}} \le 2\left(\frac{\sqrt{\bar{\kappa}_t^P} - 1}{\sqrt{\bar{\kappa}_t^P} + 1}\right)^I, \tag{131}$$

where $\bar{\kappa}_t^P \geq \kappa(\widehat{C}_t)$ is a certified predictable bound. In particular, positive $\mathcal{H}_t$-measurable certified quantities ℓ_t, L_t satisfying the spectral equivalence $\ell_t P_t \preceq \mathcal{C}_t \preceq L_t P_t$ with $0 < \ell_t \leq L_t$ give $\bar{\kappa}_t^P = L_t/\ell_t$.

Writing the left side of (131) as $\varepsilon_{t,a}^P$, the approximate width $\tilde{s}_{P,t}^2(a) = \mathbf{g}^\top \tilde{\mathbf{u}}_I$ satisfies, whenever $\varepsilon_{t,a}^P < 1$,

$$(1 - \varepsilon_{t,a}^P) s_{\mathcal{C},t}^2(a) \leq \tilde{s}_{P,t}^2(a) \leq (1 + \varepsilon_{t,a}^P) s_{\mathcal{C},t}^2(a). \tag{132}$$

There is also a directly recomputable residual certificate. For $\mathbf{r}_I = \mathbf{g} - \mathcal{C}_t \tilde{\mathbf{u}}_I$,

$$\varepsilon_{t,a}^P \leq \sqrt{\bar{\kappa}_t^P} \left(\frac{\mathbf{r}_I^\top P_t^{-1} \mathbf{r}_I}{\mathbf{g}^\top P_t^{-1} \mathbf{g}} \right)^{1/2}. \tag{133}$$

Thus stopping only after the preconditioned residual ratio is at most $\bar{\varepsilon}_t / \sqrt{\bar{\kappa}_t^P}$ certifies $\varepsilon_{t,a}^P \leq \bar{\varepsilon}_t$. For $\mathbf{g} = 0$, both widths are defined as zero. Taking $P_t = I$ recovers ordinary CG. The lemma makes no claim that a particular preconditioner improves the condition number; that requires a proved or measured spectral-equivalence bound. Iteration-varying or inexact nonlinear preconditioners are outside this fixed transformed-CG argument.

Proof. The substitutions $\mathbf{u} = P_t^{-1/2} \mathbf{y}$ and $\tilde{\mathbf{u}}_I = P_t^{-1/2} \tilde{\mathbf{y}}_I$ give

$$\|\mathbf{u} - \tilde{\mathbf{u}}_I\|_{\mathcal{C}_t}^2 = \|\mathbf{y} - \tilde{\mathbf{y}}_I\|_{\widehat{C}_t}^2, \qquad \|\mathbf{u}\|_{\mathcal{C}_t}^2 = \|\mathbf{y}\|_{\widehat{C}_t}^2.$$

The standard zero-start CG Chebyshev bound on the SPD matrix $\widehat{C}_t$ proves (131). Congruence of $\ell_t P_t \preceq \mathcal{C}_t \preceq L_t P_t$ by $P_t^{-1/2}$ puts the spectrum of $\widehat{C}_t$ in $[\ell_t, L_t]$.

Also $\mathbf{g}^\top (\tilde{\mathbf{u}}_I - \mathbf{u}) = \mathbf{u}^\top \mathcal{C}_t (\tilde{\mathbf{u}}_I - \mathbf{u})$, so Cauchy–Schwarz in the $\mathcal{C}_t$ energy norm proves (132), exactly as in Lemma 37. Finally the transformed residual is $\widehat{\mathbf{r}}_I = P_t^{-1/2} \mathbf{r}_I$. If $m_t = \lambda_{\min}(\widehat{C}_t)$ and $M_t^P = \lambda_{\max}(\widehat{C}_t)$, then

$$\|\mathbf{y} - \tilde{\mathbf{y}}_I\|_{\widehat{C}_t} \leq \frac{\|\widehat{\mathbf{r}}_I\|_2}{\sqrt{m_t}}, \qquad \|\mathbf{y}\|_{\widehat{C}_t} \geq \frac{\|\widehat{\mathbf{g}}\|_2}{\sqrt{M_t^P}}.$$

Their ratio is at most $\sqrt{\kappa(\widehat{C}_t)} \|\widehat{\mathbf{r}}_I\|_2 / \|\widehat{\mathbf{g}}\|_2$, which yields (133) after using the certified upper bound. □

The algorithmic bonus. With the *per-round* certified tolerance $\bar{\varepsilon}_t \geq \max_a \varepsilon_{t,a}$ and the one-sided transfer factor $u_t(a) \geq 1$ of Assumption 6, Algorithm 2 plays the bonus

$$w_t(a) \;:=\; \omega_t \sqrt{\frac{u_t(a)}{1 - \bar{\varepsilon}_t}} \sqrt{\tilde{s}_t^2(a)}, \tag{134}$$

where $\bar{\beta}_t \geq \beta_t$ is the predictable confidence upper bound, $\bar{\psi}_t$ the centering certificate (Assumption 5), and

$$\omega_t := \bar{\beta}_t + \bar{\psi}_t \tag{135}$$

is the combined confidence-plus-centering width factor. Define the *per-round* CG inflation factor $\alpha_t := \sqrt{(1 + \bar{\varepsilon}_t)/(1 - \bar{\varepsilon}_t)}$.

Lemma 39 (Bonus sandwiching). *Under Assumption 6 and $\max_{a \in \mathcal{A}_t} \varepsilon_{t,a} \leq \bar{\varepsilon}_t < 1$, the bonus (134) satisfies, for every $a \in \mathcal{A}_t$,*

$$\omega_t \, \bar{s}_t(a) \;\leq\; w_t(a) \;\leq\; \alpha_t \, \omega_t \, \sqrt{u_t(a)} \, s_{\mathcal{C},t}(a). \tag{136}$$

Proof. For $\mathbf{g}_t(a) = \mathbf{0}$ all three terms vanish and the bound is trivial. Otherwise Lemma 37 with $\varepsilon_{t,a} \leq \bar{\varepsilon}_t$ gives $(1 - \bar{\varepsilon}_t) s_{\mathcal{C},t}^2(a) \leq \tilde{s}_t^2(a) \leq (1 + \bar{\varepsilon}_t) s_{\mathcal{C},t}^2(a)$. The lower bound and $\omega_t \geq 0$ give $\sqrt{\tilde{s}_t^2(a)} \geq \sqrt{1 - \bar{\varepsilon}_t}\, s_{\mathcal{C},t}(a)$, so $w_t(a) \geq \omega_t \sqrt{u_t(a)}\, s_{\mathcal{C},t}(a)$; then (99) gives $\sqrt{u_t(a)}\, s_{\mathcal{C},t}(a) \geq \bar{s}_t(a)$, i.e. $w_t(a) \geq \omega_t \bar{s}_t(a)$. The upper bound gives $\sqrt{\tilde{s}_t^2(a)} \leq \sqrt{1 + \bar{\varepsilon}_t}\, s_{\mathcal{C},t}(a)$, whence $w_t(a) \leq \omega_t \sqrt{u_t(a)/(1 - \bar{\varepsilon}_t)} \sqrt{1 + \bar{\varepsilon}_t}\, s_{\mathcal{C},t}(a) = \alpha_t \omega_t \sqrt{u_t(a)}\, s_{\mathcal{C},t}(a)$. Both bounds are per-round; no horizon maximum is taken before summing. □

F.7 Confidence and Discrepancy Lemmas

Lemma 40 (Confidence for the linearized ridge estimator). *Suppose Assumptions 2 to 4 hold and $\mu^* = \mu_{\boldsymbol{\theta}^*}$ with $\|\boldsymbol{\theta}^*\| \le S$. Define the linearized ridge estimator $\hat{\boldsymbol{\theta}}_t^{\text{lin}}$ by (146), the frozen-feature variance $\bar{s}_t(a) = \sqrt{\mathbf{g}_t(a)^\top \bar{\mathbf{C}}_t^{-1} \mathbf{g}_t(a)}$, and the cumulative squared linearization error $F_t := \sum_{s=1}^{t-1} \varepsilon_{\text{lin}}(s)^2$. Then with probability at least $1-\delta$, for all $t \ge 1$ and all $a \in \mathcal{A}_t$ simultaneously,*

$$\left|\mathbf{g}_t(a)^\top \big(\hat{\boldsymbol{\theta}}_t^{\text{lin}} - \boldsymbol{\theta}^*\big)\right| \;\le\; \bar{s}_t(a)\,\beta_t, \tag{137}$$

where

$$\beta_t \;:=\; \sqrt{\gamma_{t-1} + 2\log(1/\delta)} \;+\; \sqrt{\lambda}\,S \;+\; \frac{1}{\sigma}\sqrt{F_t}. \tag{138}$$

(Here $\gamma_{t-1} = \log\det(\bar{\mathbf{C}}_t)/\det(\lambda \mathbf{I})$ is the information gain through round $t-1$.)

The quantity β_t in (138) is a theoretical radius. The algorithm may use any predictable upper bound $\bar{\beta}_t \ge \beta_t$; all results below remain valid with $\bar{\beta}_t$ in place of β_t. This is necessary because β_t depends on the unknown cumulative linearization term F_t.

Proof. From (145), $y_s = \mathbf{g}_s^\top \boldsymbol{\theta}^* + \rho_s + \eta_s$ with $|\rho_s| \le \varepsilon_{\text{lin}}(s)$. The closed form of $\hat{\boldsymbol{\theta}}_t^{\text{lin}}$ gives

$$\hat{\boldsymbol{\theta}}_t^{\text{lin}} - \boldsymbol{\theta}^* = -\lambda \bar{\mathbf{C}}_t^{-1}\boldsymbol{\theta}^* + \bar{\mathbf{C}}_t^{-1}\sigma^{-2}\sum_{s<t}\mathbf{g}_s\rho_s + \bar{\mathbf{C}}_t^{-1}\sigma^{-2}\sum_{s<t}\mathbf{g}_s\eta_s.$$

Taking the inner product with $\mathbf{g}_t(a)$ and bounding in absolute value:

(i) *Prior bias.* $|\lambda \mathbf{g}_t(a)^\top \bar{\mathbf{C}}_t^{-1}\boldsymbol{\theta}^*| \le \lambda\,\bar{s}_t(a)\,\|\boldsymbol{\theta}^*\|_{\bar{\mathbf{C}}_t^{-1}} \le \bar{s}_t(a)\sqrt{\lambda}\,S$. (Cauchy–Schwarz in the $\bar{\mathbf{C}}_t$-norm; $\lambda\|\boldsymbol{\theta}^*\|_{\bar{\mathbf{C}}_t^{-1}} \le \lambda\|\boldsymbol{\theta}^*\|/\sqrt{\lambda_{\min}(\bar{\mathbf{C}}_t)} \le \sqrt{\lambda}\,S$ since $\bar{\mathbf{C}}_t \succeq \lambda\mathbf{I}$.)

(ii) *Linearization bias.* Define $\Phi = [\mathbf{g}_1/\sigma, \ldots, \mathbf{g}_{t-1}/\sigma]$ (the $d \times (t-1)$ design matrix) and $\hat{\rho} = (\rho_1, \ldots, \rho_{t-1})^\top$. Then $\sigma^{-2}\sum \mathbf{g}_s\rho_s = \sigma^{-1}\Phi\hat{\rho}$. Since $\bar{\mathbf{C}}_t \succeq \Phi\Phi^\top$, we have $\Phi^\top\bar{\mathbf{C}}_t^{-1}\Phi \preceq \mathbf{I}$, so $\|\Phi\hat{\rho}\|_{\bar{\mathbf{C}}_t^{-1}} = \|\hat{\rho}\|_{\Phi^\top\bar{\mathbf{C}}_t^{-1}\Phi} \le \|\hat{\rho}\| = \sqrt{F_t}$. Therefore $|\mathbf{g}_t(a)^\top\bar{\mathbf{C}}_t^{-1}\sigma^{-1}\Phi\hat{\rho}| \le \bar{s}_t(a)\sigma^{-1}\sqrt{F_t}$.

(iii) *Noise.*

$$\left\|\sum_{s<t}\frac{\mathbf{g}_s}{\sigma}\frac{\eta_s}{\sigma}\right\|_{\bar{\mathbf{C}}_t^{-1}} \le \sqrt{\gamma_{t-1} + 2\log(1/\delta)}$$

holds for all $t \ge 1$ with probability at least $1-\delta$ by the time-uniform vector self-normalized martingale bound (Abbasi-Yadkori et al., 2011). Here $x_s = \mathbf{g}_s/\sigma$ is predictable with respect to the pre-reward sigma-algebra $\mathcal{G}_s$, and $\xi_s = \eta_s/\sigma$ is conditionally 1-sub-Gaussian given $\mathcal{G}_s$. For any candidate action a, Cauchy–Schwarz in the $\bar{\mathbf{C}}_t$-norm gives

$$\left|\mathbf{g}_t(a)^\top\bar{\mathbf{C}}_t^{-1}\sum_{s<t}\frac{\mathbf{g}_s\eta_s}{\sigma^2}\right| \le \bar{s}_t(a)\sqrt{\gamma_{t-1} + 2\log(1/\delta)}.$$

Summing gives (137)–(138). $\square$

Corollary 41 (Predictable confidence schedule). *Suppose a nonnegative $\mathcal{H}_t$-measurable sequence $\{\bar{F}_t\}_{t\ge1}$ satisfies $F_t \le \bar{F}_t$ for all t. Then Algorithm 2 may set*

$$\bar{\beta}_t \;:=\; \sqrt{\gamma_{t-1} + 2\log(1/\delta)} \;+\; \sqrt{\lambda}\,S \;+\; \sigma^{-1}\sqrt{\bar{F}_t},$$

which is predictable and satisfies $\bar{\beta}_t \ge \beta_t$.

Proof. $\bar{F}_t \ge F_t$ term-by-term implies each summand in (138) is dominated, so $\bar{\beta}_t \ge \beta_t$. Predictability holds because γ_{t-1}, S, $\bar{F}_t$, λ, σ, and δ are all determined before round t. $\square$

Lemma 24 supplies the computable choice $\bar{F}_t = \sum_{s<t}(\bar{\epsilon}_s^{\text{lin}})^2$; a tighter externally certified predictable sequence may replace it.

Corollary 42 (Observable information-gain surrogate via the one-sided transfer factor)**.** *Run Algorithm 2 with a predictable algorithmic curvature sequence $\mathcal{C}_s$, per-round CG tolerance with $\varepsilon_{s,a_s} \le \bar{\varepsilon}_s < 1$ for the played action, and an $\mathcal{H}_s$-measurable one-sided optimism-transfer factor $u_s(a) \ge 1$ satisfying $\bar{s}_s^2(a) \le u_s(a)\, s_{\mathcal{C},s}^2(a)$ (Assumption 6). Suppose also that the nonnegative predictable sequence $\bar{F}_t$ satisfies $\bar{F}_t \ge F_t$. Define the observable information-gain surrogate*

$$\hat{\gamma}_{t-1} \;:=\; \sum_{s=1}^{t-1} \log\Bigl(1 + \sigma^{-2}\, \frac{u_s(a_s)}{1-\bar{\varepsilon}_s}\, \tilde{s}_s^2(a_s)\Bigr),$$

where $\tilde{s}_s^2(a_s) = \mathbf{g}_s(a_s)^\top \tilde{\mathbf{u}}_s$ is the CG variance proxy computed at round s for the played action. Then $\hat{\gamma}_{t-1}$ is $\mathcal{H}_t$-measurable and $\hat{\gamma}_{t-1} \ge \gamma_{t-1}$, so

$$\hat{\beta}_t \;:=\; \sqrt{\hat{\gamma}_{t-1} + 2\log(1/\delta)} \;+\; \sqrt{\lambda}\, S \;+\; \sigma^{-1}\sqrt{\bar{F}_t}$$

is a predictable schedule with $\hat{\beta}_t \ge \beta_t$. Played-action CG accuracy is sufficient for this information comparison. If the same tolerance also holds uniformly for every candidate action before selection, then Theorem 12 holds with $\bar{\beta}_t = \hat{\beta}_t$.

Proof. The transfer factor gives $\bar{s}_s^2(a_s) \le u_s(a_s)\, s_{\mathcal{C},s}^2(a_s)$, and Lemma 37 with $\varepsilon_{s,a_s} \le \bar{\varepsilon}_s$ gives $s_{\mathcal{C},s}^2(a_s) \le \tilde{s}_s^2(a_s)/(1-\bar{\varepsilon}_s)$; hence $\bar{s}_s^2(a_s) \le u_s(a_s)\, \tilde{s}_s^2(a_s)/(1-\bar{\varepsilon}_s)$. By monotonicity of $x \mapsto \log(1+\sigma^{-2}x)$, each summand of γ_{t-1} in (70) is dominated by the corresponding summand of $\hat{\gamma}_{t-1}$. Measurability holds because $\tilde{s}_s^2(a_s)$, $u_s(a_s)$, and $\bar{\varepsilon}_s$ are all $\mathcal{H}_s$-measurable. The schedule property then follows term by term as in Corollary 41. ☐

This makes the information-gain component observable when $u_s(a_s)$ is certified. Together with Lemma 24, it renders the full confidence and centering schedules operational; it does not make them tight.

Remark (Computability and dimension dependence of β_t)**.** *Two caveats about β_t deserve emphasis. First, γ_{t-1} is defined through the frozen-feature Gram matrix $\bar{\mathbf{C}}_t$, whose per-sample Jacobians the matrix-free implementation deliberately does not store; γ_{t-1} is therefore not directly computable in the stated memory model. Second, the worst-case substitute $\gamma_{t-1} \le d\log\bigl(1+(t-1)G^2/(d\lambda\sigma^2)\bigr)$ does not exploit low realized information gain and can be order t when $d \gg t$, rendering the resulting bonus and theorem bound vacuous. Corollary 42 addresses both issues with an observable surrogate built from quantities the algorithm already computes; the resulting guarantee is data-dependent, and its tightness still depends on the realized surrogate together with the other theorem factors.*

Lemma 43 (Action-scaled centering discrepancy)**.** *Under Assumption 5, $|\mathbf{g}_t(a)^\top(\boldsymbol{\theta}_t - \hat{\boldsymbol{\theta}}_t^{\mathrm{lin}})| \le \bar{\psi}_t\, \bar{s}_t(a)$ for all $a \in \mathcal{A}_t$.*

Proof. Immediate from (79). ☐

> *Remark (status of the centering certificate $\bar{\psi}_t$).* Lemma 24 computes a predictable $\bar{\psi}_t \ge \psi_t$ from the certified optimizer residual, Welford path moments, observed collection residuals, and analytic smoothness constants. The schedule is operational but can be loose; the corrected-center variant (Corollary 21) removes it at the cost of frozen-feature access.

In the convex last-layer regime, under exact optimization, $\boldsymbol{\theta}_t = \hat{\boldsymbol{\theta}}_t^{\mathrm{lin}}$ and $\bar{\psi}_t = 0$. The full prediction error at $(\mathbf{x}_t, a)$ is bounded by the triangle inequality, with the centering discrepancy *action-scaled*:

$$|\mu_{\boldsymbol{\theta}^*}(\mathbf{x}_t, a) - \mu_{\boldsymbol{\theta}_t}(\mathbf{x}_t, a)| \;\le\; (\beta_t + \bar{\psi}_t)\, \bar{s}_t(a) \;+\; \varepsilon_{\mathrm{lin}}(t) \;\le\; \omega_t\, \bar{s}_t(a) + \varepsilon_{\mathrm{lin}}(t), \tag{139}$$

using (137), Lemma 43, the per-round linearization remainder $|\rho_t| \le \varepsilon_{\mathrm{lin}}(t)$, and $\bar{\beta}_t \ge \beta_t$.

F.8 Cost of Approximate Curvature

Proposition 44 (Unrescaled current-parameter subset)**.** *Let $\mathcal{S}_t \subseteq \{1,\ldots,t-1\}$ and $0 \le w_s \le 1$, and define the current-parameter operator*

$$\hat{\mathbf{C}}_t \;=\; \lambda\mathbf{I} + \sigma^{-2}\sum_{s\in\mathcal{S}_t} w_s\, \mathbf{g}_t(\mathbf{x}_s, a_s)\mathbf{g}_t(\mathbf{x}_s, a_s)^\top,$$

using the same relinearized Jacobians $\mathbf{g}_t(\mathbf{x}_s, a_s) = \nabla_{\boldsymbol{\theta}}\mu_{\boldsymbol{\theta}_t}(\mathbf{x}_s, a_s)$ that define $\mathbf{C}_t$ (44). Then $\hat{\mathbf{C}}_t \preceq \mathbf{C}_t$, so Theorem 45 applies with $\kappa_{+,t} = 1$ and, in the exact-transfer case, $u_t = (1+\bar{\chi}_t)^2$ (Lemma 30).

Proof. $\mathbf{C}_t - \hat{\mathbf{C}}_t = \sigma^{-2}\sum_{s\in\mathcal{S}_t}(1-w_s)\mathbf{g}_t\mathbf{g}_t^\top + \sigma^{-2}\sum_{s\notin\mathcal{S}_t}\mathbf{g}_t\mathbf{g}_t^\top$, a sum of PSD outer products (each $1-w_s \geq 0$), hence $\mathbf{C}_t - \hat{\mathbf{C}}_t \succeq 0$. With $\hat{\mathbf{C}}_t \preceq \mathbf{C}_t$ (so $\kappa_{+,t} = 1$ in (140)) and $\hat{\mathbf{C}}_t \succeq \lambda\mathbf{I}$, the hypotheses of Theorem 45 hold. □

This covers an unrescaled current-parameter sliding window, an unrescaled current-parameter subsample, and any current-parameter downweighted subset with weights in $[0, 1]$. It does *not* cover rescaled subsampling with weights $(t-1)/n > 1$ (Lemma 33), a stale periodic-refresh operator evaluated at an older parameter, or an arbitrary randomized sketch (e.g. an eigenspace Lanczos surrogate), for which a factor $\kappa_{+,t}$ must be certified separately.

Theorem 45 (Approximate curvature via one-sided transfer)**.** *Adopt the hypotheses of Theorem 12 with the algorithmic curvature $\mathcal{C}_t = \hat{\mathbf{C}}_t$, an approximate operator that is symmetric, $\mathcal{H}_t$-measurable, and satisfies $\hat{\mathbf{C}}_t \succeq \lambda\mathbf{I}$. Suppose the two* one-sided *bounds*

$$\hat{\mathbf{C}}_t \;\preceq\; \kappa_{+,t}\,\mathbf{C}_t, \qquad \mathbf{C}_t \;\preceq\; (1+\bar{\chi}_t)^2\,\bar{\mathbf{C}}_t \tag{140}$$

hold for every $t \leq T$, where $\kappa_{+,t} \geq 1$ is $\mathcal{H}_t$-measurable and $\bar{\chi}_t \geq \chi_t$ is the whitened-drift certificate of Lemma 30. Then, for every action a,

$$\bar{s}_t^2(a) \;\leq\; \kappa_{+,t}\,(1+\bar{\chi}_t)^2\,\hat{s}_t^2(a), \tag{141}$$

so Assumption 6 holds with the uniform factor $u_t = \kappa_{+,t}(1+\bar{\chi}_t)^2$ and Theorem 12 applies verbatim with $\mathcal{C}_t = \hat{\mathbf{C}}_t$ and $s_{\mathcal{C},t} = \hat{s}_t$:

$$R_T \;\leq\; 2\sqrt{(\sigma^2 + G^2/\lambda)\,\Lambda_T^{\mathcal{C}}\sum_{t=1}^{T}\alpha_t^2\,u_t(a_t)\,\omega_t^2} \;+\; 2E_T.$$

No lower *spectral factor on $\hat{\mathbf{C}}_t$ is required for optimism; the cost of an excessively small, stale, or nonmonotone approximate operator is charged through its realized width complexity $\Lambda_T^{\mathcal{C}}$ (and its variation $V_T^{\mathcal{C}}$ inside Γ_T^{dyn}), not through a lower Loewner bound.*

Proof. $\hat{\mathbf{C}}_t \preceq \kappa_{+,t}\mathbf{C}_t$ with both sides SPD inverts to $\hat{\mathbf{C}}_t^{-1} \succeq \kappa_{+,t}^{-1}\mathbf{C}_t^{-1}$, so $\hat{s}_t^2(a) \geq \kappa_{+,t}^{-1}s_t^2(a)$; and (103) gives $\bar{s}_t^2(a) \leq (1+\bar{\chi}_t)^2 s_t^2(a)$. Chaining, $\bar{s}_t^2(a) \leq (1+\bar{\chi}_t)^2\kappa_{+,t}\hat{s}_t^2(a)$, which is (141) with $s_{\mathcal{C},t} = \hat{s}_t$. Theorem 12 applies with $\mathcal{C}_t = \hat{\mathbf{C}}_t$. The two-sided $\sqrt{\rho_+/\rho_-}$ inflation—which instead assumes a lower spectral factor—is retained for comparison in Appendix J; the two results trade different assumptions (a lower spectral bound versus dynamic width complexity), and neither numerically dominates the other in general. □

Proposition 46 (Roundwise scalar-width invariance)**.** *If two width functions satisfy $\hat{s}_t(a) = c_t s_t(a)$ for every candidate action, where $c_t > 0$ is action independent, then replacing a scalar bonus coefficient b_t by b_t/c_t gives identical UCB scores and therefore the same selected action under the same tie rule.*

Proof. $\widehat{\mu}_t(a) + (b_t/c_t)\hat{s}_t(a) = \widehat{\mu}_t(a) + b_t s_t(a)$ for every a. □

This is a roundwise statement. If two policies also have identical centers, histories, optimization, and coupled exogenous randomness, if c_t is predictable, and if proportionality holds at every subsequent common history, induction gives identical actions and rewards after the coefficient adjustment. Absent those conditions the proposition makes no regret claim. Accordingly, regret differences can reflect anisotropic width distortion or a difference in damping, center, adaptive path, optimization, calibration, or solve accuracy.

Proposition 47 (Off-diagonal linear-Gram witness)**.** *Let $\sigma = 1$, $\mathbf{u} = (1,1)/\sqrt{2}$, $\mathbf{v} = (1,-1)/\sqrt{2}$, the action set be $\{\mathbf{u}, \mathbf{v}\}$, and $\boldsymbol{\theta}^* = \epsilon\mathbf{u} + \Delta\mathbf{v}$ with $\Delta > \epsilon > 0$. Rewards are noiseless, the ridge is $\lambda\mathbf{I}$, all policies use the same full-ridge mean center, and ties select $\mathbf{u}$. If*

$$\max\left\{\sqrt{\lambda(\epsilon^2+\Delta^2)}, \epsilon\sqrt{\lambda}\left(1+\sqrt{\frac{\lambda}{\lambda+1}}\right)\right\} < b \leq \left(\Delta - \frac{\epsilon}{\lambda+1}\right)\sqrt{\lambda+1}, \tag{142}$$

then full-Gram UCB plays $\mathbf{u}$ once and $\mathbf{v}$ thereafter, so $R_T^{\mathrm{full}} = \Delta - \epsilon$. The raw diagonal policy and the policy inflated by the uniform analytic factor $\kappa_n = (\lambda + n/2)/\lambda$ play $\mathbf{u}$ forever and have $R_T^{\mathrm{diag}} = T(\Delta - \epsilon)$.

Proof. After n pulls of $\mathbf{u}$ and no pull of $\mathbf{v}$,

$$C_n = \lambda \mathbf{I} + n\mathbf{u}\mathbf{u}^\top, \qquad \hat{\boldsymbol{\theta}}_n = \frac{n\epsilon}{\lambda + n}\mathbf{u}.$$

Thus the full scores are $n\epsilon/(\lambda+n) + b/\sqrt{\lambda+n}$ and $b/\sqrt{\lambda}$. The strict lower inequality in (142) makes the second score larger at $n = 1$. The confidence lower bound also implies $b > \Delta\sqrt{\lambda}$; after any number of $\mathbf{v}$ pulls its score is strictly larger than Δ, whereas the upper inequality makes the fixed $\mathbf{u}$ score at most Δ. Hence there is no return. In contrast, $D_n = \mathrm{diag}(C_n) = (\lambda + n/2)\mathbf{I}$, so both diagonal widths are equal and the center gives $\mathbf{u}$ a strict advantage after round one. Moreover $D_n \preceq \kappa_n C_n$ analytically, but multiplying both diagonal widths by $\sqrt{\kappa_n}$ leaves them equal; uniform certified inflation therefore does not change the diagonal action. For example, $(\lambda, \epsilon, \Delta, b) = (1, 0.1, 1, 1.1)$ satisfies the interval. $\square$

This is a two-dimensional linear-Gram existence witness. For a linear predictor with Gaussian squared loss, this Gram is exactly the GGN/Fisher operator, so the construction isolates a GGN-relevant geometric mechanism. It is neither a nonlinear-network result nor a uniform dominance result. A dense action-dependent factor recovers the full widths, but it uses the reference geometry that the diagonal approximation was meant to avoid.

Composing subsampling with feature drift. Lemma 33 guarantees a sandwich for the *relinearized* $\mathbf{C}_t$: $(1-\varepsilon)\mathbf{C}_t \preceq \hat{\mathbf{C}}_t \preceq (1+\varepsilon)\mathbf{C}_t$. Only the *upper* half is needed here: $\hat{\mathbf{C}}_t \preceq (1+\varepsilon)\mathbf{C}_t$ gives $\kappa_{+,t} = 1 + \varepsilon$, and composing with the one-sided whitened bound $\mathbf{C}_t \preceq (1 + \bar{\chi}_t)^2 \bar{\mathbf{C}}_t$ (Lemma 30) yields $u_t = (1 + \varepsilon)(1 + \bar{\chi}_t)^2$ in Theorem 45. A union bound over $t \leq T$ (replacing δ' by δ'/T in Lemma 33) makes the subsampling event time-uniform and contributes to δ_{cert}.

For an *unrescaled current-parameter window* (or any current-parameter downweighted subset with weights in $[0, 1]$), Proposition 44 gives the *deterministic* upper factor $\kappa_{+,t} = 1$ with *no* distributional assumption, so Theorem 45 applies directly with $u_t = (1 + \bar{\chi}_t)^2$. Only the *stale periodic-refresh* operator (§F.3(iii)), which reuses Jacobians from an older parameter, lacks a deterministic $\kappa_{+,t}$; there Theorem 45 provides a guarantee only when a one-sided bound $\hat{\mathbf{C}}_t \preceq \kappa_{+,t}\mathbf{C}_t$ is separately certified.

What remains open. The path schedules close the computability gap, but their constants can be extremely loose. The approximate-rank result replaces exact rank by a realized or predictably bounded spectral tail, but does not show that this tail or feature drift remains small during unrestricted training. Useful control of these two quantities for a genuinely adapting backbone, tighter smoothness/residual certificates, and verified numerical enclosures for floating-point solver and spectral calculations remain open.

G Related Work

Neural and linear contextual bandits. LinUCB (Abbasi-Yadkori et al., 2011) and Thompson sampling (Thompson, 1933; Agrawal and Goyal, 2013) provide the analytical backbone for contextual bandit exploration. Neural extensions—NeuralUCB (Zhou et al., 2020), NeuralTS (Zhang et al., 2021), neural-linear models (Riquelme et al., 2018)—use different neural feature covariances. NeuralUCB and NeuralTS accumulate full neural-gradient Grams in their stated algorithms, while NeuralLinear maintains Bayesian linear uncertainty over learned last-layer features. CC-UCB instead re-evaluates every replayed Jacobian at the current parameter and accesses the resulting full operator implicitly. Neural-linear designs that train a deep representation but explore only in the last layer (Xu et al., 2022a) occupy the opposite design point. Recent alternatives sharpen different parts of this landscape. Salgia (2023) analyze overparameterized smooth-activation networks through non-asymptotic network–NTK approximation and give an efficient sublinear-regret algorithm. Deb et al. (2024) instead reduce contextual bandits to online neural regression, using perturbed predictions to obtain their regression guarantees. Bui et al. (2025) combine a learned deep representation with a variance-aware linear-UCB exploration layer. These are, respectively, an NTK analysis, a regression-oracle route, and a learned last-layer geometry; none analyzes the dynamically relinearized full GGN and its CG error. Our low-rank/excitation result is correspondingly narrower than generic neural regression, but controls that current-parameter operator.

Online Bayesian and EKF-style neural bandits. Durán-Martín et al. (2022) apply extended Kalman filter (EKF) updates in a learned or random low-dimensional affine parameter subspace. They maintain a dense or diagonal covariance in that subspace, inducing a low-rank covariance over parameters spanning all network layers, with memory independent of the number of rounds. This differs from applying exact curvature–vector products in the full parameter space. More directly, Chang et al. (2023) maintain a diagonal-plus-low-rank approximation to the full-parameter posterior precision through recursive EKF and online-SVD updates, and use it for Thompson sampling in an MNIST contextual bandit. Their LO-FI method is a one-pass compressed-posterior approach with fixed-rank cost linear in parameter count and no replay of past data. CC-UCB instead relinearizes the full replay history at the current parameter and accesses the uncompressed GGN through residual-checked CG. Their bandit study is empirical; our conditional UCB analysis explicitly accounts for CG truncation, but does not establish a practical advantage over LO-FI.

Last-layer and structured Laplace approximations. Last-layer Laplace fixes the backbone and restricts the approximate posterior to the final layer (Kristiadi et al., 2020), omitting backbone-weight uncertainty by construction. This is only one scalable option: all-layer block-diagonal Kronecker curvature has also been used (Ritter et al., 2018), and Daxberger et al. (2021) systematize all-weight, subnetwork, and last-layer variants with full, low-rank, KFAC, or diagonal curvature. CC-UCB uses the uncompressed current-parameter GGN as its reference operator without materializing the covariance matrix.

Sampling-based full-parameter exploration. Langevin Monte Carlo Thompson sampling (LMC-TS) (Xu et al., 2022b) explores with approximate samples of a loss-defined parameter posterior, obtained by noisy gradient steps on the regularized loss; like CC-UCB it never materializes a covariance matrix. Its regret guarantee is for linear contextual bandits; its neural-bandit evidence is empirical. The two routes pay for exploration differently: LMC-TS through the mixing of a per-round sampling chain, CC-UCB through CG truncation and spectral distortion of an explicit curvature metric, which yields deterministic widths, an optimism-style analysis, and a fixed curvature operator reused across separate per-action CG solves.

Efficient nonlinear UCB. Kassraie and Krause (2022) analyze NTK-UCB and propose NN-UCB and its convolutional variant CNN-UCB. NN-UCB combines a trained-network mean with confidence widths from the empirical NTK feature map at initialization. They prove sublinear regret for the SupNN-UCB variant under NTK-RKHS, sufficient-exploration, and network-width assumptions. Their fixed-feature uncertainty differs from CC-UCB's replayed GGN/Fisher curvature at the current parameter $\boldsymbol{\theta}_t$, accessed matrix-free via CG. CC-UCB thus targets the regime where the practitioner has a specific pretrained model and wants to use its curvature as-is, rather than constructing a surrogate NTK-based estimator.

Spectral sketching for linear bandits. Kuzborskij et al. (2019) analyze linear bandits where the Gram matrix is replaced by a Frequent Directions sketch, deriving regret bounds that depend on the spectral approximation quality. Wen et al. (2026) show that fixed-size sketch regret bounds can degenerate to the trivial linear rate under heavy spectral tails, and propose an adaptive dyadic block sketch. Our spectral-distortion theorem (Theorem 45) instantiates the same conceptual lever in the matrix-free GGN/CVP setting by exposing transfer and realized-width terms along the realized trajectory. We do not claim novelty for the sandwich-to-regret reduction itself; the contribution is adapting it to the CG/GGN context with explicit CG truncation and feature-drift factors.

Laplace approximation in deep learning. The Laplace approximation (MacKay, 1992; Ritter et al., 2018) provides a Gaussian posterior centered at a MAP estimate with covariance given by the inverse curvature. Scalable variants use diagonal, KFAC (Martens and Grosse, 2015), or subnetwork approximations (Daxberger et al., 2021). Matrix-free full-Jacobian/Fisher predictive variance using implicit products and CG predates this work (Maddox et al., 2021); CC-UCB adapts that computational route to an online, current-relinearized GGN operator and supplies bandit-specific transfer and solver-error accounting.

GGN–vector products, Hessian spectral analysis, and influence functions. Pearlmutter's algorithm (Pearlmutter, 1994) computes exact Hessian–vector products; the analogous GGN–vector product has the same Jacobian-transpose-times-Jacobian-times-vector structure (Schraudolph, 2002). Recent work on Krylov methods for Hessian spectral analysis (Granziol and Juarev, 2026) demonstrates related distributed HVP/Krylov infrastructure at $> 10^8$-parameter scale, but does not benchmark the exact per-example GGN operator or end-to-end CC-UCB latency. Influence functions (Koh and Liang, 2017) use inverse-curvature–vector products to attribute predictions to training points; the CG-based implicit solve we employ is the same computational primitive.

Optimal experimental design. Information-theoretic exploration objectives—such as maximizing $\log\det(\bar{\mathbf{C}}_t)$ or minimizing the trace of $\bar{\mathbf{C}}_t^{-1}$—connect bandit exploration to the classical D-optimal and A-optimal design criteria (Valko et al., 2013). CC-UCB's use of the full GGN/Fisher curvature enables stochastic estimates of these quantities via Hutchinson-type trace estimators (Ubaru et al., 2017), without forming the matrix.

Filtration. We separate *pre-action* randomness, revealed before the agent commits to a_t, from *post-reward* randomness, drawn only after r_t is observed. Let Ω_t collect all pre-action algorithmic randomness used at round t: curvature subsampling indices, randomized sketches, and stochastic solver seeds. The randomized sketch or subsample that defines the curvature operator is drawn into Ω_t and *fixed before action selection*; it is not re-randomized during any conjugate-gradient (CG) solve at round t. (Standard CG requires a fixed symmetric positive definite matrix–vector oracle across its iterations and does not apply to a stochastic oracle that changes between iterations.) Let U_{t+1} collect the *post-reward* randomness used to form the next parameter estimate $\boldsymbol{\theta}_{t+1}$ (minibatch order, dropout, perturbed warm starts). Define $\mathcal{F}_0 = \sigma(\boldsymbol{\theta}_1)$ and, for $t \geq 1$, the recursively built σ-algebra

$$\mathcal{F}_t \;:=\; \mathcal{F}_{t-1} \vee \sigma\big(\mathbf{x}_t, \mathcal{A}_t, \Omega_t, a_t, r_t, U_{t+1}\big), \tag{143}$$

so that $\boldsymbol{\theta}_{t+1}$—computed from the data through round t together with the post-reward randomness U_{t+1}—is $\mathcal{F}_t$-measurable. Define the *pre-action history* σ-algebra and the pre-reward σ-algebra

$$\mathcal{H}_t^- \;:=\; \mathcal{F}_{t-1} \vee \sigma\big(\mathbf{x}_t, \mathcal{A}_t\big), \qquad \mathcal{H}_t \;:=\; \mathcal{H}_t^- \vee \sigma(\Omega_t), \qquad \mathcal{G}_t \;:=\; \mathcal{H}_t \vee \sigma(a_t). \tag{144}$$

At round t the context $\mathbf{x}_t$, the finite action set $\mathcal{A}_t$, and the pre-action randomness Ω_t are revealed; the agent selects a_t; then the reward $r_t = \mu^*(\mathbf{x}_t, a_t) + \eta_t$ is observed; finally U_{t+1} is drawn and $\boldsymbol{\theta}_{t+1}$ is computed. $\mathcal{H}_t^-$ is the history immediately before the randomized certificate draw; after that draw, its realized operator and schedules are $\mathcal{H}_t$-measurable. By construction of Algorithm 2 the selection rule is a deterministic function of $\mathcal{H}_t$-measurable quantities, so the chosen action a_t is itself $\mathcal{H}_t$-measurable and hence $\mathcal{G}_t = \mathcal{H}_t$; we retain the symbol $\mathcal{G}_t$ where the played action is used. The following are all $\mathcal{H}_t$-measurable (i.e. determined before the reward r_t): the candidate action set $\mathcal{A}_t$ and every candidate feature $\mathbf{g}_t(\mathbf{x}_t, a)$, $a \in \mathcal{A}_t$; the chosen action a_t; the algorithmic curvature operator $\mathcal{C}_t$ (Appendix C); the confidence schedule $\bar{\beta}_t$; the centering certificate $\bar{\psi}_t$ (Assumption 5); the per-round CG tolerance $\bar{\varepsilon}_t$ together with any condition-number certificate used by the CG stopping rule; and the one-sided optimism-transfer factor $u_t(a)$ (Assumption 6). Since $\boldsymbol{\theta}_t$ is $\mathcal{F}_{t-1} \subseteq \mathcal{H}_t$-measurable, each candidate design vector $\mathbf{g}_t(\mathbf{x}_t, a)$ is $\mathcal{H}_t$-measurable and the realized design vector $\mathbf{g}_t(\mathbf{x}_t, a_t)$ is $\mathcal{G}_t$-measurable. We assume $\{U_s\}_{s\geq 1}$ is independent of the reward noise $\{\eta_s\}_{s\geq 1}$, so that revealing U_{t+1} and forming $\boldsymbol{\theta}_{t+1}$ after r_t preserves the conditional sub-Gaussian property of Assumption 2 for the future noise $\{\eta_s\}_{s>t}$. Index every randomized certificate source (operator sketch, transfer bound, condition bound, or other randomized oracle) by $j \in \mathcal{J}$. For its round-t validity event $\mathcal{E}_{j,t}$ choose in advance $\delta_{j,t} \geq 0$ such that its stated conditional guarantee gives $\Pr(\mathcal{E}_{j,t}^c \mid \mathcal{H}_t^-) \leq \delta_{j,t}$ and

$$\sum_{j\in\mathcal{J}} \sum_{t=1}^{T} \delta_{j,t} \leq \delta_{\mathrm{cert}}, \qquad \mathcal{E}_{\mathrm{cert}} := \bigcap_{j\in\mathcal{J}} \bigcap_{t=1}^{T} \mathcal{E}_{j,t}.$$

The tower property and union bound yield $\Pr(\mathcal{E}_{\mathrm{cert}}) \geq 1 - \delta_{\mathrm{cert}}$. Deterministic or almost-sure certificates use $\delta_{j,t} = 0$. The self-normalized confidence event is separate and receives probability δ. When Lemma 23 supplies the residual-energy envelope, include its time-uniform event $\mathcal{E}_R$ as one certificate source and reserve δ_R inside δ_{cert} (equivalently, require $\delta_R + \sum_{j,t} \delta_{j,t} \leq \delta_{\mathrm{cert}}$ for the remaining sources). The residual event is never counted again in the self-normalized failure probability δ.

Auxiliary linearized ridge estimator. For the analysis we define a *linearized ridge estimator* based on frozen features. Set the predictable intercept $b_s := \mu_{\boldsymbol{\theta}_s}(\mathbf{x}_s, a_s) - \mathbf{g}_s(\mathbf{x}_s, a_s)^\top \boldsymbol{\theta}_s$ and the pseudo-response $y_s := r_s - b_s$. Under Assumption 4 and realizability ($\mu^* = \mu_{\boldsymbol{\theta}^*}$),

$$y_s \;=\; \mathbf{g}_s(\mathbf{x}_s, a_s)^\top \boldsymbol{\theta}^* \;+\; \rho_s(\mathbf{x}_s, a_s, \boldsymbol{\theta}^*) \;+\; \eta_s, \tag{145}$$

where $|\rho_s| \leq \varepsilon_{\mathrm{lin}}(s)$. The linearized estimator is

$$\hat{\boldsymbol{\theta}}_t^{\mathrm{lin}} \;:=\; \arg\min_{\boldsymbol{\theta}} \left[\frac{1}{2\sigma^2} \sum_{s=1}^{t-1} \big(y_s - \mathbf{g}_s(\mathbf{x}_s, a_s)^\top \boldsymbol{\theta}\big)^2 \;+\; \frac{\lambda}{2} \|\boldsymbol{\theta}\|^2 \right] \;=\; \bar{\mathbf{C}}_t^{-1} \, \frac{1}{\sigma^2} \sum_{s=1}^{t-1} \mathbf{g}_s(\mathbf{x}_s, a_s)\, y_s. \tag{146}$$

This is a standard linear ridge regression with predictable design vectors $\mathbf{g}_s/\sigma$ and Gram matrix $\bar{\mathbf{C}}_t$, so the self-normalized martingale bound (Abbasi-Yadkori et al., 2011) applies to it directly (Lemma 40).

GGN versus Hessian (context only). The regret theorem uses the GGN/Fisher curvature as the exploration metric. It does not require the GGN to be an accurate Hessian approximation. For squared loss (42) the full Hessian of the data-fit term is

$$\mathbf{H}_t \;=\; \frac{1}{\sigma^2}\sum_{s=1}^{t-1}\Big[\mathbf{g}_t(\mathbf{x}_s,a_s)\mathbf{g}_t(\mathbf{x}_s,a_s)^\top \;-\; \big(r_s-\mu_{\boldsymbol{\theta}_t}(\mathbf{x}_s,a_s)\big)\nabla^2\mu_{\boldsymbol{\theta}_t}(\mathbf{x}_s,a_s)\Big] \;+\; \lambda\mathbf{I}.$$

The curvature matrix $\mathbf{C}_t$ drops the residual-times-second-derivative term, retaining only the outer-product part that is guaranteed PSD. The dropped residual-times-second-derivative term $D_t = \sigma^{-2}\sum_{s<t}(r_s-\mu_{\boldsymbol{\theta}_t})\nabla^2\mu_{\boldsymbol{\theta}_t}$ has operator norm at most $(t-1)\epsilon_r H/\sigma^2$, where ϵ_r bounds the residuals and H the Hessian norms. When this quantity is small relative to λ, $\mathbf{C}_t$ approximates the full Hessian in the Loewner sense (a formal statement with proof is given in Appendix K). This comparison is *not* invoked in the regret analysis (Theorem 12 uses $\mathbf{C}_t$ directly as the exploration metric); it serves only to contextualize $\mathbf{C}_t$ as a proxy for the full Hessian. Outside the small-residual regime, the GGN is not an exact Hessian approximation; it is the PSD Fisher/GGN metric used by the algorithm.

Remark (Fisher, empirical Fisher, and GGN). Three curvature objects appear in the literature and must be distinguished. The *empirical Fisher* is the realized score outer product $\sum_s \nabla_{\boldsymbol{\theta}}\log p(r_s\,|\,\mathbf{x}_s,a_s;\boldsymbol{\theta})\,\nabla_{\boldsymbol{\theta}}\log p(r_s\,|\,\mathbf{x}_s,a_s;\boldsymbol{\theta})^\top$; it depends on observed rewards and is *not* the operator used by CC-UCB. The *true (model) Fisher* is the model expectation of the score outer product, $F(\boldsymbol{\theta}) = \mathbb{E}_{r\sim p(\cdot|\mathbf{x},a;\boldsymbol{\theta})}[\nabla_{\boldsymbol{\theta}}\log p\;\nabla_{\boldsymbol{\theta}}\log p^\top]$. The *generalized Gauss–Newton* (GGN) drops the residual-times-Hessian term from the full Hessian, retaining only the outer-product Jacobian part that is guaranteed PSD (Martens, 2020; Schraudolph, 2002). For the fixed-variance Gaussian working model underlying the squared-loss quasi-likelihood used here, the model Fisher and GGN are identical, and both reduce to the outer-product curvature $\sigma^{-2}\sum_s \mathbf{g}_s\mathbf{g}_s^\top$ used in (44). More generally, for exponential-family negative log-likelihoods parameterized by natural parameters $z_{\boldsymbol{\theta}}(\mathbf{x},a)$, the true Fisher equals the GGN: $J_{\boldsymbol{\theta}}^\top\nabla^2 A(z)\,J_{\boldsymbol{\theta}}$, where $J_{\boldsymbol{\theta}} = \nabla_{\boldsymbol{\theta}} z_{\boldsymbol{\theta}}$ and $\nabla^2 A$ is the Hessian of the log-partition function (Martens, 2020; Pascanu and Bengio, 2013). Bernoulli rewards yield scalar weight $p(1-p)$; categorical rewards yield $\mathrm{Diag}(\mathbf{p}) - \mathbf{p}\mathbf{p}^\top$. See Appendix L for the extension.

We access $\mathbf{C}_t$ only through curvature–vector products $\mathbf{v}\mapsto\mathbf{C}_t\,\mathbf{v}$, computed via forward-mode Jacobian–vector products composed with reverse-mode vector–Jacobian products (Pearlmutter, 1994; Schraudolph, 2002).

Extension to exponential-family likelihoods. The same matrix-free CG/CVP machinery extends algorithmically to exponential-family likelihoods (e.g. Bernoulli or categorical rewards) via the Fisher/GGN curvature $J_t^\top(\nabla^2 A)\,J_t$, but no regret guarantee beyond Gaussian squared loss is claimed in this draft. Details of the curvature definitions are in Appendix L.

Per-round CG tolerance $\bar{\varepsilon}_t$. The algorithm uses a *known* per-round tolerance $\bar{\varepsilon}_t\in(0,1)$ in the UCB bonus (line 8), not the true per-action energy-norm errors $\varepsilon_{t,a}$ (which are unobservable without the exact solves). It suffices that $\max_{a\in\mathcal{A}_t}\varepsilon_{t,a}\le\bar{\varepsilon}_t$ at every round ((49); zero-gradient actions have $\varepsilon_{t,a}=0$ by convention and are trivially within tolerance); the bonus is then conservative. Two strategies ensure this:

1. *Adaptive residual stopping (recommended).* In exact arithmetic, for each action monitor the computable Euclidean relative residual $\|r_I\|/\|\mathbf{g}_t(a)\|$ at each CG step (skipping actions with $\mathbf{g}_t(a)=\mathbf{0}$). Since $\varepsilon_{t,a}\le\sqrt{\bar{\kappa}_t}\cdot\|r_I\|/\|\mathbf{g}_t(a)\|$ (standard CG bound, with $\bar{\kappa}_t$ a certified upper bound on the condition number of $\mathcal{C}_t$), stopping that solve when $\|r_I\|/\|\mathbf{g}_t(a)\|\le\bar{\varepsilon}_t/\sqrt{\bar{\kappa}_t}$ guarantees $\varepsilon_{t,a}\le\bar{\varepsilon}_t$, hence (49). This is conservative by the factor $\sqrt{\bar{\kappa}_t}$ but requires no access to $\mathcal{C}_t^{-1}$ and adapts automatically to the per-round condition number. A floating-point implementation needs a verified residual enclosure to inherit this certification literally.
2. *Fixed budget (requires a uniform condition-number bound).* If an a priori bound $\bar{\kappa}_t\le\kappa_{\max}$ holds for *all rounds* $t\le T$, choose I via Eq. (151) at $\kappa_{\max}$ and a fixed target $\bar{\varepsilon}_t\equiv\bar{\varepsilon}$. For a *nonnegative weighted outer-product* operator $\mathcal{C}_t=\lambda\mathbf{I}+\sigma^{-2}\sum_{s\in\mathcal{S}_t}w_s\,\mathbf{g}_t(\mathbf{x}_s,a_s)\mathbf{g}_t(\mathbf{x}_s,a_s)^\top$ the condition number is bounded in closed form, $\bar{\kappa}_t\le 1+W_tG^2/(\lambda\sigma^2)$ with $W_t=\sum_{s\in\mathcal{S}_t}w_s$; a fixed buffer of size n ($W_t=n$, unrescaled) makes this independent of T. For the full-history operator ($W_t=t-1$), and for rescaled uniform subsampling with $w_s=(t-1)/n$ (so $W_t=\sum_{s\in\mathcal{S}_t}w_s=t-1$), the bound grows with t. For a known finite horizon, a fixed budget at $\kappa_{\max}=1+(T-1)G^2/(\lambda\sigma^2)$ is valid but scales as $O(\sqrt{T}\log(1/\bar{\varepsilon}))$; a horizon-independent fixed budget is not uniform, so adaptive stopping or an increasing I_t is preferable (see Appendix I). For a *general* SPD operator that is not a nonnegative weighted outer product (e.g. a randomized sketch), the closed-form bound $1+W_tG^2/(\lambda\sigma^2)$ does not apply, and a separately certified $\mathcal{H}_t$-measurable condition-number upper bound $\bar{\kappa}_t$ must be supplied.

In the regret bound (Theorem 12), the CG inflation enters through the per-round factor $\alpha_t = \sqrt{(1+\bar{\varepsilon}_t)/(1-\bar{\varepsilon}_t)}$, inside the sum of squares $\sum_t \alpha_t^2 u_t(a_t)\omega_t^2$; no horizon maximum over $\bar{\varepsilon}_t$ is taken.

Action screening (practical heuristic). When the action set $\mathcal{A}_t$ is large, computing the CG solve for every action is prohibitive ($|\mathcal{A}_t| \times I$ CG steps per round). A practical heuristic is to screen the top-K actions by mean prediction $\widehat{\mu}_t(a)$ (forward passes only), then compute UCB widths for the K candidates. This reduces cost to $K \times I$ CG solves. *The regret guarantee in Theorem 12 analyzes the full-enumeration case ($K = |\mathcal{A}_t|$); screening can drop the optimal arm and is not covered by the theory unless one adds an assumption that $a_t^* \in \mathcal{A}_t^K$ for all t.*

Algorithm 3 Operational CC-UCB with path certificates

Require: Reward predictor $\mu_{\boldsymbol{\theta}}$, damping $\lambda > 0$, noise proxy σ, confidence level δ, radius S, analytic constants G, L_g, L_μ; algorithmic curvature mode $\mathcal{C}_t$ (exact full-history $\mathbf{C}_t$ or approximate $\hat{\mathbf{C}}_t$) with certified one-sided factor $\kappa_{+,t}$ and condition upper bound $\bar{\kappa}_t$; per-round CG tolerance $\bar{\varepsilon}_t < 1$

Initialize $\boldsymbol{\theta}_1$, certified $\zeta_1 \geq \|\nabla \mathcal{L}_1(\boldsymbol{\theta}_1)\|$, $n \leftarrow 0$, $\bar{\boldsymbol{\theta}} \leftarrow \mathbf{0}$, $J \leftarrow 0$, $R^{\text{col}} \leftarrow 0$, $\bar{F} \leftarrow 0$, $\hat{\gamma} \leftarrow 0$, and $\mathcal{D}_0 \leftarrow \emptyset$

for $t = 1, 2, \ldots, T$ **do**

  Observe context $\mathbf{x}_t$ and action set $\mathcal{A}_t$; draw pre-action randomness Ω_t {$\mathcal{H}_t$ of (144)}

  $Q_t \leftarrow J + n\|\boldsymbol{\theta}_t - \bar{\boldsymbol{\theta}}\|^2$; $\bar{\chi}_t \leftarrow L_g\sqrt{Q_t}/(\sigma\sqrt{\lambda})$; $u_t \leftarrow \kappa_{+,t}(1+\bar{\chi}_t)^2$

  $\bar{M}_t \leftarrow \sigma^{-2}[L_g\sqrt{R^{\text{col}}Q_t} + (3GL_g/2)Q_t + (L_g^2/2)Q_t^{3/2}]$; $\bar{\psi}_t \leftarrow (\zeta_t + \bar{M}_t)/\sqrt{\lambda}$ {in a radius-R region use $L_g^2 R Q_t$ for the last term}

  $\bar{\epsilon}_t^{\text{lin}} \leftarrow (L_\mu/2)(S + \|\boldsymbol{\theta}_t\|)^2$; $\bar{\beta}_t \leftarrow \sqrt{\hat{\gamma} + 2\log(1/\delta)} + \sqrt{\lambda}S + \sigma^{-1}\sqrt{\bar{F}}$; $\omega_t \leftarrow \bar{\beta}_t + \bar{\psi}_t$ {corrected center uses $\omega_t = \bar{\beta}_t$}

  Form or receive a fixed $\mathcal{H}_t$-measurable SPD matrix–vector oracle $\mathbf{v} \mapsto \mathcal{C}_t\mathbf{v}$, $\mathcal{C}_t \succeq \lambda\mathbf{I}$, held fixed across all CG solves this round {exact/buffered special case $\mathcal{C}_t\mathbf{v} = \lambda\mathbf{v} + \sigma^{-2}\sum_{s\in\mathcal{S}_t} w_s\, \mathbf{g}_t(\mathbf{x}_s, a_s)[\mathbf{g}_t(\mathbf{x}_s, a_s)^\top\mathbf{v}]$, weights w_s per §F.3}

  for each $a \in \mathcal{A}_t$ **do**

    $\widehat{\mu}_t(a) \leftarrow \mu_{\boldsymbol{\theta}_t}(\mathbf{x}_t, a)$ {mean prediction (or corrected center (75))}

    $\mathbf{g}_t(a) \leftarrow \nabla_{\boldsymbol{\theta}}\, \mu_{\boldsymbol{\theta}_t}(\mathbf{x}_t, a)$ {gradient feature}

    if $\mathbf{g}_t(a) = \mathbf{0}$ **then**

      $\tilde{s}_t^2(a) \leftarrow 0$ {skip CG; no residual ratio}

    else

      $\tilde{\mathbf{u}} \leftarrow \text{CG}(\mathcal{C}_t,\ \mathbf{g}_t(a))$; stop only once $\|r_I\|/\|\mathbf{g}_t(a)\| \leq \bar{\varepsilon}_t/\sqrt{\bar{\kappa}_t}$, or use a certified budget I_t from (151) {recompute the original residual in floating point; a point check is not an enclosure}

      $\tilde{s}_t^2(a) \leftarrow \mathbf{g}_t(a)^\top\tilde{\mathbf{u}}$ {approx. predictive variance}

    end if

  end for

  $a_t \leftarrow \arg\max_{a\in\mathcal{A}_t}\left[\widehat{\mu}_t(a) + \omega_t\sqrt{\dfrac{u_t(a)}{1-\bar{\varepsilon}_t}}\sqrt{\tilde{s}_t^2(a)}\right]$ {bonus (134); ties broken by a fixed rule (e.g. lowest index) or by Ω_t}

  Play a_t, observe r_t

  $\mathcal{D}_t \leftarrow \mathcal{D}_{t-1} \cup \{(\mathbf{x}_t, a_t, r_t)\}$

  $c_t \leftarrow \mu_{\boldsymbol{\theta}_t}(\mathbf{x}_t, a_t) - r_t$; $R^{\text{col}} \leftarrow R^{\text{col}} + c_t^2$; $\bar{F} \leftarrow \bar{F} + (\bar{\epsilon}_t^{\text{lin}})^2$; $\hat{\gamma} \leftarrow \hat{\gamma} + log[1 + \sigma^{-2}u_t\tilde{s}_t^2(a_t)/(1-\bar{\varepsilon}_t)]$

  $n \leftarrow n+1$; $d \leftarrow \boldsymbol{\theta}_t - \bar{\boldsymbol{\theta}}$; $\bar{\boldsymbol{\theta}} \leftarrow \bar{\boldsymbol{\theta}} + d/n$; $J \leftarrow J + d^\top(\boldsymbol{\theta}_t - \bar{\boldsymbol{\theta}})$ {vector Welford update}

  Run J_{t+1}^{opt} warm-started optimizer steps on the *full* data set $\mathcal{D}_t$ to obtain $\boldsymbol{\theta}_{t+1}$, projecting to the certified region if required

  Certify $\zeta_{t+1} \geq \|\nabla\mathcal{L}_{t+1}(\boldsymbol{\theta}_{t+1})\|$; unless the terminal optimizer evaluation supplies this quantity, this requires one additional full-data gradient evaluation

end for

For the exact-curvature theorem, Algorithm 2 is run with $\mathcal{S}_t = \{1, \ldots, t-1\}$, $w_s = 1$, full action enumeration, $\mathcal{C}_t = \mathbf{C}_t$, and the transfer factor $u_t = (1+\bar{\chi}_t)^2$ from Lemma 30. For an approximate operator $\mathcal{C}_t = \hat{\mathbf{C}}_t$, use $u_t = \kappa_{+,t}(1+\bar{\chi}_t)^2$ from Theorem 45. The selection rule needs only the *one-sided* upper transfer factor $u_t(a)$ for optimism; no lower spectral factor is required (the cost of the operator is charged through the realized dynamic complexity $\Lambda_T^{\mathcal{C}}$). Algorithm 3 computes u_t, $\bar{\psi}_t$, and $\bar{\beta}_t$ from pre-action state. The operator-specific inputs are the proved factor $\kappa_{+,t}$ and condition upper bound $\bar{\kappa}_t$; exact full and unrescaled-subset operators use $\kappa_{+,t} = 1$.

Windowed/refresh curvature and action screening are practical heuristics unless the corresponding one-sided transfer factor or optimal-arm-survival condition is separately verified.

Corrected-center remark. Replacing the center $\widehat{\mu}_t(a)$ by the corrected center $\widehat{\mu}_t^{\mathrm{corr}}(a)$ of (75) sets $\bar{\psi}_t = 0$ (so $\omega_t = \bar{\beta}_t$) and removes the centering-stability assumption exactly (Corollary 21), at the cost of frozen-feature access for $\hat{\boldsymbol{\theta}}_t^{\mathrm{lin}}$; this departs from the $O(d)$ additional curvature-state-memory, no-stored-feature design unless a frozen-feature oracle, replay mechanism, or certified sketch is supplied (see the remark after Corollary 21).

The remainder of this legacy appendix analyzes the optimizer-centered CC-UCB specialization.

How to read the legacy one-sided theorem. The theorem is a transfer result for a general predictable SPD curvature sequence. Lemma 24 and Corollary 16 instantiate its nonlinear confidence, centering, and feature-transfer inputs from $O(d)$ path state, observed collection residuals, analytic smoothness constants, and a certified optimizer residual. These schedules are executable but can be very loose. A sublinear guarantee from this bound still requires control of the realized dynamic width complexity and a restricted parameter path; unrestricted end-to-end fine-tuning is not covered. No horizon-uniform lower transfer is required by Theorem 12; the lower sandwich appears only in the explicit near-linear corollary.

Discussion of terms.

- *Exploration term.* Write $S_T := \sum_{t=1}^{T} \alpha_t^2\, u_t(a_t)\, \omega_t^2$ for the sum of squares of the per-round factors. The exploration term is $2\sqrt{(\sigma^2 + G^2/\lambda)\, \Lambda_T^{\mathcal{C}}\, S_T}$. A sufficient condition for sublinear regret is

$$\Lambda_T^{\mathcal{C}}\, S_T = o(T^2), \qquad E_T = o(T).$$

  Concretely this holds whenever the per-round factors are polylogarithmically bounded ($\sup_t \alpha_t^2 u_t(a_t)\omega_t^2 = O(\operatorname{polylog} T)$, so $S_T = O(T \operatorname{polylog} T)$) and the realized dynamic complexity is small ($\Lambda_T^{\mathcal{C}} = O(\operatorname{polylog} T)$, e.g. via $\Lambda_T^{\mathcal{C}} \le \Gamma_T^{\mathrm{dyn}}$ with both the variation charge $V_T^{\mathcal{C}}$ and the endpoint $\log\det(\mathcal{C}_{T+1}/\lambda\mathbf{I})$ polylogarithmically bounded). Note ω_t^2 contains the cumulative squared linearization error through $\bar{F}_t$ (via β_t) and the centering certificate $\bar{\psi}_t$, so a small $\Lambda_T^{\mathcal{C}}$ alone is insufficient. α_t is bounded whenever the CG tolerance $\bar{\varepsilon}_t$ is bounded away from 1; in the exact-curvature case $u_t = (1 + \bar{\chi}_t)^2$ is bounded whenever the whitened-drift certificate $\bar{\chi}_t$ is bounded (Lemma 30); the whitened ℓ_2 accumulation $\chi_t \le \|\mathsf{D}_t\|_F$ makes this a near-stationarity condition on replayed-Jacobian drift. The unstable-curvature cost is isolated in $\Lambda_T^{\mathcal{C}}$ (through $V_T^{\mathcal{C}}$), not hidden in a spectral constant. The factor $\sqrt{\sigma^2 + G^2/\lambda}$ arises from the width-sum bound of the dynamic width-complexity lemma (Lemma 34, Eq. (122)); when $G^2 \le \lambda\sigma^2$ it is at most $\sigma\sqrt{2}$.
- *Linearization remainder.* The bound involves $E_T = \sum_t \varepsilon_{\mathrm{lin}}(t)$, not $T\varepsilon_{\mathrm{lin}}$. Near-linear regimes also require the squared-error sum $F_T = \sum_t \varepsilon_{\mathrm{lin}}(t)^2$ (through $\bar{F}_t$, hence ω_t) to be controlled. $\|\boldsymbol{\theta}^* - \boldsymbol{\theta}_t\| = O(1/t)$ gives $E_T = O(1)$; $\|\boldsymbol{\theta}^* - \boldsymbol{\theta}_t\| = O(t^{-1/4})$ gives $E_T = O(\sqrt{T})$. The additive role of E_T mirrors misspecified linear bandits, where a uniformly bounded per-round model error commonly contributes an additive regret term that scales linearly in the cumulative approximation error (Ghosh et al., 2017; Lattimore et al., 2020). The displayed bound requires $E_T/T \to 0$; a nonvanishing uniform per-round error bound alone yields an $O(T)$ additive term.
- *Centering.* The centering certificate enters *multiplicatively*, through $\omega_t = \bar{\beta}_t + \bar{\psi}_t$ inside the sum of squares S_T, rather than as a separate additive term. It vanishes ($\bar{\psi}_t = 0$) when $\boldsymbol{\theta}_t = \hat{\boldsymbol{\theta}}_t^{\mathrm{lin}}$ (convex last-layer models with exact optimization), and the corrected-center variant (Corollary 21) removes it in general at the cost of a frozen-feature solve.
- *Regret improvement over diagonal/last-layer surrogates.* The bound itself does not prove that full curvature yields strictly lower regret than diagonal or last-layer methods in any specific regime; it provides an upper bound with explicit per-round factors. *Demonstrating regret improvement requires a separate empirical study.* No such confirmatory study is reported here. The theorem exposes the transfer factor u_t and realized width complexity $\Lambda_T^{\mathcal{C}}$ as operator-dependent terms along the realized trajectory; it does not identify a causal regret cost because the adaptive path and other factors may also change.
- *Scope.* The bound is proved for Gaussian/squared-loss rewards. Extending it to exponential-family likelihoods (Bernoulli, categorical) would require a weighted frozen-feature analysis accounting for the data-dependent Fisher weights $\nabla^2 A(z)$ and is left as future work (Appendix L).

Corollary 48 (Legacy one-sided sanity check: frozen features recover linear UCB)**.** *Suppose the backbone is frozen and the reward head is linear in its parameters, the ridge objective is optimized exactly, and take the algorithmic curvature to be the sequential frozen Gram matrix, $\mathcal{C}_t = \bar{\mathbf{C}}_t$. Then: $\varepsilon_{\mathrm{lin}}(t) = 0$ for all t ($L = 0$ in Assumption 4), so $E_T = F_T = 0$; $\bar{\psi}_t = 0$ (under exact optimization the MAP and linearized estimators coincide, so Assumption 5 holds with equality), giving $\omega_t = \bar{\beta}_t$; $u_t(a) = 1$ for all a (indeed $\bar{s}_t^2(a) = s_{\mathcal{C},t}^2(a)$, so Assumption 6 holds with equality); and, since $\bar{\mathbf{C}}_{t+1} = \bar{\mathbf{C}}_t + \sigma^{-2}\mathbf{g}_t(\mathbf{x}_t, a_t)\mathbf{g}_t(\mathbf{x}_t, a_t)^\top$ by (46), the update is exactly the rank-one step, i.e. $\mathcal{C}_{t+1} = \mathcal{C}_t^+$, so $\Xi_t = 0$, $V_T^{\mathcal{C}} = 0$, and $\Lambda_T^{\mathcal{C}} = \gamma_T$. Then Theorem 12 gives*

$$R_T \;\le\; 2\sqrt{(\sigma^2 + G^2/\lambda)\,\gamma_T \sum_{t=1}^{T} \alpha_t^2\,\bar{\beta}_t^2}.$$

If in addition $\bar{\varepsilon}_t \le \bar{\varepsilon}$ so that $\alpha_t \le \alpha_I := \sqrt{(1+\bar{\varepsilon})/(1-\bar{\varepsilon})}$, the looser Corollary 49 recovers the familiar LinUCB form $R_T \le 2\,\alpha_I\,\bar{\beta}_{\max,T}\sqrt{(\sigma^2 + G^2/\lambda)\,T\,\gamma_T}$, where $\bar{\beta}_{\max,T} := \max_{t\le T}\bar{\beta}_t$.

Proof. Substitute $\varepsilon_{\mathrm{lin}}(t) = 0$, $\bar{\psi}_t = 0$, $u_t(a) = 1$, and $\Lambda_T^{\mathcal{C}} = \gamma_T$, $V_T^{\mathcal{C}} = 0$ into (54); the frozen Gram is monotone, so $\Xi_t = 0$ and the dynamic potential collapses to the information gain γ_T. The LinUCB form is Corollary 49 with $\alpha_t \le \alpha_I$, $u_t = 1$, $\omega_t = \bar{\beta}_t \le \bar{\beta}_{\max,T}$. ☐

Corollary 49 (Looser worst-case reduction)**.** *If $\alpha_t \le \alpha_{\max}$, $u_t(a_t) \le u_{\max}$, and $\omega_t \le \omega_{\max}$ for all $t \le T$, then $S_T \le \alpha_{\max}^2 u_{\max}\omega_{\max}^2 T$ and*

$$R_T \;\le\; 2\alpha_{\max}\omega_{\max}\sqrt{u_{\max}}\sqrt{(\sigma^2 + G^2/\lambda)T\Lambda_T^{\mathcal{C}}} + 2E_T.$$

This $\sqrt{T\Lambda_T^{\mathcal{C}}}$ form is weakly looser than (54), with equality when the per-round factors are constant. It is recorded only as a legacy reduction, not as the main result.

H Proof of the one-sided dynamic theorem

We recall the key objects: $\mathbf{g}_s(\mathbf{x}, a) = \nabla_{\boldsymbol{\theta}}\mu_{\boldsymbol{\theta}_s}(\mathbf{x}, a)$ (frozen features), $\bar{\mathbf{C}}_t = \lambda\mathbf{I} + \sigma^{-2}\sum_{s<t}\mathbf{g}_s(\mathbf{x}_s, a_s)\mathbf{g}_s(\mathbf{x}_s, a_s)^\top$ (frozen-feature Gram matrix), $\bar{s}_t(a) = \sqrt{\mathbf{g}_t(a)^\top\bar{\mathbf{C}}_t^{-1}\mathbf{g}_t(a)}$, the algorithmic curvature $\mathcal{C}_t$ with width $s_{\mathcal{C},t}(a) = \sqrt{\mathbf{g}_t(a)^\top\mathcal{C}_t^{-1}\mathbf{g}_t(a)}$, the linearized ridge estimator $\hat{\boldsymbol{\theta}}_t^{\mathrm{lin}}$ (Eq. 146), the combined width factor $\omega_t = \bar{\beta}_t + \bar{\psi}_t$, the per-round CG inflation $\alpha_t = \sqrt{(1+\bar{\varepsilon}_t)/(1-\bar{\varepsilon}_t)}$, and the realized dynamic complexity $\Lambda_T^{\mathcal{C}}$ (Eq. 118). Write $\widehat{\mu}_t(a) = \mu_{\boldsymbol{\theta}_t}(\mathbf{x}_t, a)$.

Step 1: Confidence, centering, and prediction-error bound. By Lemma 40, with probability $\ge 1-\delta$,

$$\left|\mathbf{g}_t(a)^\top(\hat{\boldsymbol{\theta}}_t^{\mathrm{lin}} - \boldsymbol{\theta}^*)\right| \;\le\; \bar{s}_t(a)\,\beta_t, \qquad \forall\, t,\, a, \tag{147}$$

where β_t is as in (138). The self-normalized martingale inequality (Abbasi-Yadkori et al., 2011) is applied to the 1-sub-Gaussian sequence $\xi_s = \eta_s/\sigma$ with the predictable design $\mathbf{g}_s/\sigma$ and Gram matrix $\bar{\mathbf{C}}_t$. Each realized design vector $\mathbf{g}_s(\mathbf{x}_s, a_s)$ is $\mathcal{G}_s$-measurable (where $\mathcal{G}_s = \mathcal{H}_s \vee \sigma(a_s)$ is the pre-reward sigma-algebra of (144)), and the sequential update (46) ensures the hypotheses of the self-normalized bound are satisfied. By Lemma 43 (Assumption 5), $|\mathbf{g}_t(a)^\top(\boldsymbol{\theta}_t - \hat{\boldsymbol{\theta}}_t^{\mathrm{lin}})| \le \bar{\psi}_t\,\bar{s}_t(a)$. Combined with $|\rho_t| \le \varepsilon_{\mathrm{lin}}(t)$ and $\bar{\beta}_t \ge \beta_t$, the triangle inequality gives the prediction-error bound (139)

$$|\mu_{\boldsymbol{\theta}^*}(\mathbf{x}_t, a) - \widehat{\mu}_t(a)| \;\le\; (\beta_t + \bar{\psi}_t)\,\bar{s}_t(a) + \varepsilon_{\mathrm{lin}}(t) \;\le\; \omega_t\,\bar{s}_t(a) + \varepsilon_{\mathrm{lin}}(t), \qquad \forall\, a. \tag{148}$$

(The corrected-center variant of Corollary 21 reaches (148) with $\omega_t = \bar{\beta}_t$ and without Assumption 5.)

Step 2: Bonus sandwiching. By Lemma 39 (which uses Lemma 37 with $\max_{a\in\mathcal{A}_t}\varepsilon_{t,a} \le \bar{\varepsilon}_t$ and the one-sided transfer (99)), for all a,

$$\omega_t\,\bar{s}_t(a) \;\le\; w_t(a) \;\le\; \alpha_t\,\omega_t\,\sqrt{u_t(a)}\,s_{\mathcal{C},t}(a). \tag{149}$$

The lower bound and (148) give $\mu^*(\mathbf{x}_t, a) \le \widehat{\mu}_t(a) + w_t(a) + \varepsilon_{\mathrm{lin}}(t)$ for all a: the UCB covers μ^* up to the additive $\varepsilon_{\mathrm{lin}}(t)$ model-mismatch term.

Step 3: Instantaneous regret. From the UCB rule $\widehat{\mu}_t(a_t) + w_t(a_t) \geq \widehat{\mu}_t(a_t^*) + w_t(a_t^*)$, and using $\mu^*(a_t^*) \leq \widehat{\mu}_t(a_t^*) + w_t(a_t^*) + \varepsilon_{\text{lin}}(t)$ (Step 2 at a_t^*, valid since $a_t^* \in \mathcal{A}_t$),

$$\begin{aligned} \mu^*(a_t^*) - \mu^*(a_t) &\leq [\widehat{\mu}_t(a_t^*) + w_t(a_t^*) + \varepsilon_{\text{lin}}(t)] - \mu^*(a_t) \\ &\leq [\widehat{\mu}_t(a_t) + w_t(a_t) + \varepsilon_{\text{lin}}(t)] - \mu^*(a_t) \quad \text{(UCB selection)} \\ &= w_t(a_t) + [\widehat{\mu}_t(a_t) - \mu^*(a_t)] + \varepsilon_{\text{lin}}(t) \\ &\leq w_t(a_t) + \omega_t\, \bar{s}_t(a_t) + 2\varepsilon_{\text{lin}}(t) \quad \text{((148) at } a_t) \\ &\leq 2\, w_t(a_t) + 2\varepsilon_{\text{lin}}(t) \quad \text{(lower bound in (149)).} \end{aligned}$$

Step 4: Upper bonus bound. The upper bound in (149) gives

$$\mu^*(a_t^*) - \mu^*(a_t) \;\leq\; 2\,\alpha_t\,\omega_t\,\sqrt{u_t(a_t)}\, s_{\mathcal{C},t}(a_t) \;+\; 2\varepsilon_{\text{lin}}(t). \tag{150}$$

Step 5: Cauchy–Schwarz after summing. Summing (150) over $t = 1, \ldots, T$ and applying Cauchy–Schwarz to the sum (*not* per round),

$$R_T \;\leq\; 2\sum_{t=1}^{T} \alpha_t \omega_t \sqrt{u_t(a_t)}\, s_{\mathcal{C},t}(a_t) + 2E_T \;\leq\; 2\sqrt{\sum_{t=1}^{T} \alpha_t^2 \omega_t^2 u_t(a_t)}\,\sqrt{\sum_{t=1}^{T} s_{\mathcal{C},t}^2(a_t)} + 2E_T.$$

Step 6: Dynamic potential and collection. Lemma 34 gives $\sum_{t=1}^{T} s_{\mathcal{C},t}^2(a_t) \leq (\sigma^2 + G^2/\lambda)\,\Lambda_T^{\mathcal{C}}$. Substituting,

$$R_T \;\leq\; 2\sqrt{(\sigma^2 + G^2/\lambda)\,\Lambda_T^{\mathcal{C}} \sum_{t=1}^{T} \alpha_t^2\, u_t(a_t)\,\omega_t^2} \;+\; 2E_T,$$

which is (54). The interpretable and observable versions (74) follow from $\Lambda_T^{\mathcal{C}} \leq \Gamma_T^{\text{dyn}}$ (Lemma 34) and $\Lambda_T^{\mathcal{C}} \leq \widehat{\Lambda^{\mathcal{C}}}_T$ (Corollary 36). The Cauchy–Schwarz step is applied only after summing, so the time-varying factors $\alpha_t, u_t(a_t), \omega_t$ enter through a sum of squares; no horizon maximum is taken before Cauchy–Schwarz, and the usual $\sqrt{T}$ scaling is recovered when the per-round factors are uniformly bounded (Corollary 49). The stated probability follows from a union bound over the self-normalized confidence event $\mathcal{E}_{\text{conf}}$ ($\geq 1 - \delta$) and the certificate event $\mathcal{E}_{\text{cert}}$ ($\geq 1 - \delta_{\text{cert}}$), giving $\geq 1 - \delta - \delta_{\text{cert}}$. □

I CG Convergence Analysis

Fix a round t and an action a with $\mathbf{g} := \mathbf{g}_t(a) \neq \mathbf{0}$. After $I_{t,a}$ iterations of CG on the SPD system $\mathcal{C}_t\mathbf{u} = \mathbf{g}$ (starting from $\tilde{\mathbf{u}}_0 = \mathbf{0}$), the standard energy-norm bound gives

$$\varepsilon_{t,a} \;=\; \frac{\|\tilde{\mathbf{u}}_{I_{t,a}} - \mathcal{C}_t^{-1}\mathbf{g}\|_{\mathcal{C}_t}}{\|\mathcal{C}_t^{-1}\mathbf{g}\|_{\mathcal{C}_t}} \;\leq\; 2\left(\frac{\sqrt{\kappa_t} - 1}{\sqrt{\kappa_t} + 1}\right)^{I_{t,a}},$$

where $\kappa_t = \lambda_{\max}(\mathcal{C}_t)/\lambda_{\min}(\mathcal{C}_t)$ is the true condition number. The bound is right-hand-side independent, so a common iteration budget bounds $\max_a \varepsilon_{t,a}$. This zero-start form is what the fixed-budget certificate (151) uses. For a *warm start* $\tilde{\mathbf{u}}_0 \neq \mathbf{0}$, energy-norm minimization over the shifted Krylov space instead gives

$$\frac{\|\tilde{\mathbf{u}}_{I_{t,a}} - \mathcal{C}_t^{-1}\mathbf{g}\|_{\mathcal{C}_t}}{\|\mathcal{C}_t^{-1}\mathbf{g}\|_{\mathcal{C}_t}} \;\leq\; 2\,q^{\,I_{t,a}}\,\frac{\|\tilde{\mathbf{u}}_0 - \mathcal{C}_t^{-1}\mathbf{g}\|_{\mathcal{C}_t}}{\|\mathcal{C}_t^{-1}\mathbf{g}\|_{\mathcal{C}_t}}, \qquad q = \frac{\sqrt{\kappa_t} - 1}{\sqrt{\kappa_t} + 1},$$

i.e. the extra factor is the initial relative energy error $\|e_0\|_{\mathcal{C}_t}/\|\mathcal{C}_t^{-1}\mathbf{g}\|_{\mathcal{C}_t}$ (equal to one at $\tilde{\mathbf{u}}_0 = \mathbf{0}$, recovering the zero-start bound). Under the same certified condition-number bound, adaptive residual stopping (line 5) remains valid with a warm start because it certifies the realized residual directly; but the fixed-budget count (151) is valid *as written* only for zero initialization, and a fixed-budget warm-start certificate additionally requires a bound on that initial relative energy error (the zero-start iteration count does not by itself certify a warm-started solve). Any certified upper bound $\bar{\kappa}_t \geq \kappa_t$ may replace κ_t; for a nonnegative weighted outer-product operator, $\lambda_{\min}(\mathcal{C}_t) \geq \lambda$ and $\lambda_{\max}(\mathcal{C}_t) \leq \lambda + W_t G^2/\sigma^2$ (where $W_t = \sum_{s \in \mathcal{S}_t} w_s$) give

$$\kappa_t \leq \frac{\lambda_{\max}(\mathcal{C}_t)}{\lambda} \leq 1 + \frac{W_t G^2}{\lambda \sigma^2} =: \bar{\kappa}_t,$$

whereas a general SPD operator requires a separately certified $\bar{\kappa}_t$.

Two cases at value one must be distinguished. The true condition number $\kappa_t = 1$ forces all eigenvalues equal, i.e. $\mathcal{C}_t = c_t \mathbf{I}$ for some scalar $c_t > 0$, and one CG iteration from $\tilde{\mathbf{u}}_0 = \mathbf{0}$ is *exact* ($\varepsilon_{t,a} = 0$) for every $\mathbf{g} \neq \mathbf{0}$; this does *not* by itself imply $c_t = \lambda$. With the specific certificate used here, $\bar{\kappa}_t = 1$ additionally forces $\lambda_{\max}(\mathcal{C}_t) = \lambda$, which together with $\mathcal{C}_t \succeq \lambda \mathbf{I}$ gives $\mathcal{C}_t = \lambda \mathbf{I}$ exactly. In either reading the single-step solve is exact, so we keep $\bar{\kappa}_t = 1$ as a separate branch and never evaluate the logarithmic budget below at its zero denominator. For $\bar{\kappa}_t > 1$, setting the right-hand side of the rate equal to $\bar{\varepsilon}_{\mathrm{CG}}$ and requiring $\bar{\varepsilon}_{\mathrm{CG}} < 1$ (where $\bar{\varepsilon}_{\mathrm{CG}}$ denotes the round-t target tolerance $\bar{\varepsilon}_t$) gives the per-solve budget

$$I_{\min}(\bar{\kappa}_t, \bar{\varepsilon}_{\mathrm{CG}}) \;=\; \left\lceil \frac{\log(2/\bar{\varepsilon}_{\mathrm{CG}})}{\log\big((\sqrt{\bar{\kappa}_t}+1)/(\sqrt{\bar{\kappa}_t}-1)\big)} \right\rceil \;=\; O\big(\sqrt{\bar{\kappa}_t}\,\log(1/\bar{\varepsilon}_{\mathrm{CG}})\big), \tag{151}$$

which, applied to every scored action, ensures $\max_a \varepsilon_{t,a} \leq \bar{\varepsilon}_{\mathrm{CG}} = \bar{\varepsilon}_t$. For $W_t = n$ (unrescaled fixed buffer size), $\bar{\kappa}_t \leq 1 + nG^2/(\lambda\sigma^2)$ is bounded independently of T, so $I_{\min}$ is constant. For full history, and for rescaled uniform subsampling with $w_s = (t-1)/n$, we have $W_t = t-1$, so the condition-number *bound* $\bar{\kappa}_t = O(tG^2/(\lambda\sigma^2))$ grows with t, and the worst-case certificate is then obtained with $I_{\min} = \Theta\big(G\sqrt{t/(\lambda\sigma^2)}\,\log(1/\bar{\varepsilon}_{\mathrm{CG}})\big)$ iterations per solve—i.e. $I = O(\sqrt{t}\,\log(1/\bar{\varepsilon}_{\mathrm{CG}}))$ suffices under this bound (this is an upper-bound certificate, not a lower bound on the number of iterations CG actually needs). CG may converge faster than this worst-case certificate suggests under favorable spectral clustering of $\mathcal{C}_t$; the theorem does not predict realized iteration counts.

J Two-Sided Spectral-Distortion Corollary

The main-text Theorem 45 uses only a one-sided upper transfer factor. For comparison we record the earlier two-sided result, which instead assumes both an upper *and* a lower spectral factor and yields the familiar $\sqrt{\rho_+/\rho_-}$ inflation. The two results trade different assumptions—the corollary below needs a lower spectral bound; the dynamic Theorem 12 needs the dynamic width complexity $\Lambda_T^{\mathcal{C}}$ instead—and neither numerically dominates the other in general.

Corollary 50 (Two-sided spectral-distortion inflation)**.** *Adopt the hypotheses of Theorem 12 with $\mathcal{C}_t = \hat{\mathbf{C}}_t$, retaining the* original nonlinear center *(so $\omega_t = \bar{\beta}_t + \bar{\psi}_t$; in the convex last-layer regime under exact optimization, $\bar{\psi}_t = 0$ and $\omega_t = \bar{\beta}_t$), and a uniform CG tolerance $\bar{\varepsilon}_t \leq \bar{\varepsilon}$ so that $\alpha_t \leq \alpha_I := \sqrt{(1+\bar{\varepsilon})/(1-\bar{\varepsilon})}$. Suppose predictable factors $\rho_{-,t}, \rho_{+,t}$ are certified before action selection and the* two-sided *sandwich $\rho_{-,t}\bar{\mathbf{C}}_t \preceq \hat{\mathbf{C}}_t \preceq \rho_{+,t}\bar{\mathbf{C}}_t$ holds for all $t \leq T$ with $0 < \rho_{-,t} \leq \rho_{+,t} < \infty$; set $\rho_- = \min_t \rho_{-,t}$, $\rho_+ = \max_t \rho_{+,t}$, and (to satisfy the convention $u_t \geq 1$ of Assumption 6) take the transfer factor $u_t := \max\{1, \rho_{+,t}\} \leq \max\{1, \rho_+\}$. Write $\omega_{\max,T} := \max_{t \leq T}(\bar{\beta}_t + \bar{\psi}_t)$ and $\rho_+^\star := \max\{1, \rho_+\}$. Then, on the intersection of the confidence event $\mathcal{E}_{\mathrm{conf}}$ ($\geq 1-\delta$) and the certificate event $\mathcal{E}_{\mathrm{cert}}$ (which now also contains the two-sided sandwich event; $\geq 1-\delta_{\mathrm{cert}}$),*

$$R_T \;\leq\; 2\,\alpha_I\,\sqrt{\rho_+^\star/\rho_-}\;\omega_{\max,T}\sqrt{(\sigma^2 + G^2/\lambda)\,T\,\gamma_T} \;+\; 2E_T$$

with probability at least $1-\delta-\delta_{\mathrm{cert}}$, where γ_T is the frozen-feature information gain (70)*. Under the standard normalization $\rho_+ \geq 1$ (achievable by rescaling the sandwich so its upper factor is at least one), $\rho_+^\star = \rho_+$ and the inflation is the familiar $\sqrt{\rho_+/\rho_-}$.*

Proof. This is a *direct* argument through the monotone frozen Gram $\bar{\mathbf{C}}_t$; it does *not* substitute $\hat{s}_t$ for the dynamic width in Corollary 49. *Step 1 (variance sandwich).* Inverting the two-sided sandwich gives $\rho_{+,t}^{-1}\bar{\mathbf{C}}_t^{-1} \preceq \hat{\mathbf{C}}_t^{-1} \preceq \rho_{-,t}^{-1}\bar{\mathbf{C}}_t^{-1}$, so, evaluating the quadratic form at $\mathbf{g}_t(a)$, $\rho_{+,t}^{-1/2}\,\bar{s}_t(a) \leq \hat{s}_t(a) \leq \rho_{-,t}^{-1/2}\,\bar{s}_t(a)$. The lower half gives $\bar{s}_t^2(a) \leq \rho_{+,t}\hat{s}_t^2(a) \leq u_t \hat{s}_t^2(a)$, so Assumption 6 holds with $u_t = \max\{1, \rho_{+,t}\}$. *Step 2 (per-round regret).* With $s_{\mathcal{C},t} = \hat{s}_t$, Theorem 12 Steps 1–4 give $\mathrm{regret}_t \leq 2\,\alpha_t\,\omega_t\,\sqrt{u_t}\,\hat{s}_t(a_t) + 2\varepsilon_{\mathrm{lin}}(t)$; using $\hat{s}_t(a_t) \leq \rho_{-,t}^{-1/2}\bar{s}_t(a_t)$, $u_t = \max\{1, \rho_{+,t}\} \leq \rho_+^\star$, $\alpha_t \leq \alpha_I$, and $\omega_t \leq \omega_{\max,T}$,

$$\mathrm{regret}_t \;\leq\; 2\,\alpha_I\,\omega_{\max,T}\,\sqrt{\rho_+^\star/\rho_{-,t}}\;\bar{s}_t(a_t) + 2\varepsilon_{\mathrm{lin}}(t) \;\leq\; 2\,\alpha_I\,\omega_{\max,T}\,\sqrt{\rho_+^\star/\rho_-}\;\bar{s}_t(a_t) + 2\varepsilon_{\mathrm{lin}}(t).$$

Step 3 (frozen-feature elliptic potential). Sum over t and apply Cauchy–Schwarz, then the classical elliptic-potential identity for the *monotone* Gram $\bar{\mathbf{C}}_t$ ((46)), $\sum_t \bar{s}_t^2(a_t) \le (\sigma^2 + G^2/\lambda) \sum_t \log(1 + \sigma^{-2}\bar{s}_t^2(a_t)) = (\sigma^2 + G^2/\lambda)\,\gamma_T$:

$$R_T \le 2\,\alpha_I\,\omega_{\max,T}\,\sqrt{\rho_+^\star/\rho_-}\sqrt{T\sum_t \bar{s}_t^2(a_t)} + 2E_T \le 2\,\alpha_I\,\omega_{\max,T}\,\sqrt{\rho_+^\star/\rho_-}\sqrt{(\sigma^2 + G^2/\lambda)\,T\,\gamma_T} + 2E_T.$$

The probability statement is the union bound over $\mathcal{E}_{\text{conf}}$ and $\mathcal{E}_{\text{cert}}$. □

For Lemma 33, allocate δ_{cert}/T per round and use the worst-case $m = T{-}1$ in (117) to obtain the required time-uniform sandwich.

K GGN Approximation Error

Lemma 51 (GGN approximation error)**.** *Suppose $\mu_{\boldsymbol{\theta}}(\mathbf{x}, a)$ is twice differentiable in $\boldsymbol{\theta}$. Suppose $|r_s - \mu_{\boldsymbol{\theta}_t}(\mathbf{x}_s, a_s)| \le \epsilon_r$ and $\|\nabla^2\mu_{\boldsymbol{\theta}_t}(\mathbf{x}_s, a_s)\|_{\text{op}} \le H$ for all $s < t$. Define $\zeta := (t{-}1)\,\epsilon_r\,H/(\sigma^2\lambda)$. Then:*

(i) $\|\mathbf{H}_t - \mathbf{C}_t\|_{\text{op}} \le (t{-}1)\,\epsilon_r\,H/\sigma^2 = \zeta\lambda$.
(ii) If $\zeta < 1$, then $(1{-}\zeta)\mathbf{C}_t \preceq \mathbf{H}_t \preceq (1{+}\zeta)\mathbf{C}_t$.

Proof. For $D_t = \sigma^{-2}\sum_{s<t}(r_s - \mu_{\boldsymbol{\theta}_t})\nabla^2\mu_{\boldsymbol{\theta}_t}$, the triangle inequality gives $\|D_t\|_{\text{op}} \le \zeta\lambda$, proving (i). Combining $-\|D_t\|\mathbf{I} \preceq -D_t \preceq \|D_t\|\mathbf{I}$ with $\mathbf{C}_t \succeq \lambda\mathbf{I}$ proves (ii). □

L Exponential-Family Curvature Definitions

For a regular exponential-family negative log-likelihood with natural parameter $z_{\boldsymbol{\theta}}(\mathbf{x}, a) \in \mathbb{R}^k$ (scalar for Bernoulli, k-dimensional for categorical) and log-partition function A, the GGN/Fisher curvature matrices are

$$\mathbf{C}_t^{\exp} \;=\; \lambda\mathbf{I} \;+\; \sum_{s=1}^{t-1} J_t(\mathbf{x}_s, a_s)^\top\,\nabla^2 A\big(z_t(\mathbf{x}_s, a_s)\big)\,J_t(\mathbf{x}_s, a_s), \tag{152}$$

$$\bar{\mathbf{C}}_t^{\exp} \;=\; \lambda\mathbf{I} \;+\; \sum_{s=1}^{t-1} J_s(\mathbf{x}_s, a_s)^\top\,\nabla^2 A\big(z_s(\mathbf{x}_s, a_s)\big)\,J_s(\mathbf{x}_s, a_s), \tag{153}$$

where $J_t(\mathbf{x}, a) = \nabla_{\boldsymbol{\theta}} z_{\boldsymbol{\theta}_t}(\mathbf{x}, a) \in \mathbb{R}^{k\times d}$ is the Jacobian of the natural parameter and $\nabla^2 A(z)$ is the output-space Hessian of the log-partition function. For Bernoulli rewards $\nabla^2 A(z) = p(1{-}p)$ (scalar); for categorical rewards $\nabla^2 A(z) = \mathrm{Diag}(\mathbf{p}) - \mathbf{p}\mathbf{p}^\top$). Setting $k = 1$ and $\nabla^2 A = \sigma^{-2}$ recovers the Gaussian curvature (44)–(45).

M Legacy empirical evidence and scope

Earlier repository revisions contained several studies designed for the optimizer-centered, one-sided dynamic theorem. They are summarized here as a scientific record, not as empirical validation of Theorem 1. None executes the headline exponential transported score while recording a predictable $\bar{D}_t$, a certified two-sided operator pair $\kappa_{-,t}, \kappa_{+,t}$, all-domain solver upper validity, played-action sharpness, and the resulting new regret bound. Most detailed figures, comparison tables, and protocol chronologies have therefore been removed from the manuscript. The two compact tables below retain provenance for their source aggregates.

Corrected-center and local-transport traces. The scaled-tanh and bounded-tanh studies record corrected centers, frozen information gain, path statistic Q_t, exact-current widths, and dense or CG solver diagnostics. They used scalar-link or additive local scores rather than the exponential score in (10). Their reported PASS/FAIL fields refer only to the legacy theorem event. In particular, optimizer and small-drift fields removed from the new headline theorem cannot be used to declare the revised event failed. The historical scaled-tanh aggregate was insufficient for a revised corrected-center analysis because the required per-round payload was stored in an external raw bundle. Restoring that bundle would permit a post-hoc audit of the local-small-drift premises and of whether the executed score matches the revised local score. It would not establish that compatibility by itself. The headline exponential score requires a new policy execution.

Method	$T = 250$	$T = 500$	$T = 1000$
full CG	375.7	438.1	479.3
full dense	361.5	422.4	461.7
diagonal	4120.8	7710.5	13456.0
Lanczos–Ritz	495.7	496.4	509.3
subsample 64	339.7	373.1	396.6
refresh 20	370.3	465.1	526.7
window 64	252.0	255.3	260.2

Table 4: Legacy fixed-reference linear right-hand-side to regret ratios. These values audit the earlier theorem and show numerical vacuity; they are not evaluations of the revised bound. Rows using unenclosed spectral or condition-number point estimates are not rigorous legacy certificates either.

Table 5: Legacy real-autodiff resource audit at history size 512, ten actions, and iterative-solver target original-system residual 10^{-3}. Values summarize the isolated primitive; no dense-reference accuracy is imputed for the larger model.

Model	Method	Time (s)	Peak GiB	Sample-CVPs	Max residual
mlp_100k	Separate CG	0.2101	0.108	25600	1.40e-04
mlp_100k	Batched CG	0.0395	0.126	25600	1.37e-04
mlp_100k	Jacobi-PCG	0.05416	0.141	28570	9.44e-04
mlp_100k	Diagonal	0.4428	0.086	0	9.30e+00
mlp_100k	Last layer	0.002994	0.097	0	–
mlp_100k	Explicit dense reference	0.02338	1.477	0	0.00e+00
mlp_10m	Separate CG	0.7532	2.844	61440	6.17e-04
mlp_10m	Batched CG	0.5674	4.219	61440	6.60e-04
mlp_10m	Jacobi-PCG	0.5825	5.370	61440	7.65e-04
mlp_10m	Diagonal	0.4889	1.839	0	2.76e+02
mlp_10m	Last layer	0.002785	2.387	0	–
mlp_10m	Explicit dense reference	skipped	–	–	–

Exact-linear and solver sanity checks. In the fixed-feature linear studies, the dense exact-current and full-CG rows exercise the specialization $\bar{D}_t = 0$. The dense row checks the LinUCB reduction; the full-CG row checks original-operator residual logic and float64 all-action width agreement against dense solves. These are algebra and implementation diagnostics, not verified numerical enclosures. They do not test nontrivial metric transport. Approximate-operator rows record upper comparisons or point generalized eigenvalues, not a reportable positive lower certificate $\kappa_{-,t}$.

Nonlinear path and operator mechanisms. The nonlinear drift audit directly checks the corrected-center identity and records replay-path statistics, but its schedules fail and its scores predate the revised theorem. Off-diagonal and operator-ablation studies provide diagnostic examples in which operator approximation error, width distortion, action disagreement, and regret do not share a total ordering. They motivate separating the three metrics in Section 2; they are not theorem instantiations. The spectral-tail study's `rank_truncation` control explicitly sets discarded-coordinate widths to zero. That control cannot be represented by a finite SPD operator satisfying (8).

Autodiff feasibility. The real-`torch.func` study demonstrates JVP/VJP application of a GGN primitive and records residual, operator-call, time, and memory diagnostics. It measures isolated linear-algebra primitives rather than a transported-UCB policy. The larger model has no dense reference oracle. The study therefore supports implementation feasibility only.

Negative and incomplete benchmarks. The balanced synthetic comparison does not establish a full-curvature advantage. The MNIST study remains near its context-free baseline, and Covertype fails its contextual baseline gate. None of these three benchmarks executes the corrected transported score or records the revised certificates. The Wheel study terminated during tuning and had no reportable evaluation. The detailed legacy outputs are not part of the compact checkout; Git history preserves them. Their plots and tables do not contribute to the present theory-first paper.

N Broader Impact

Exploration can expose users to under-vetted actions and amplify feedback loops. Deployment requires safety constraints, action filters, offline evaluation, and online monitoring; the regret certificates here do not establish safety.

O Data and Code Availability

The repository contains the manuscript, theory derivations, implementation, tests, experiment configurations, artifact generators, and the selected compact diagnostic aggregates used by the retained appendix tables. The manuscript does not use the legacy experiments as evidence for Theorem 1; their scope is summarized in Appendix M. Detailed legacy figures, generated tables, raw-bundle inventories, and derived benchmark outputs are not part of the compact checkout; Git history preserves them. No public raw-bundle URL is claimed.